Model Decay & Cold Start

A machine learning model is not a statue; it is a living organism that begins to die the moment it is deployed. This session explores the 'Entropy of Intelligence'—understanding why models fail in the wild (Model Decay), the struggle of first-day performance (The Cold Start Problem), and the technical architecture required to keep a system relevant through robust monitoring and feedback loops.

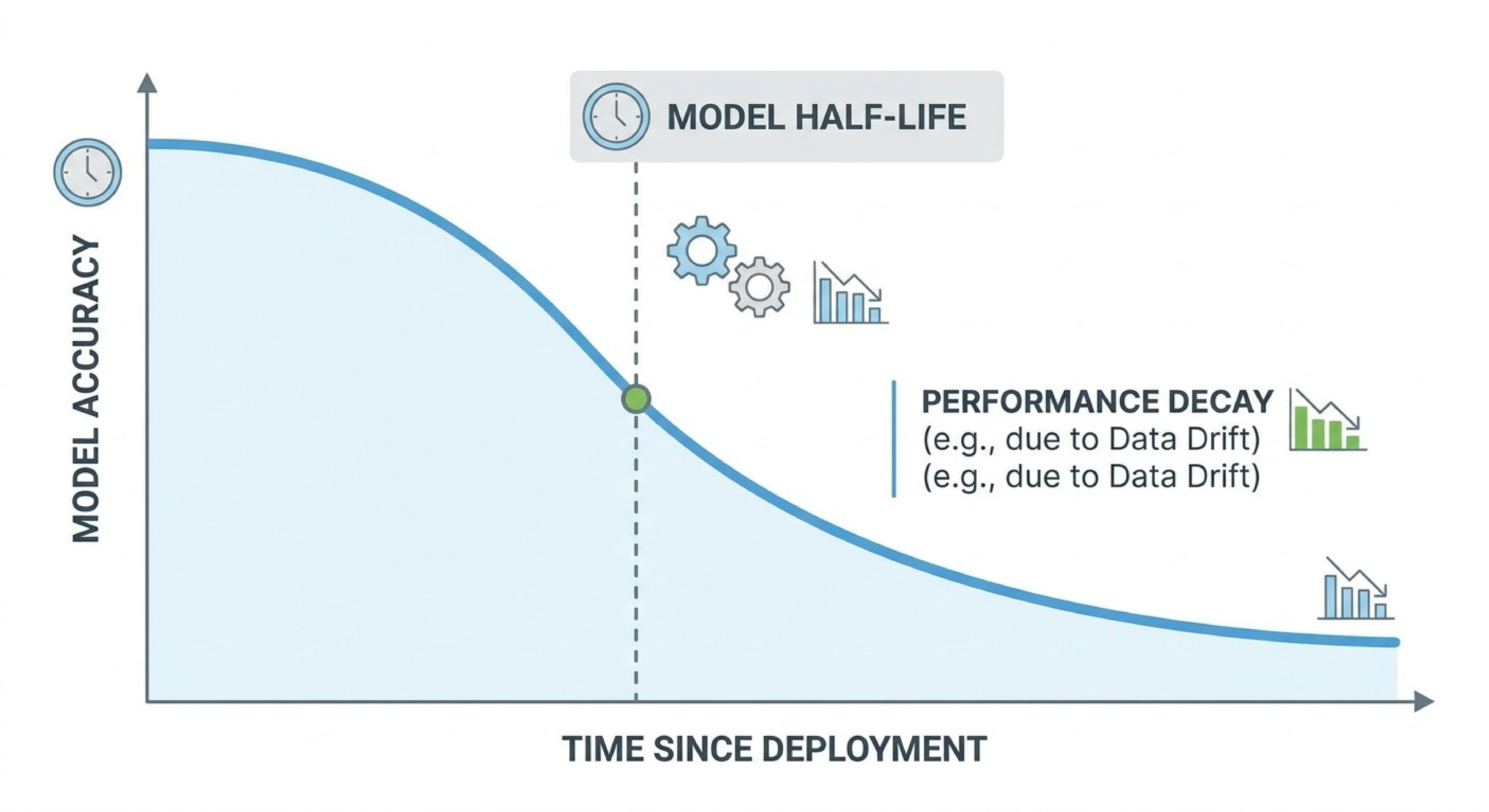
ECON484 Machine Learning

Bora GÜNGÖREN

The Entropy of Intelligence (Model Decay)

Why Models Begin to Die at Deployment

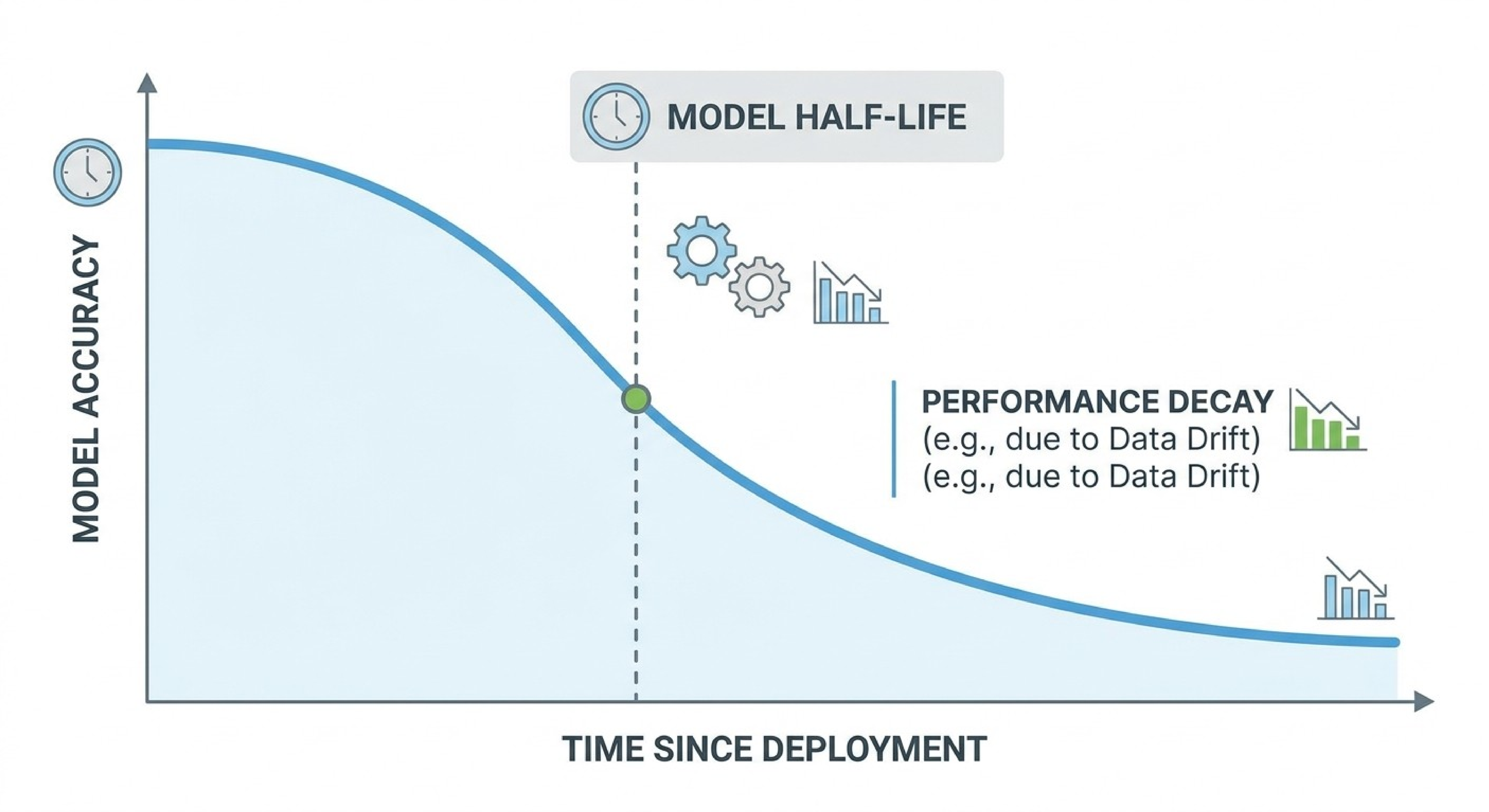
The Static vs. Dynamic Paradox: A model is a frozen mathematical snapshot of a specific moment in time. The world, however, is non-stationary.



The Entropy of Intelligence (Model Decay)

The "Freshness" Half-Life: Just like radioactive isotopes, every model has a half-life where its predictive power drops by 50%.

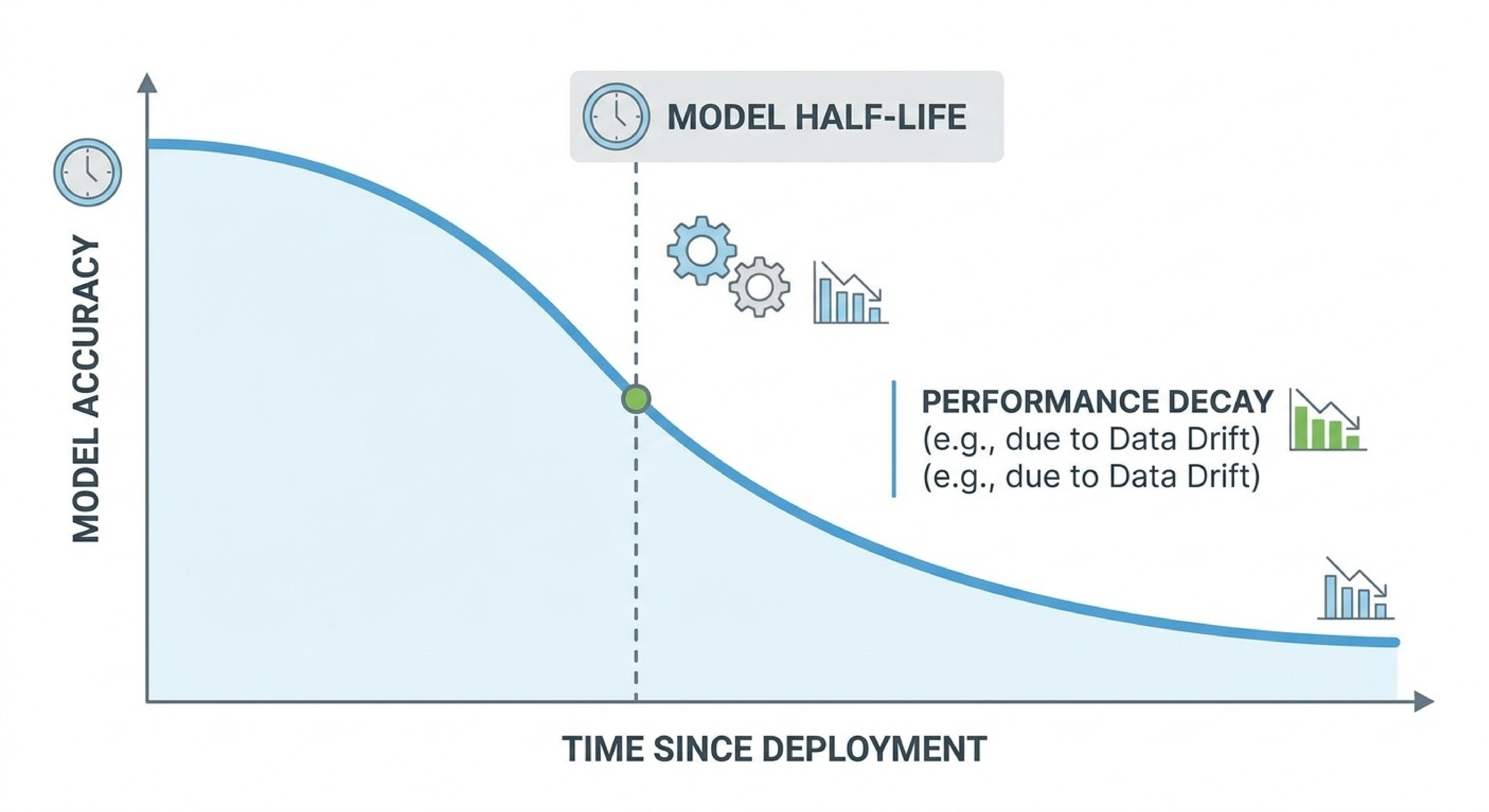
Operational Reality: Deployment is not the finish line; it is the start of a countdown toward obsolescence.



The Entropy of Intelligence (Model Decay)

Data Drift (Feature Drift) starts when the input starts to change.

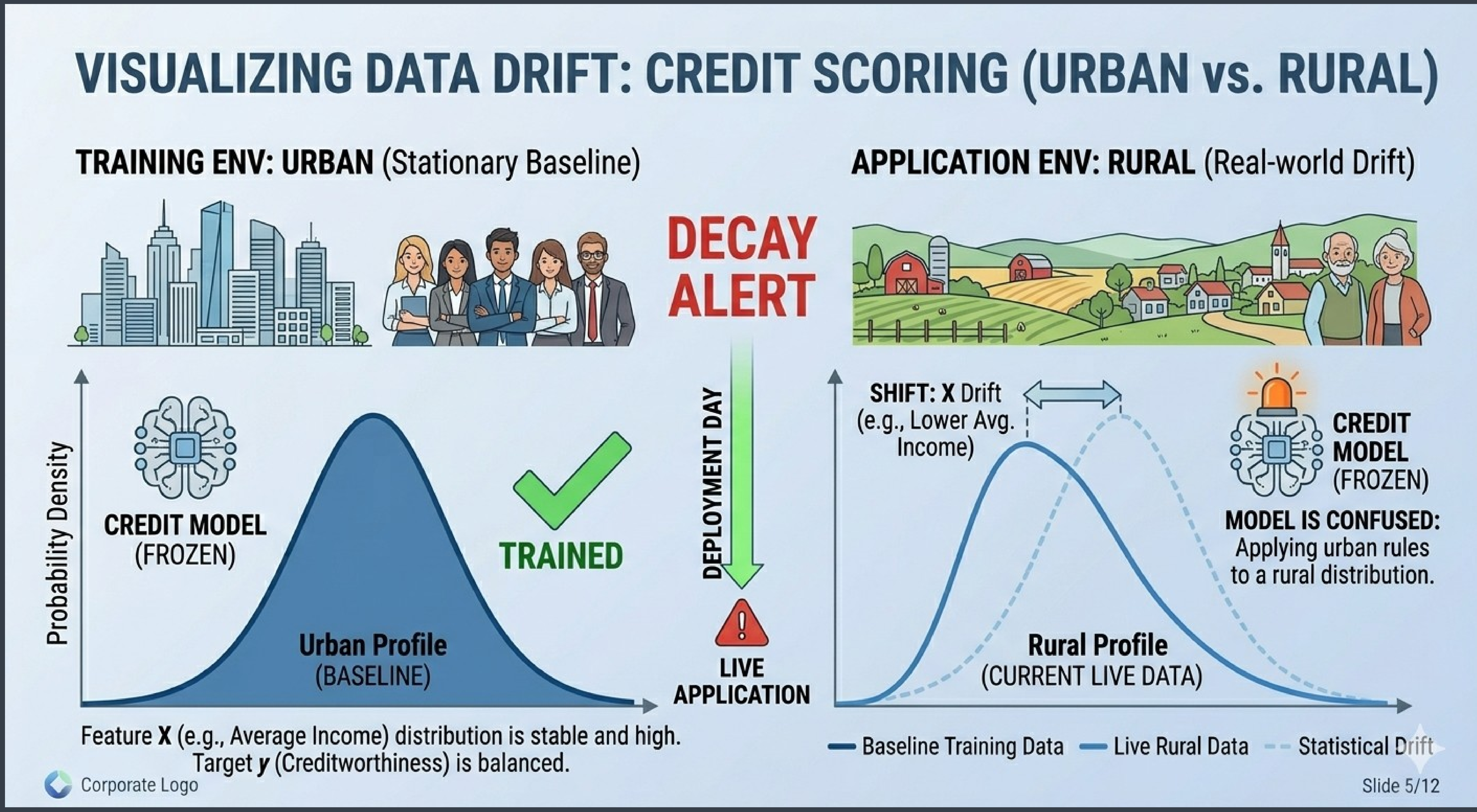
Formal Definition: The statistical properties of your input data (X) change, while the relationship to the target (y) remains the same.



The Entropy of Intelligence (Model Decay)

Example: A credit scoring model trained on urban data is suddenly applied to a rural demographic. The features (income, spending) have different distributions

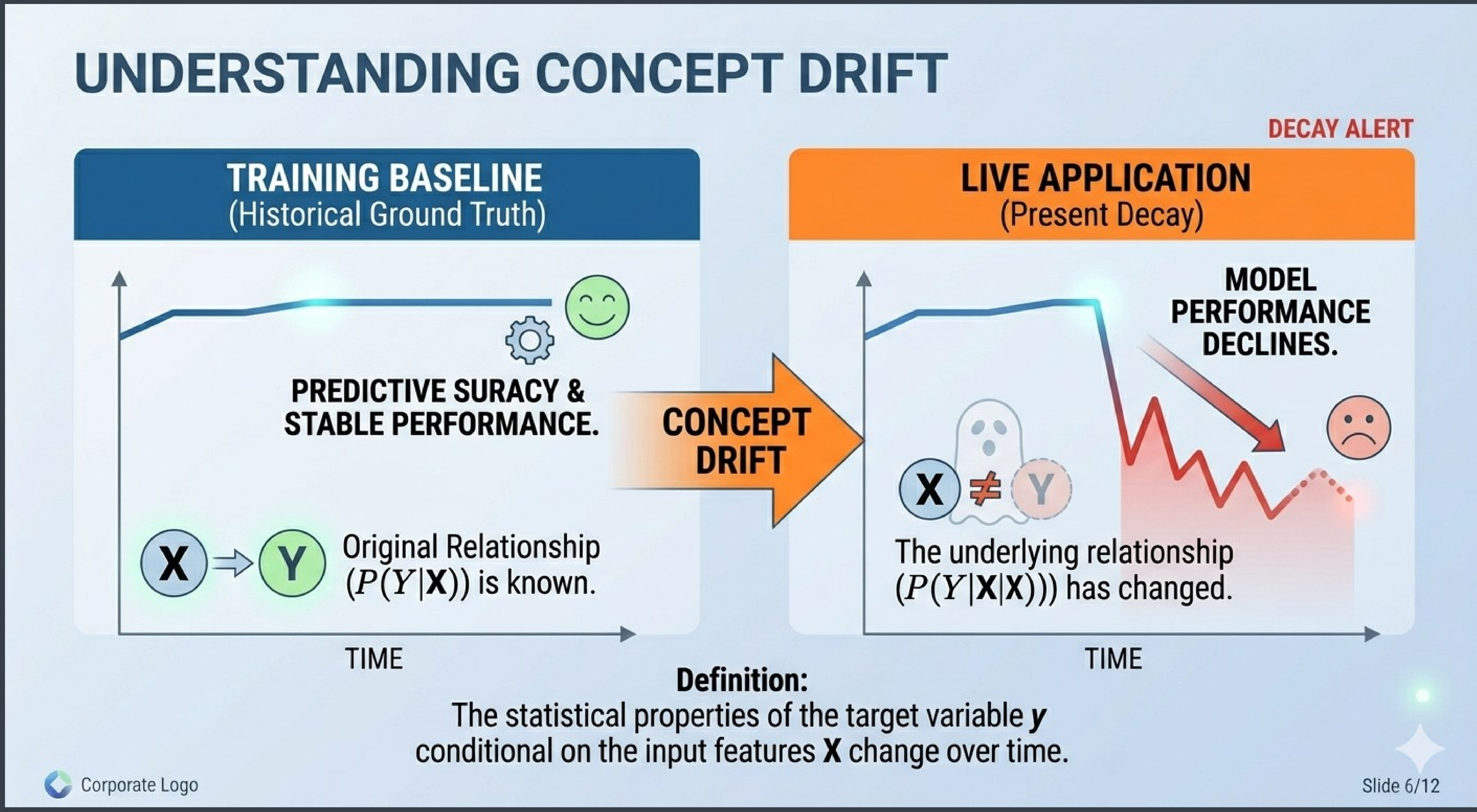
Visualizing the Shift: Use Population Stability Index (PSI) or Kullback-Leibler (KL) divergence to measure how far your live data has drifted from your training baseline



The Entropy of Intelligence (Model Decay)

Concept Drift happens when the rules change.

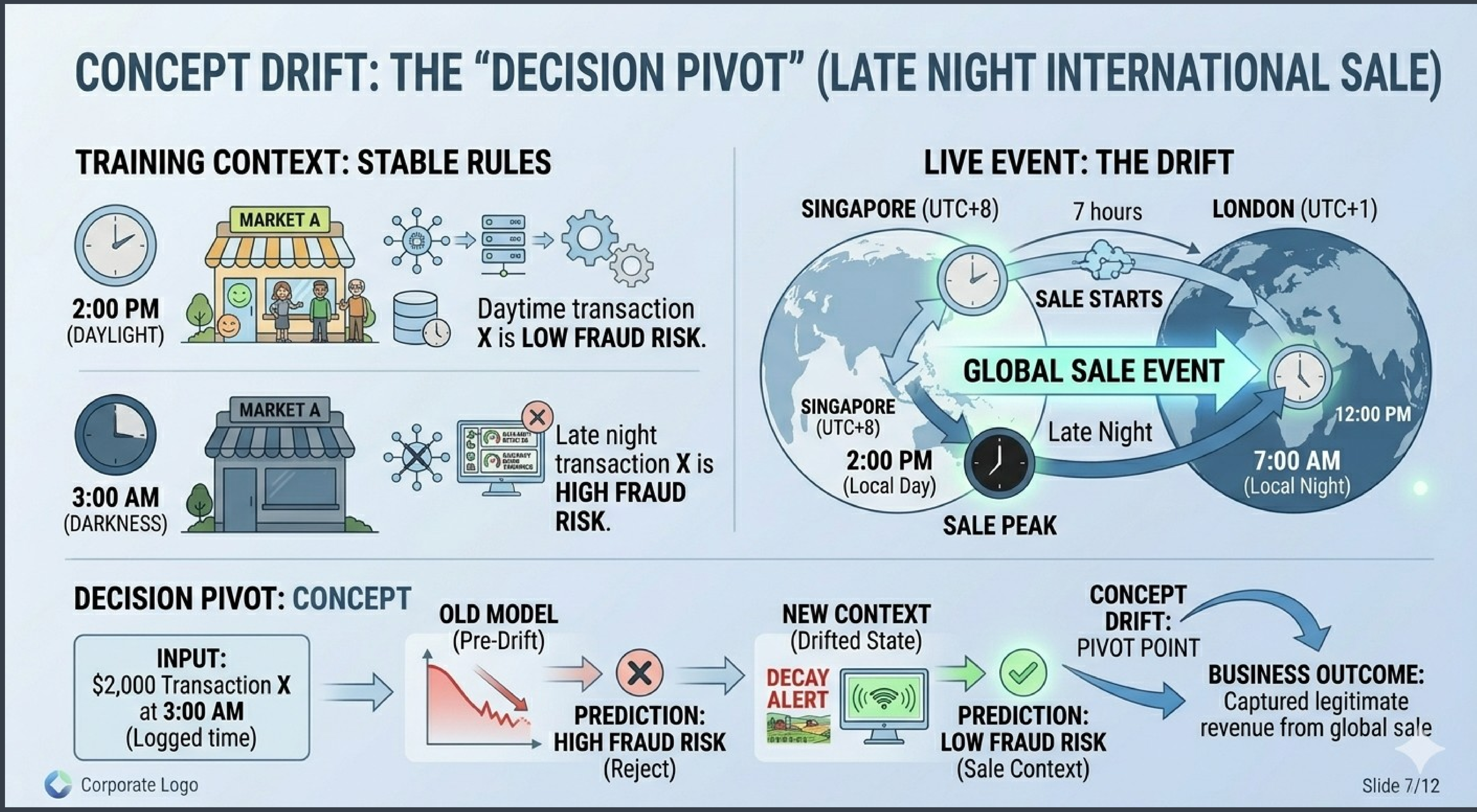
Formal Definition: The fundamental relationship between the input and the target (P(y | X)) changes. The "meaning" of the data has evolved.



The Entropy of Intelligence (Model Decay)

Example (Decision Pivot): Yesterday, a $5,000 transaction at 3 AM was "Fraud." Today, during a global sale event, it is "Normal." Tomorrow will it be “Fraud” again?

Practical (Hidden) Danger: Unlike Data Drift, Concept Drift can happen even if your input distributions look perfectly normal, making it harder to detect without ground truth.

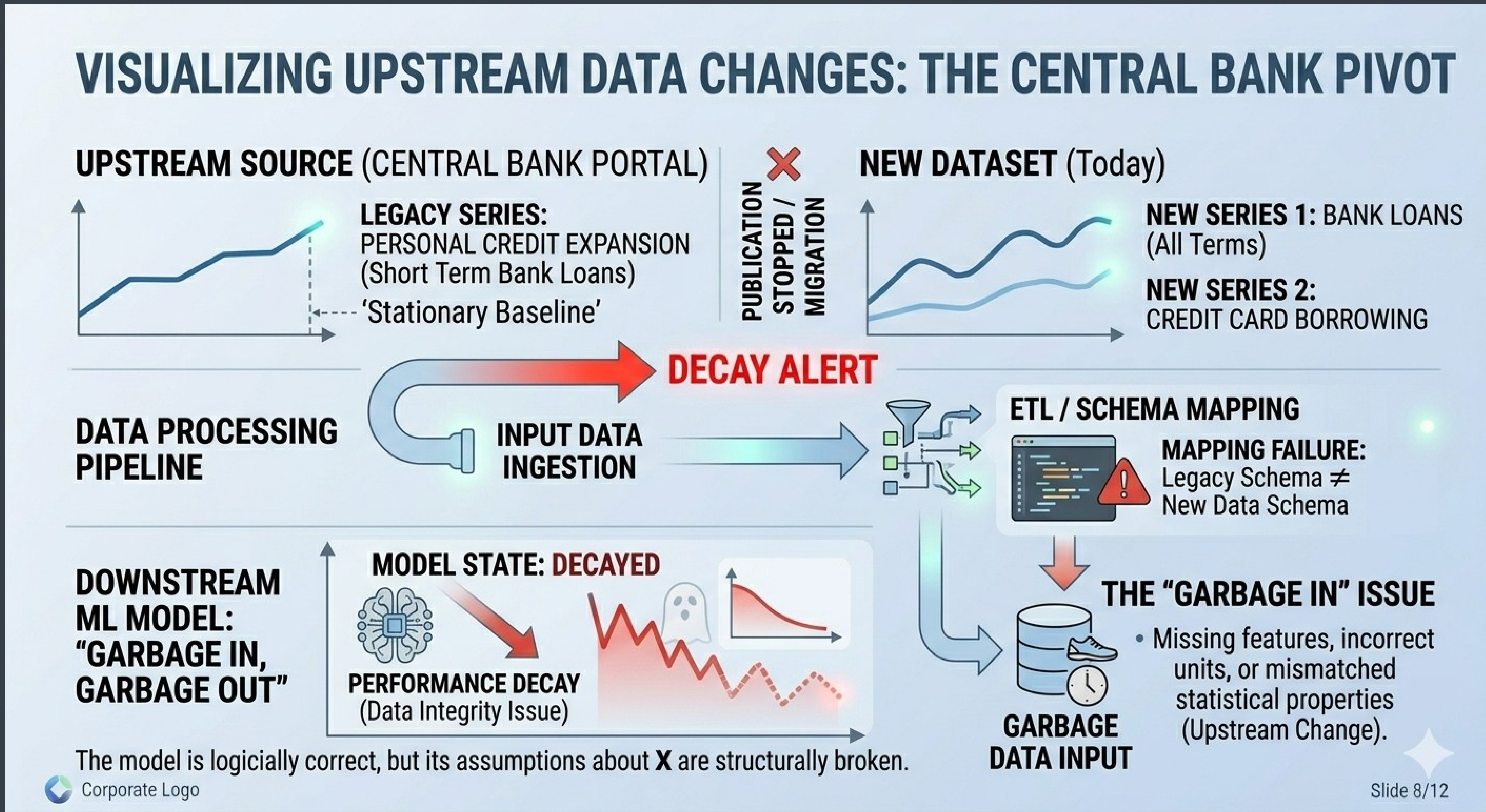


The Entropy of Intelligence (Model Decay)

Upstream data changes often happen without your consent or prior notice.

Example: A change in a third-party API, a sensor malfunction, updates to a device firmware, or a database schema update that isn't communicated to the ML team.

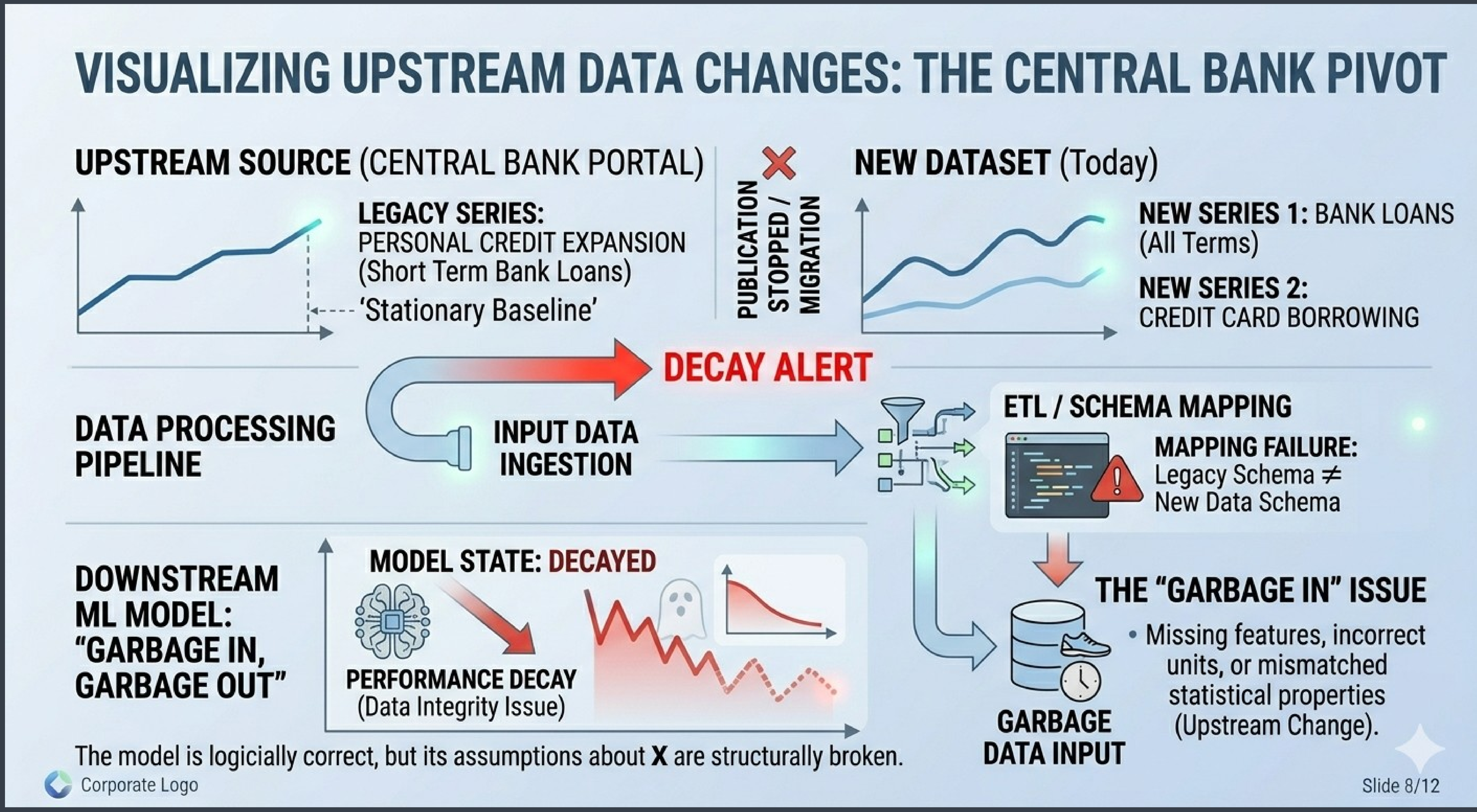
Where you source your inputs is considered “upstream” (like a river).



The Entropy of Intelligence (Model Decay)

Data Integrity vs. Model Accuracy: From your point of view, the model isn't wrong in its logic; it is being fed "garbage" due to structural changes.

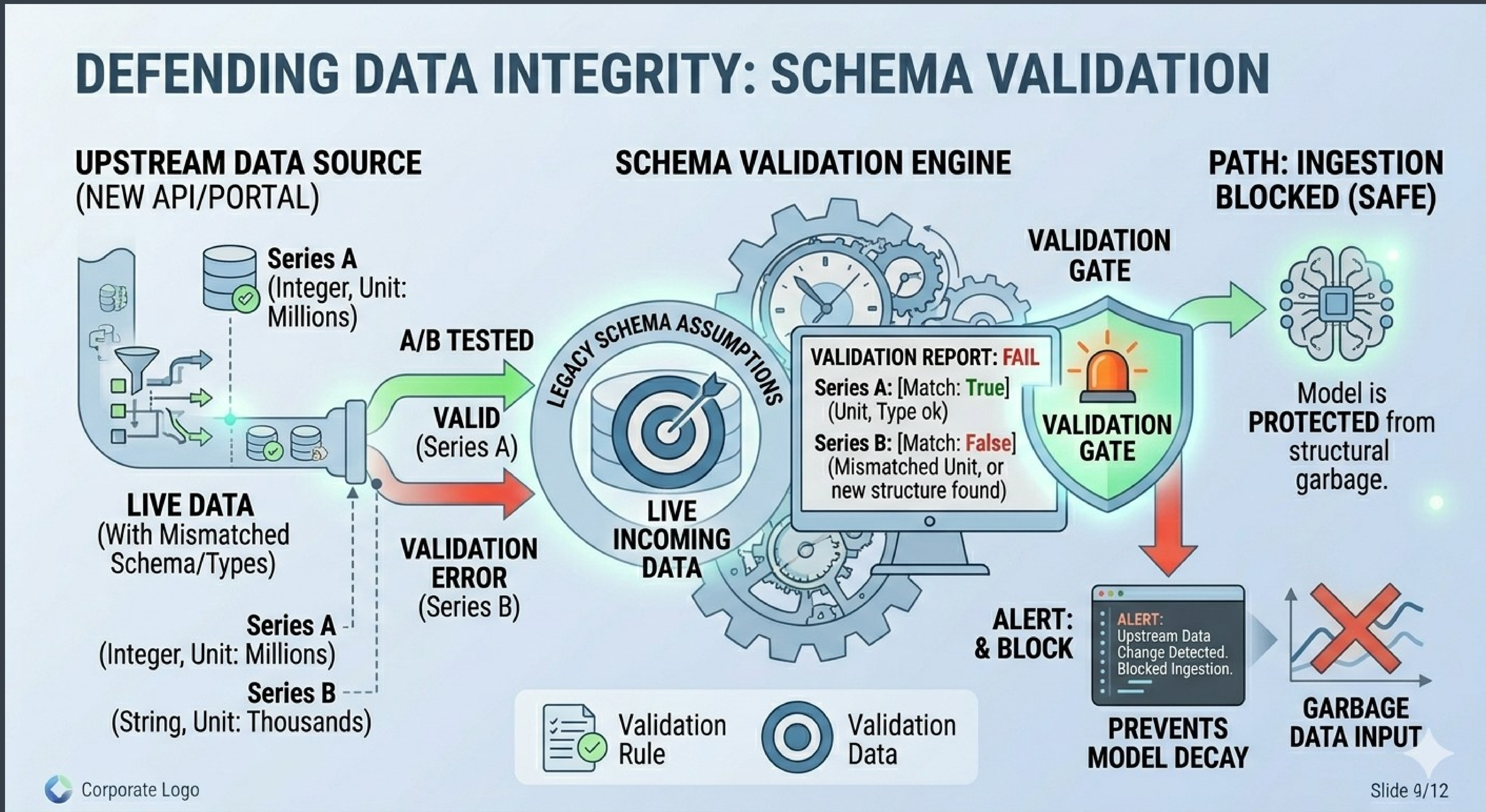
On the other hand, upstream changes are bound to happen and you have to adapt.



The Entropy of Intelligence (Model Decay)

You have to detect upstream changes yourselves. You cannot rely solely on notifications.

Change Detection: Schema validation and null-value monitoring are the first line of defense before statistical drift analysis.

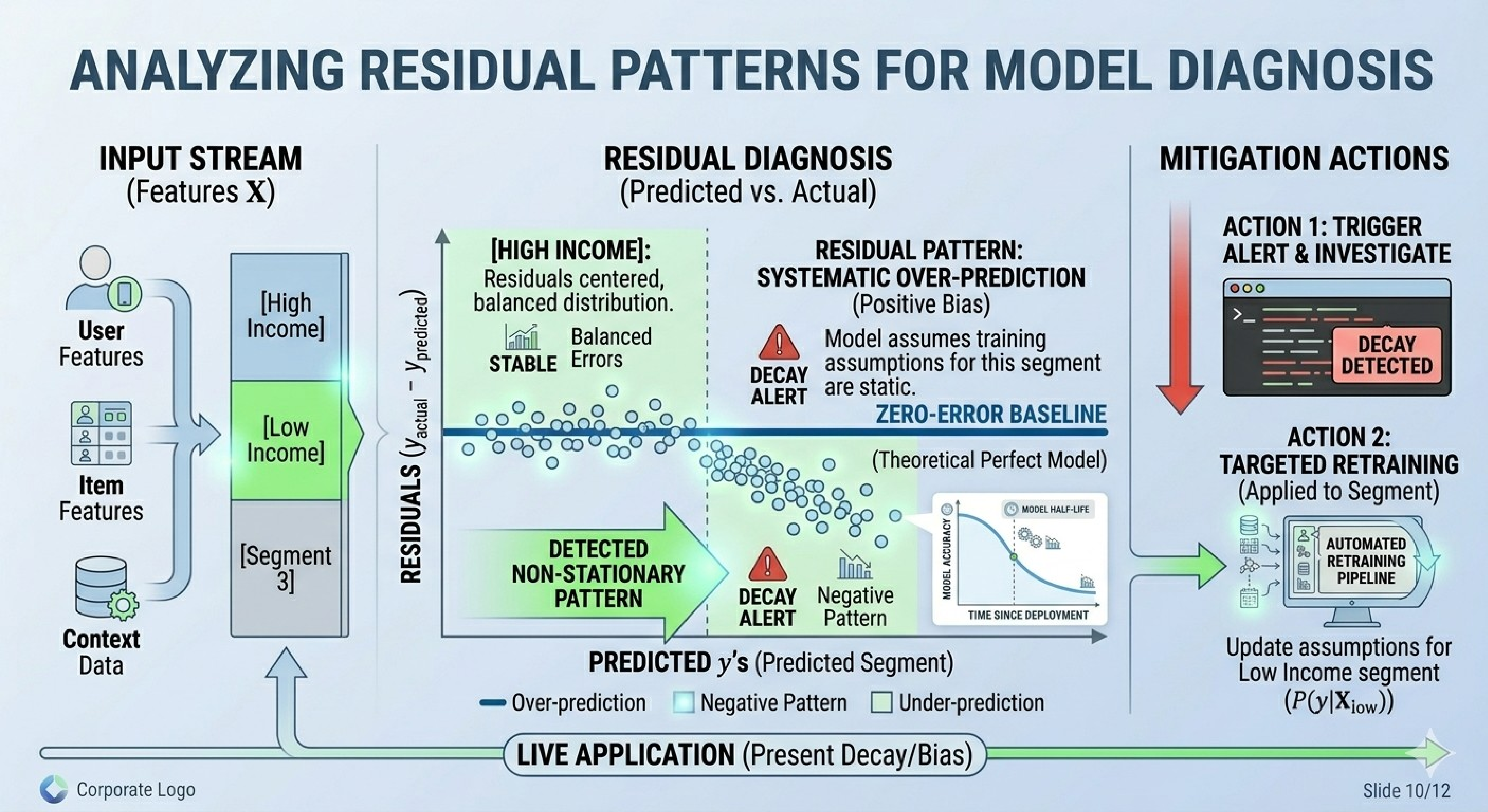


The Entropy of Intelligence (Model Decay)

Often a “Residual Audit” is conducted to measure the decay. Measurements on aggregate accuracy cannot be used alone.

Pattern Detection (Residual Analysis):
Look at the patterns of the errors.

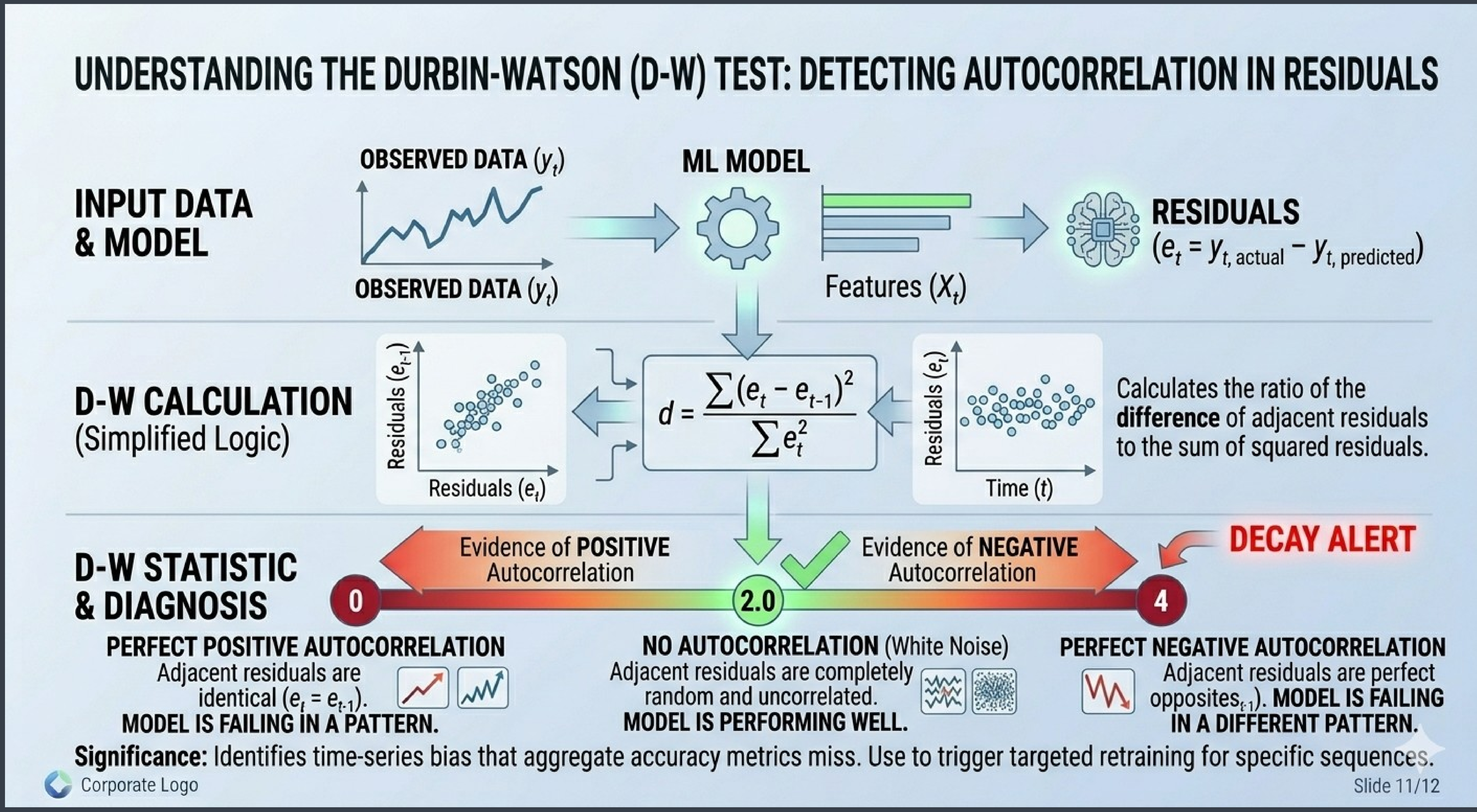
We will summarize some methods in class but not discuss in details.



The Entropy of Intelligence (Model Decay)

Detecting Autocorrelation (Time/Sequence Patterns)

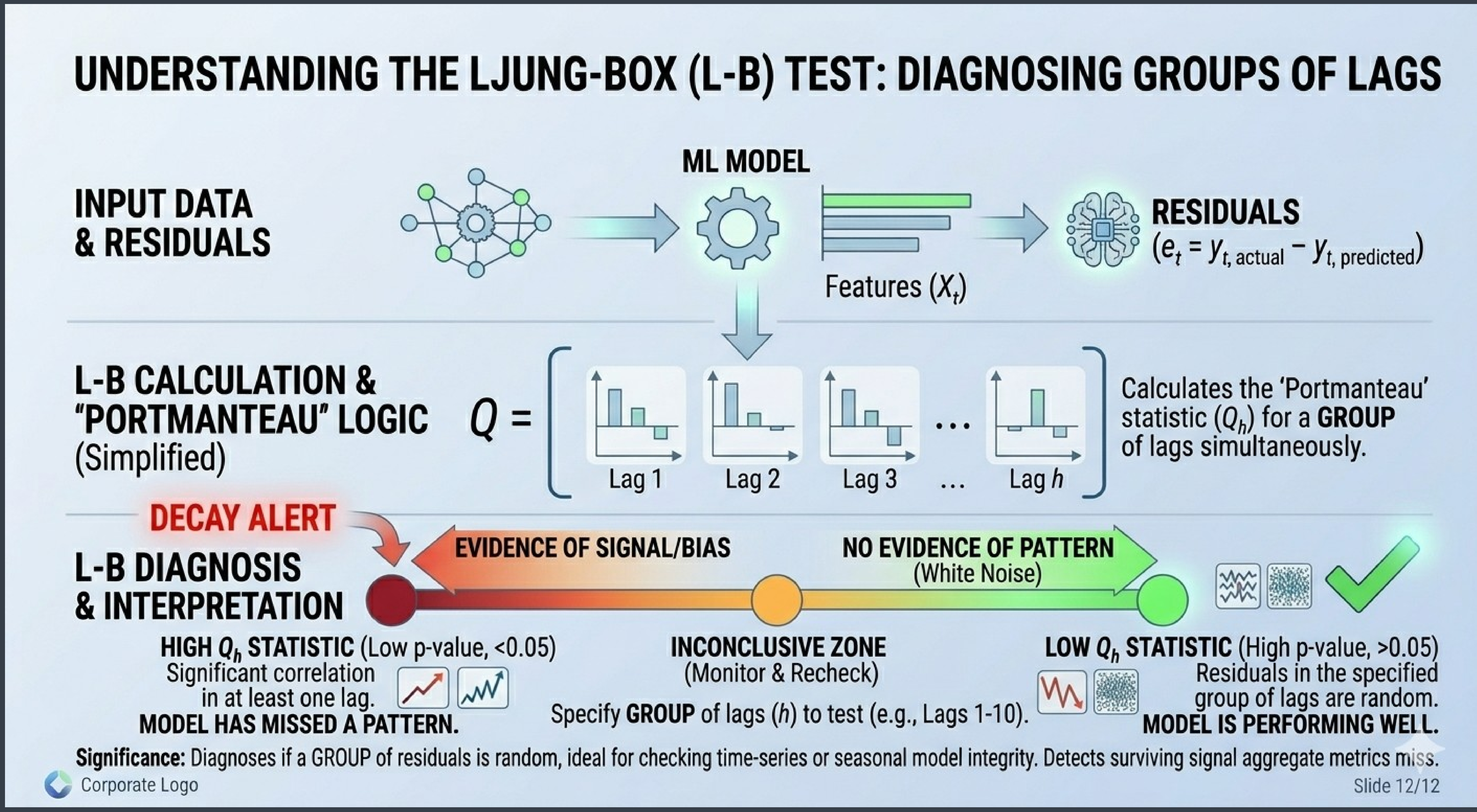
Durbin-Watson Test: A classic test that produces a statistic between 0 and 4. A value of 2.0 means no autocorrelation. Values approaching 0 indicate positive correlation (runs of similar errors), while values approaching 4 indicate negative correlation.



The Entropy of Intelligence (Model Decay)

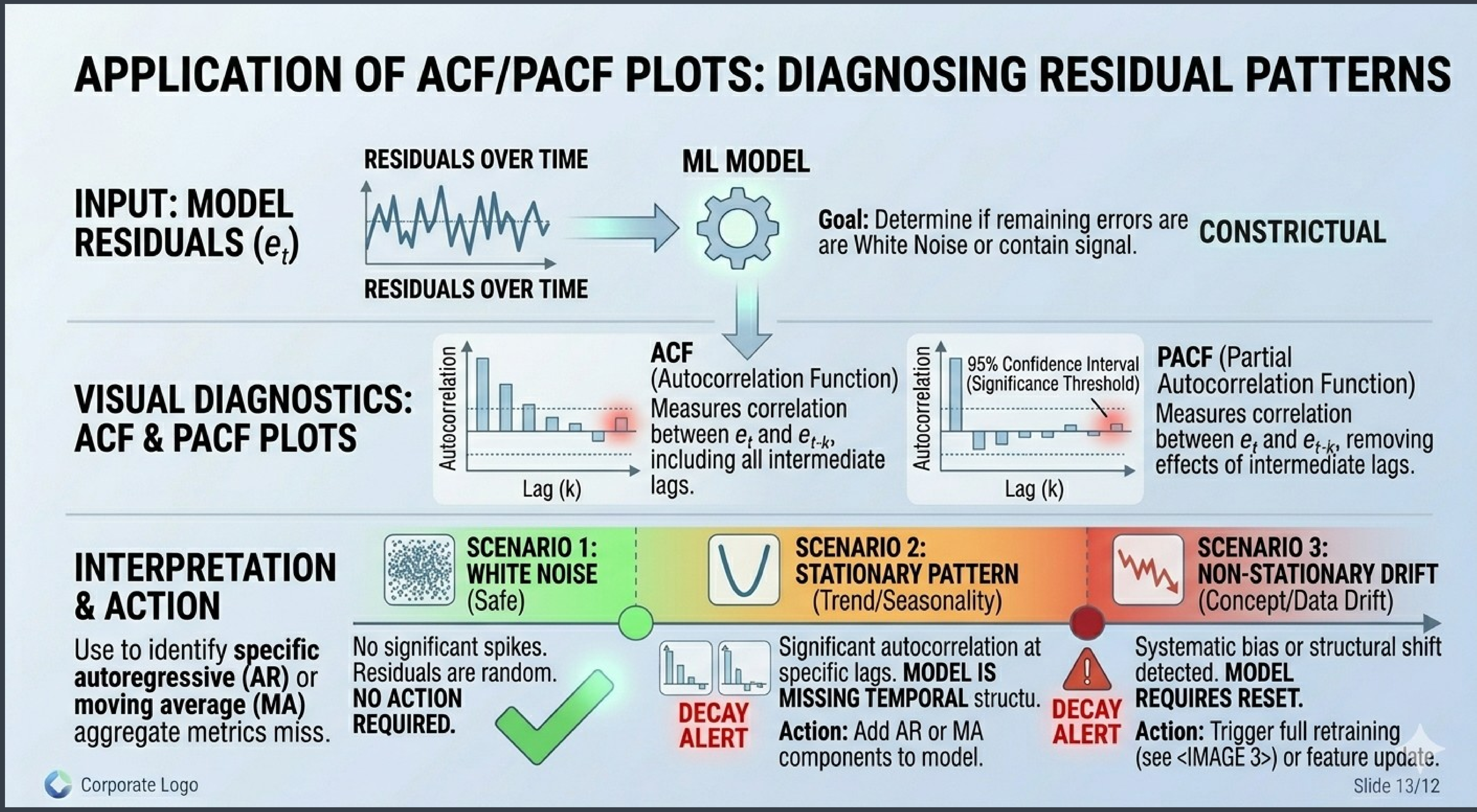
Ljung-Box Test: Often used in time-series (ARIMA) modeling. It examines a "portmanteau" (group) of lags at once to see if any significant correlation exists in the residuals.

A lag represents a fixed time delay between an observation and a previous data point.Sample autocorrelation for each (assumed) lag tells us how much the residual at time t correlates with the residual at t−k.



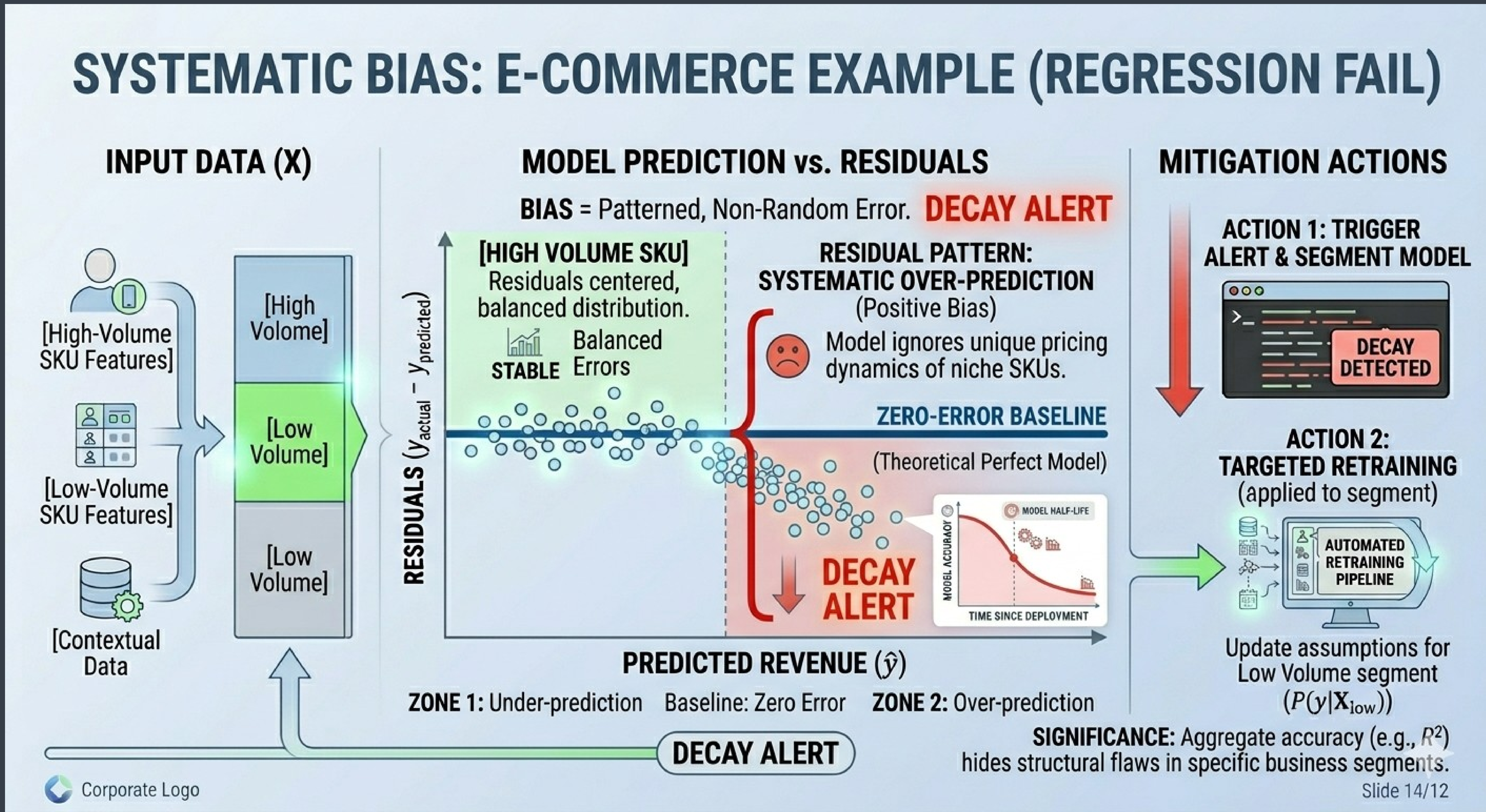
The Entropy of Intelligence (Model Decay)

ACF/PACF Plots: Visualizing the Auto-Correlation Function. If any bars cross the "blue" significance threshold, your model is likely missing a seasonal or trend component.



The Entropy of Intelligence (Model Decay)

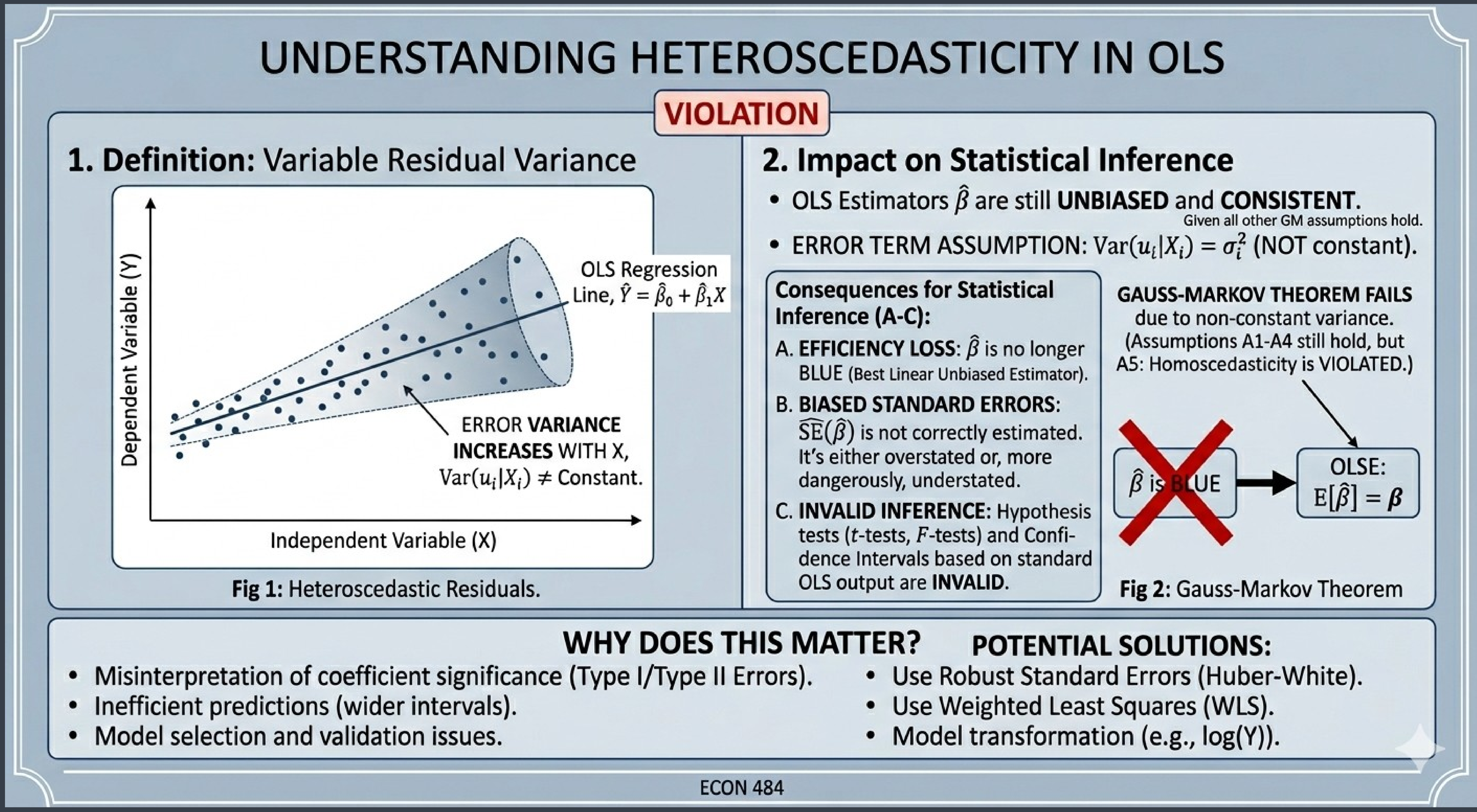
Systematic Bias: If the model starts consistently over-predicting or under-predicting in a specific segment, decay is occurring.



The Entropy of Intelligence (Model Decay)

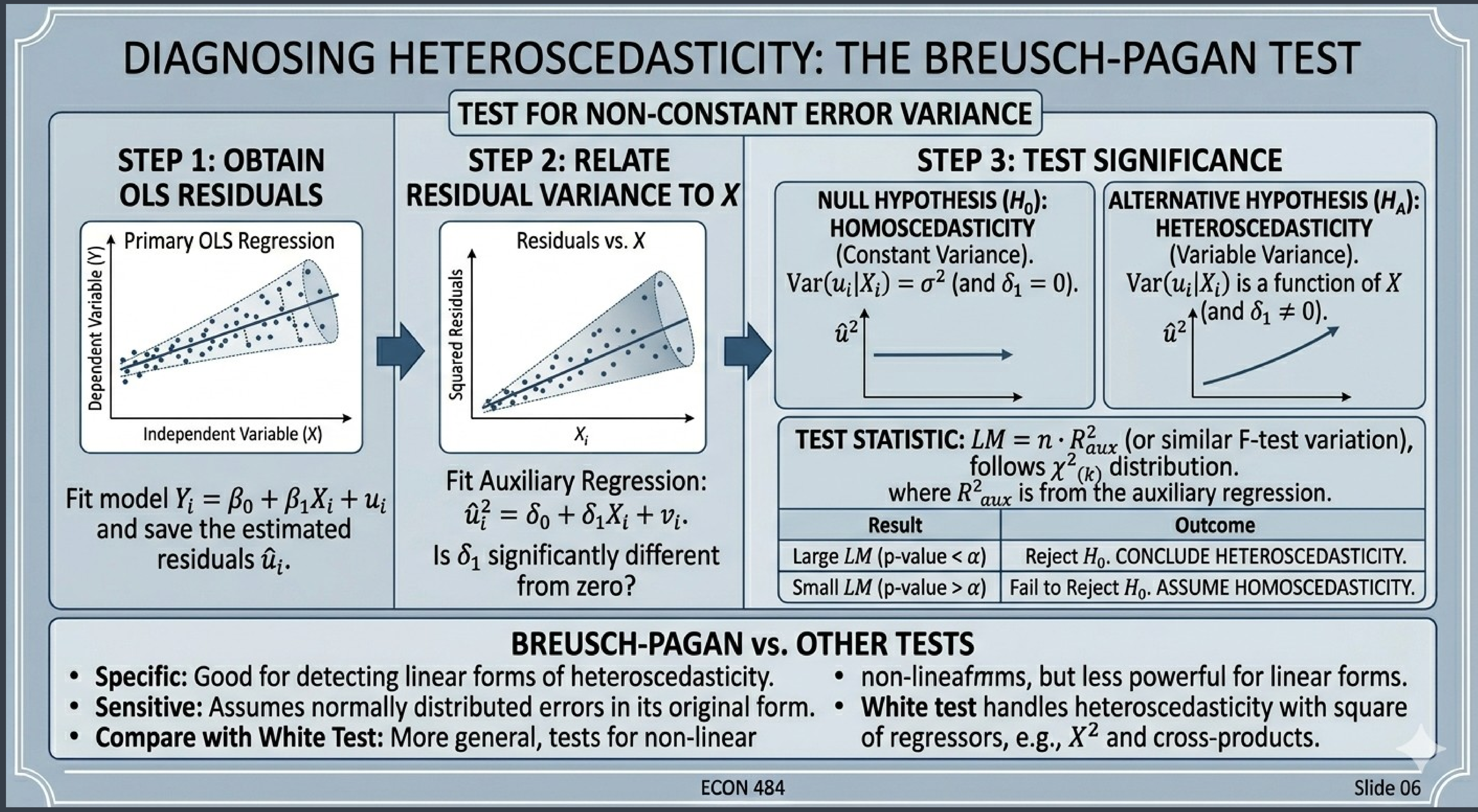
Detecting Heteroscedasticity (Variance Patterns)

This occurs when the "spread" of your errors changes as the input values change (e.g., your model is very accurate for low-income users but wildly inaccurate for high-income users).



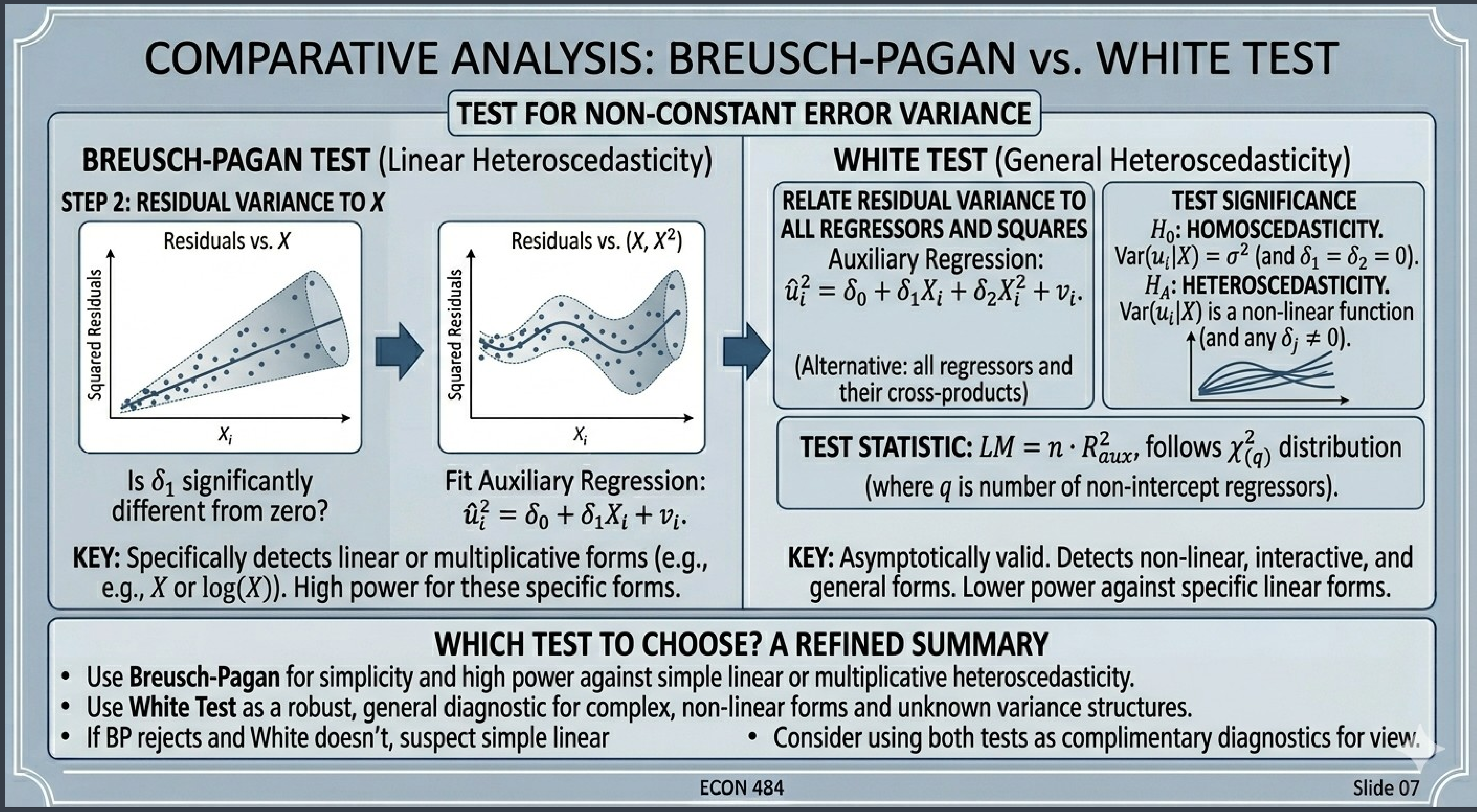
The Entropy of Intelligence (Model Decay)

Breusch-Pagan Test: Regresses the squared residuals against the independent variables. If the resulting p-value is low, you reject the null hypothesis of constant variance (homoscedasticity).



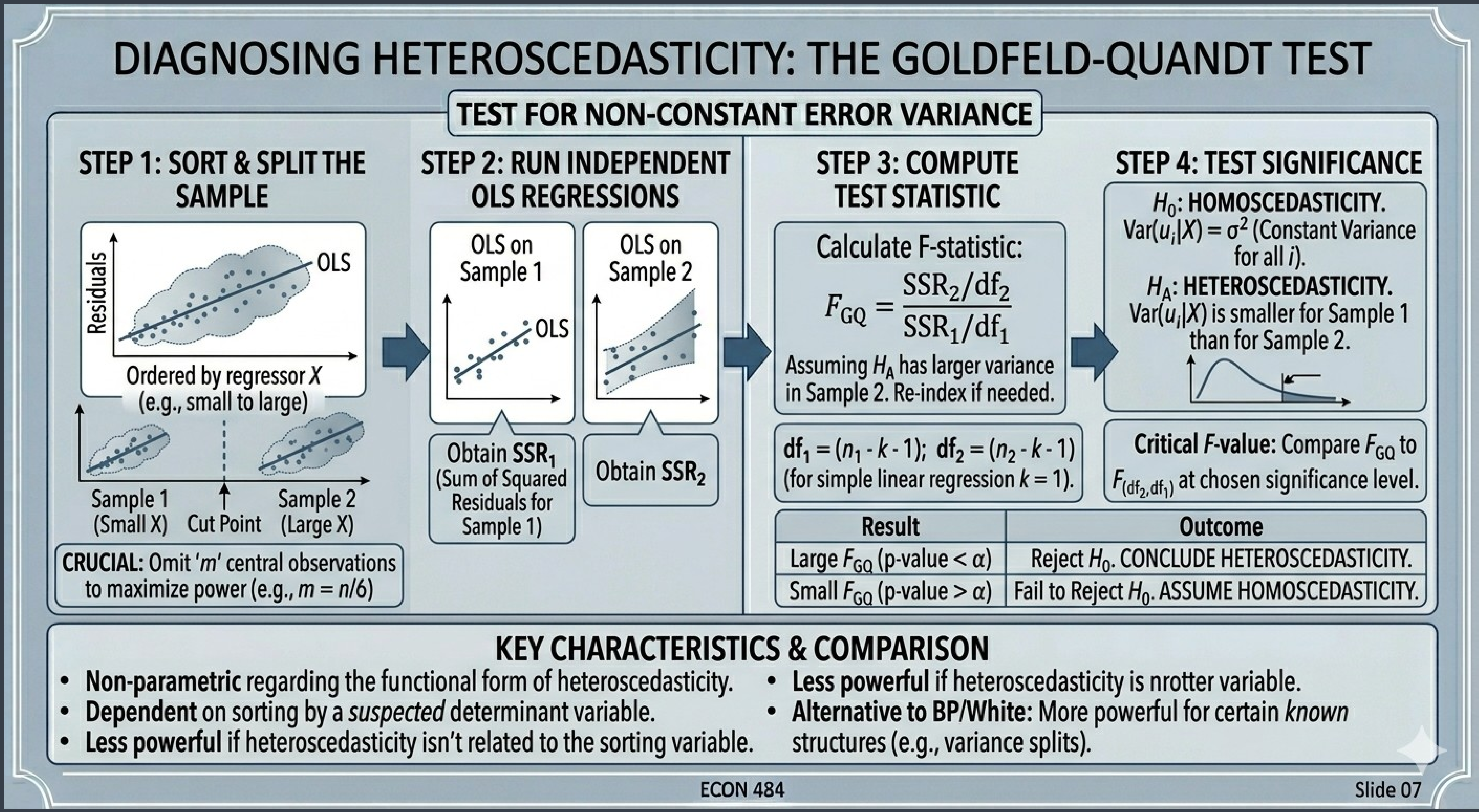
The Entropy of Intelligence (Model Decay)

White Test: A more general version of the Breusch-Pagan test that also accounts for non-linearities and interactions in the error variance.



The Entropy of Intelligence (Model Decay)

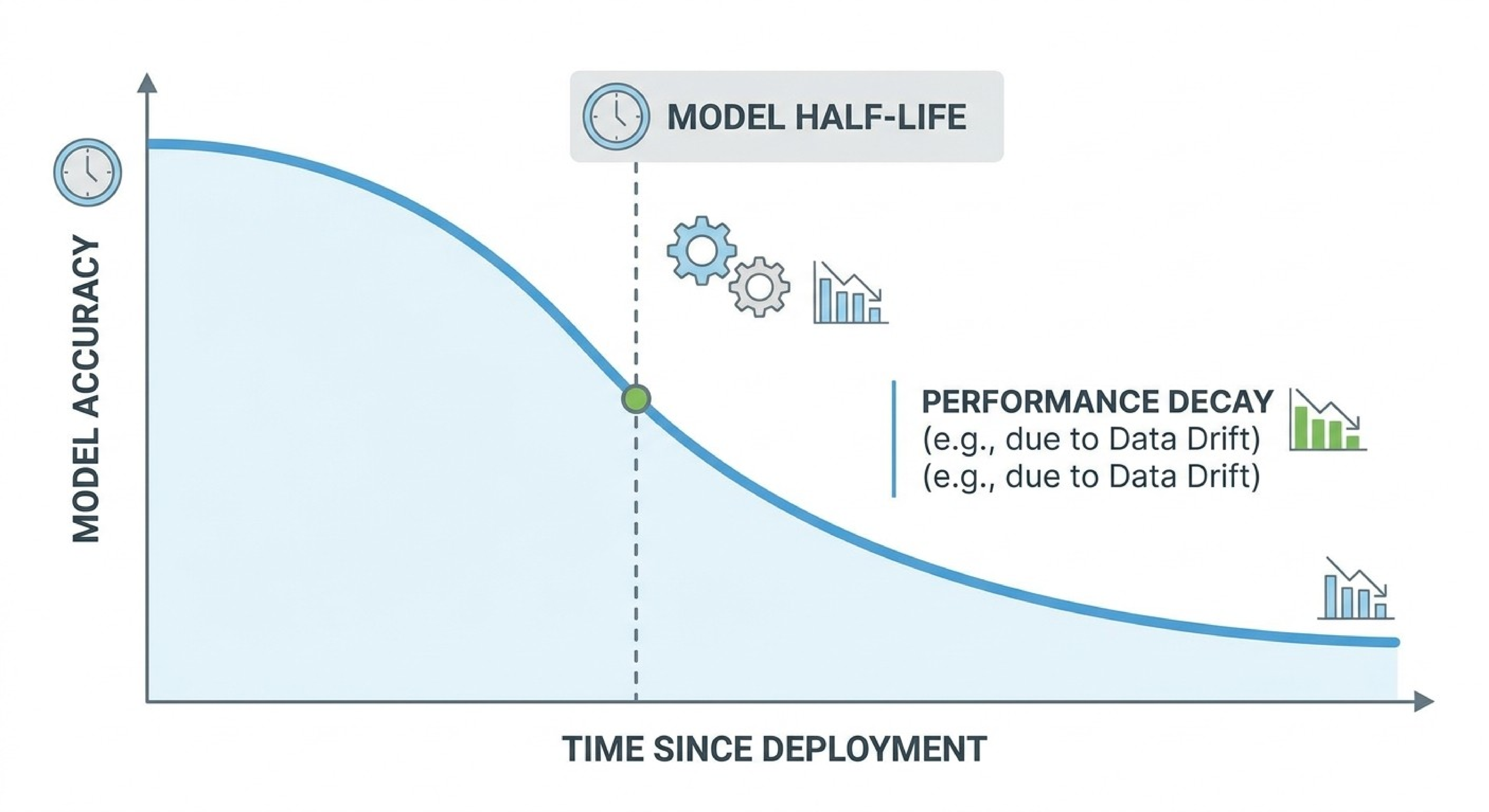
Goldfeld-Quandt Test: Useful for checking if the variance differs between two specific subsets of your data (e.g., Rural vs. Urban)



The Entropy of Intelligence (Model Decay)

The Decay Curve: Plotting performance over time to identify the exact "inflection point" where the model’s business value becomes negative.

Producing this curve in real-time requires a specific data architecture that bridges the gap between the moment a prediction is made and the moment the actual outcome (the "ground truth") is known.



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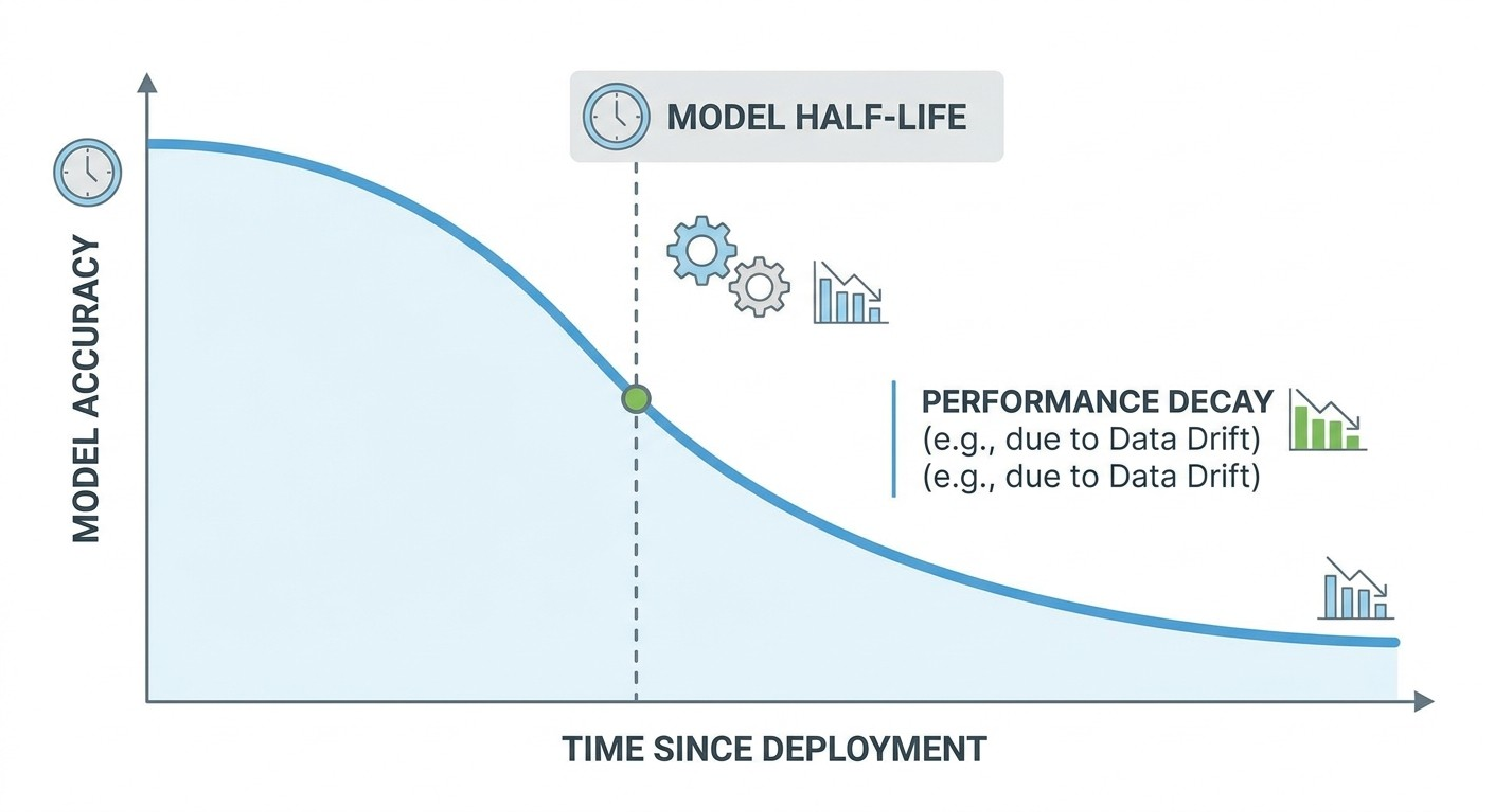
Prediction Logging (The Timestamp):
Every time the model makes a prediction in production, the system logs the input features, the model version, and a precise timestamp (Tpred).

Ground Truth Capture: The system waits for the actual outcome to occur. For example, in a "delivery time" model, this happens when the package is scanned at the door. In a "loan default" model, this might take months.

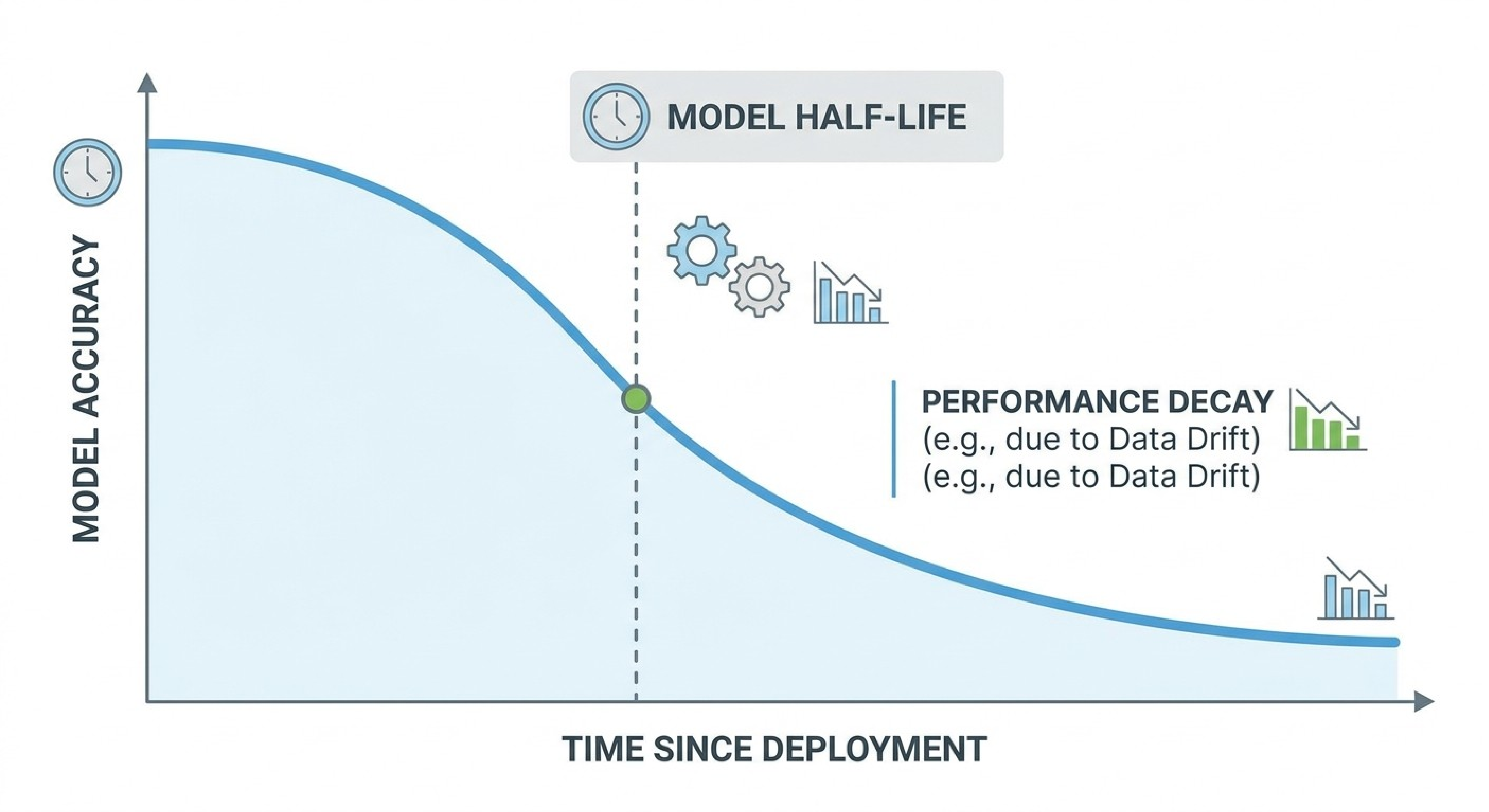


The Entropy of Intelligence (Model Decay)

The Feedback Loop (Join): A monitoring service performs a join between the logged prediction and the late-arriving ground truth using a unique Request ID.

Error Calculation: The system calculates the error metric (e.g., RMSE, Log-Loss, or Accuracy) for that specific observation.

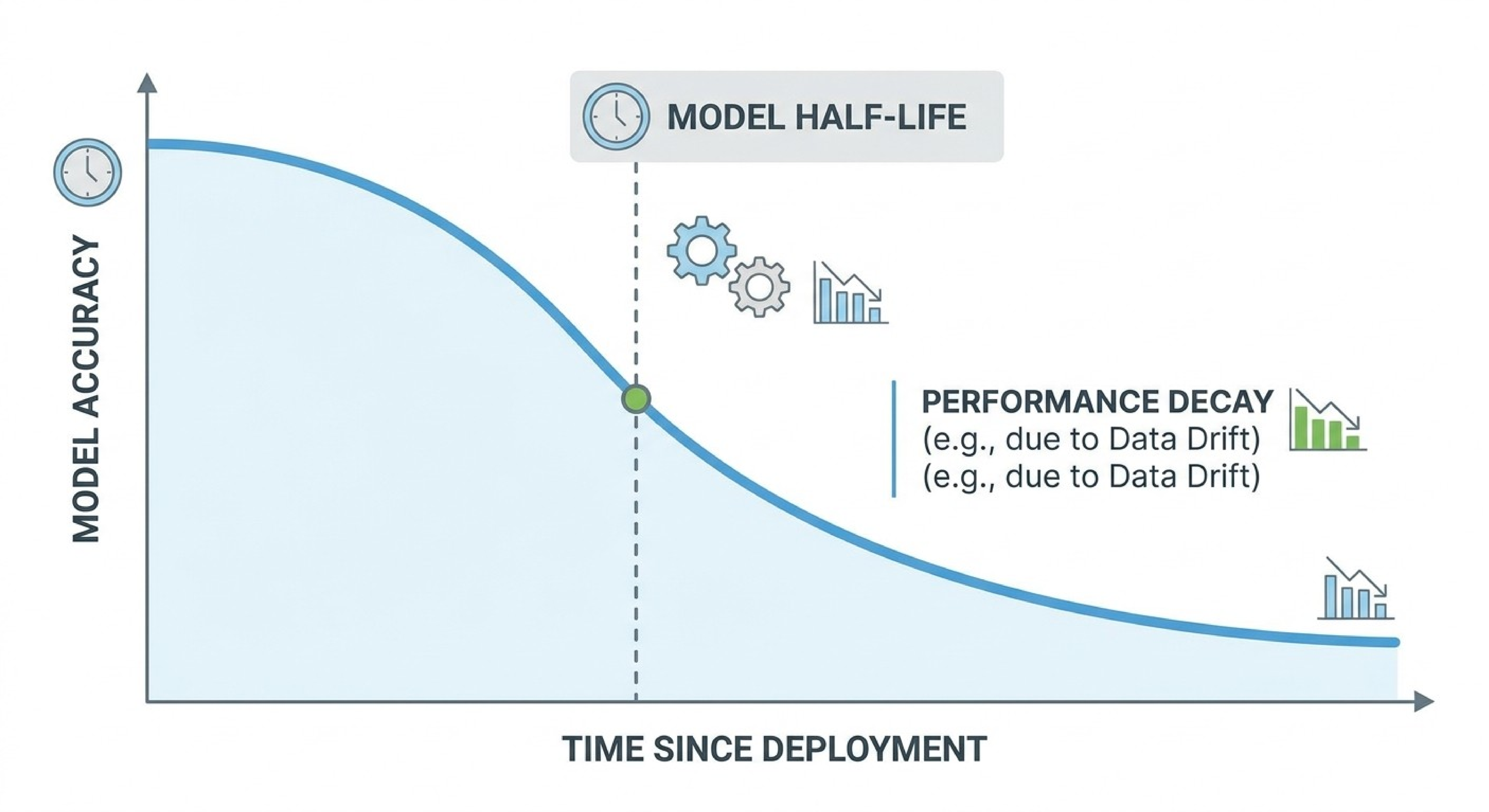
Temporal Aggregation: These errors are grouped by the "Age of the Model" (Current Time - Deployment Time).



The Entropy of Intelligence (Model Decay)

As the data flows through this pipeline, specific tests are used to "verify" the points on the decay curve and determine if the drop in performance is statistically significant.

PSI (Population Stability Index): Used to check if the distribution of the current incoming data has shifted compared to the training data. If PSI is high, you can predict the decay curve will drop before you even see the ground truth.



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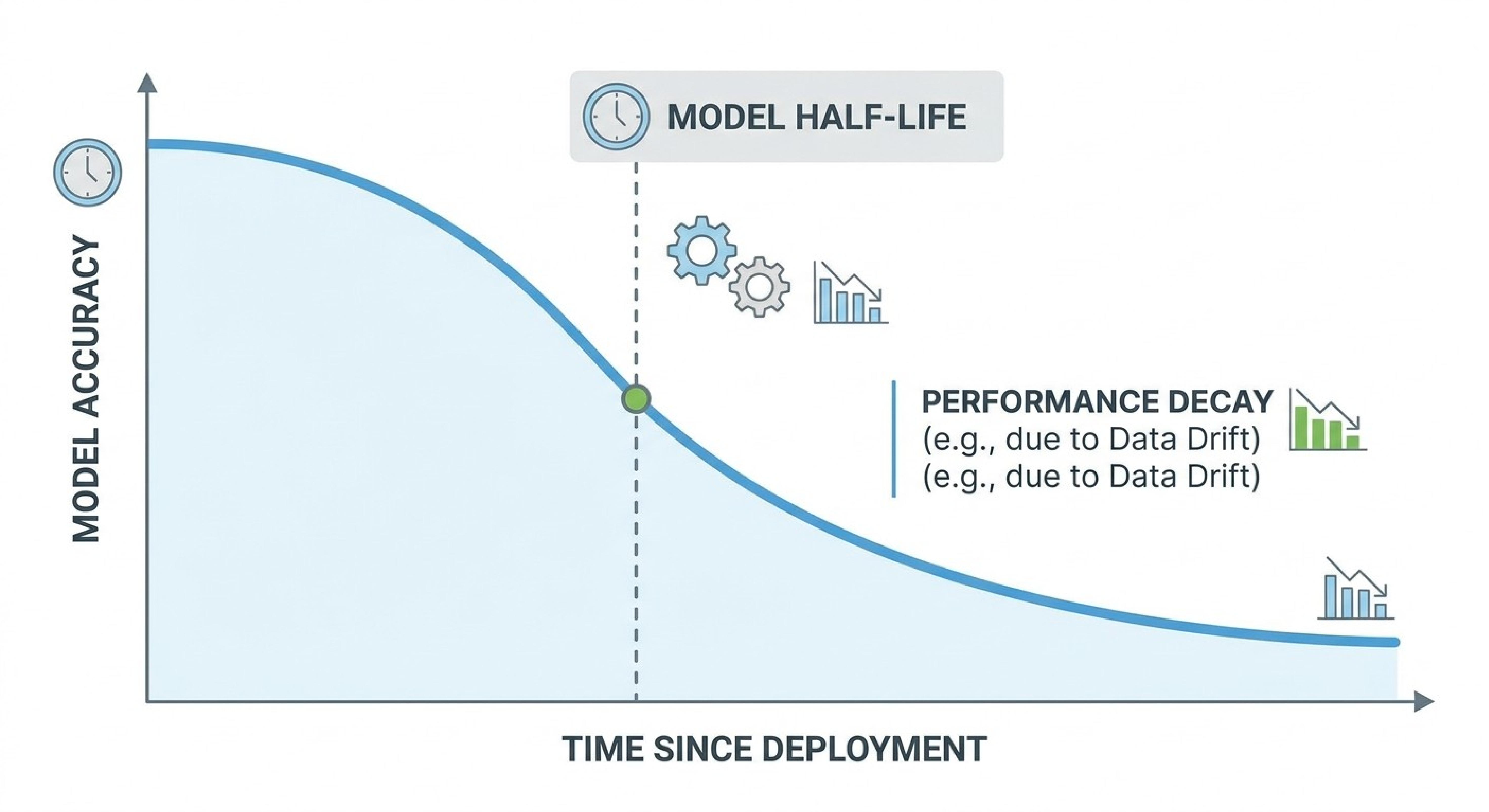
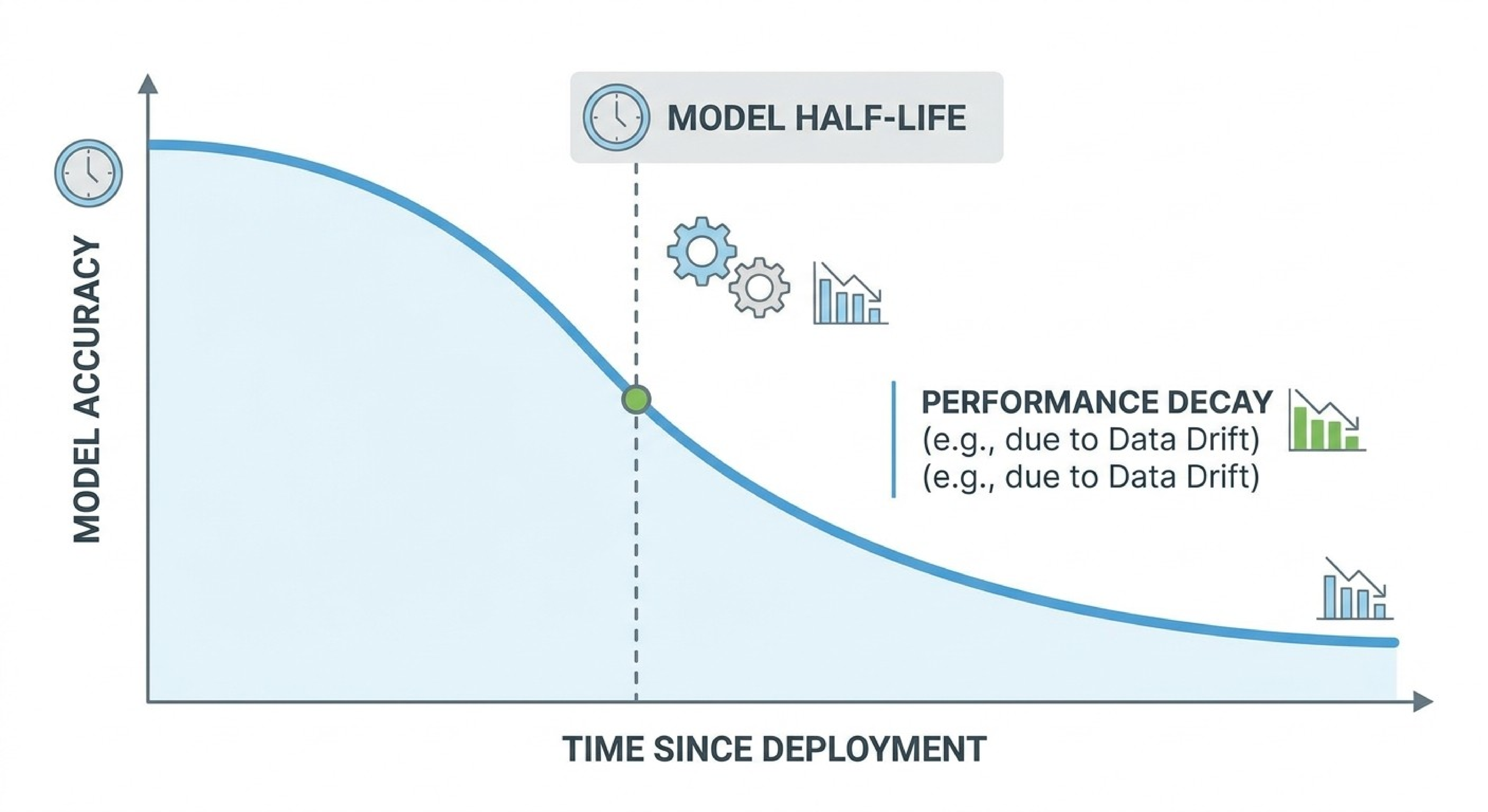


The Entropy of Intelligence (Model Decay)

Model decay has a significant effect on the life-time and thus “life-time value” of a model. It has to be addressed in your decision defense.

Define a Threshold: At what exact point of decay do we "pull the plug" and revert to a naive heuristic?

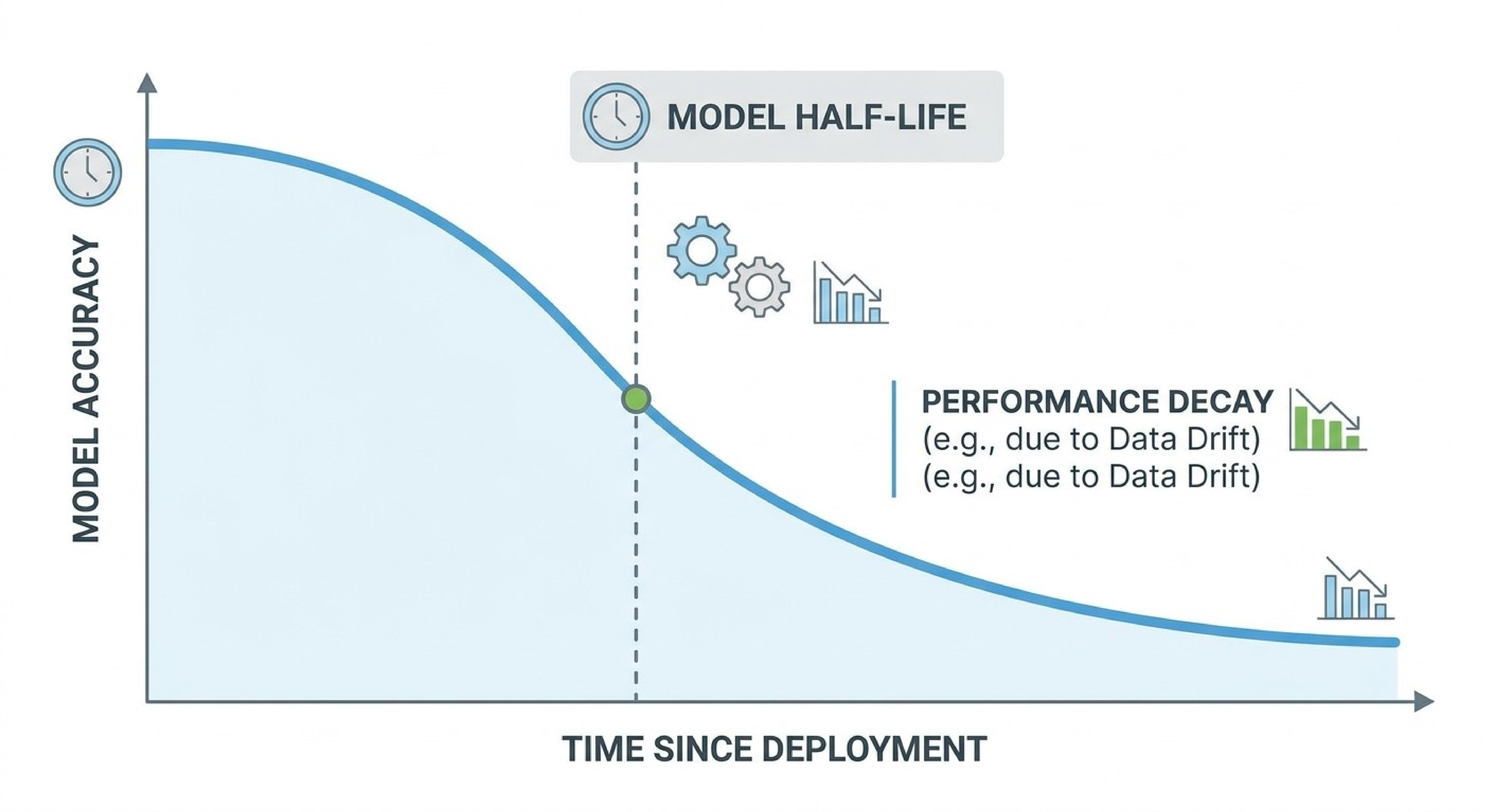


The Entropy of Intelligence (Model Decay)

Address Risks: A decaying model is more dangerous than no model because it provides a "confident" but wrong answer.

Ethical Perspective: For ethically sensitive decisions, decay becomes more important.

Example: If a healthcare or lending model decays, it doesn't just lose money; it creates biased or harmful outcomes for real people.

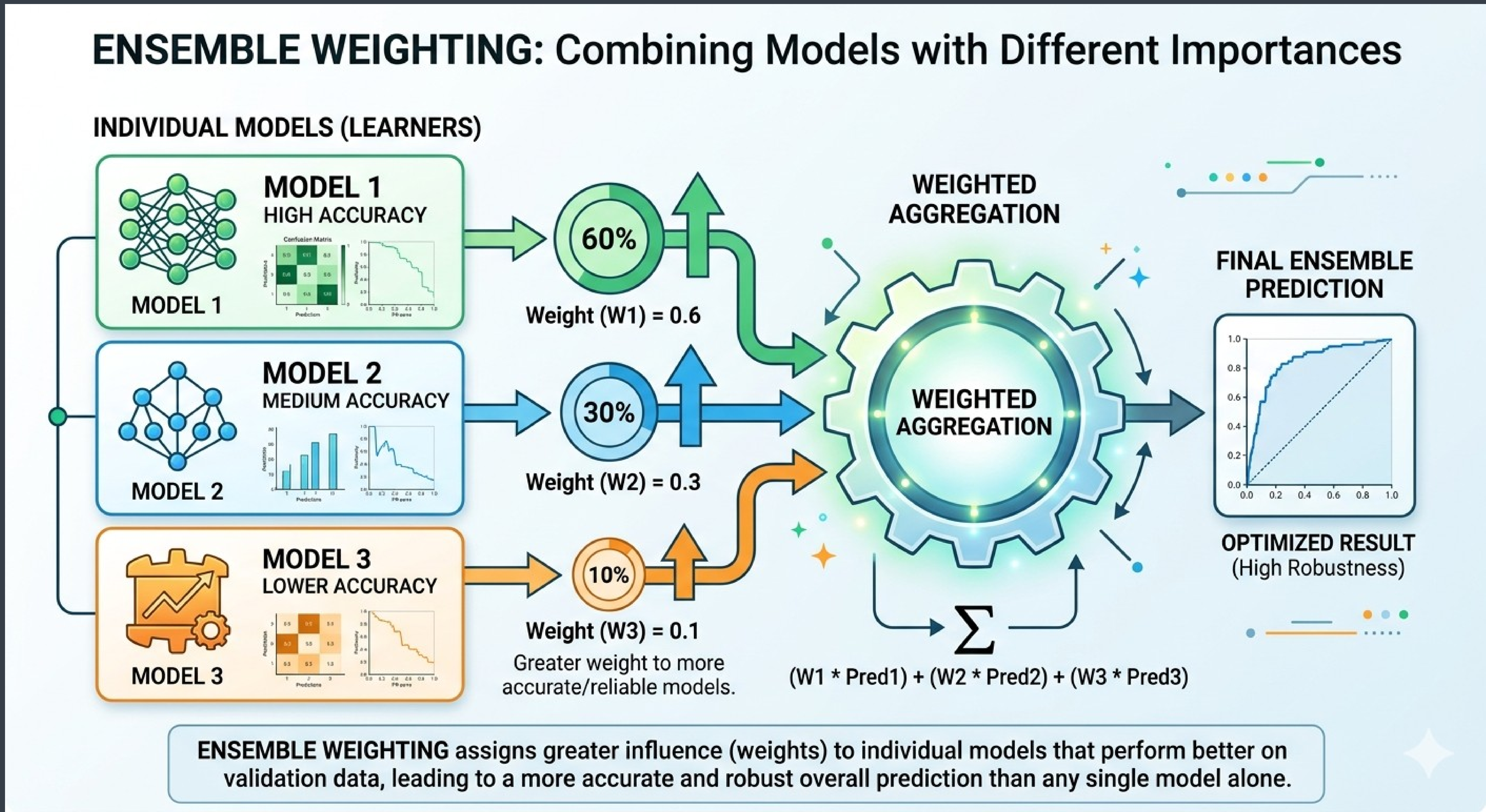


The Entropy of Intelligence (Model Decay)

Mitigation strategies exist.

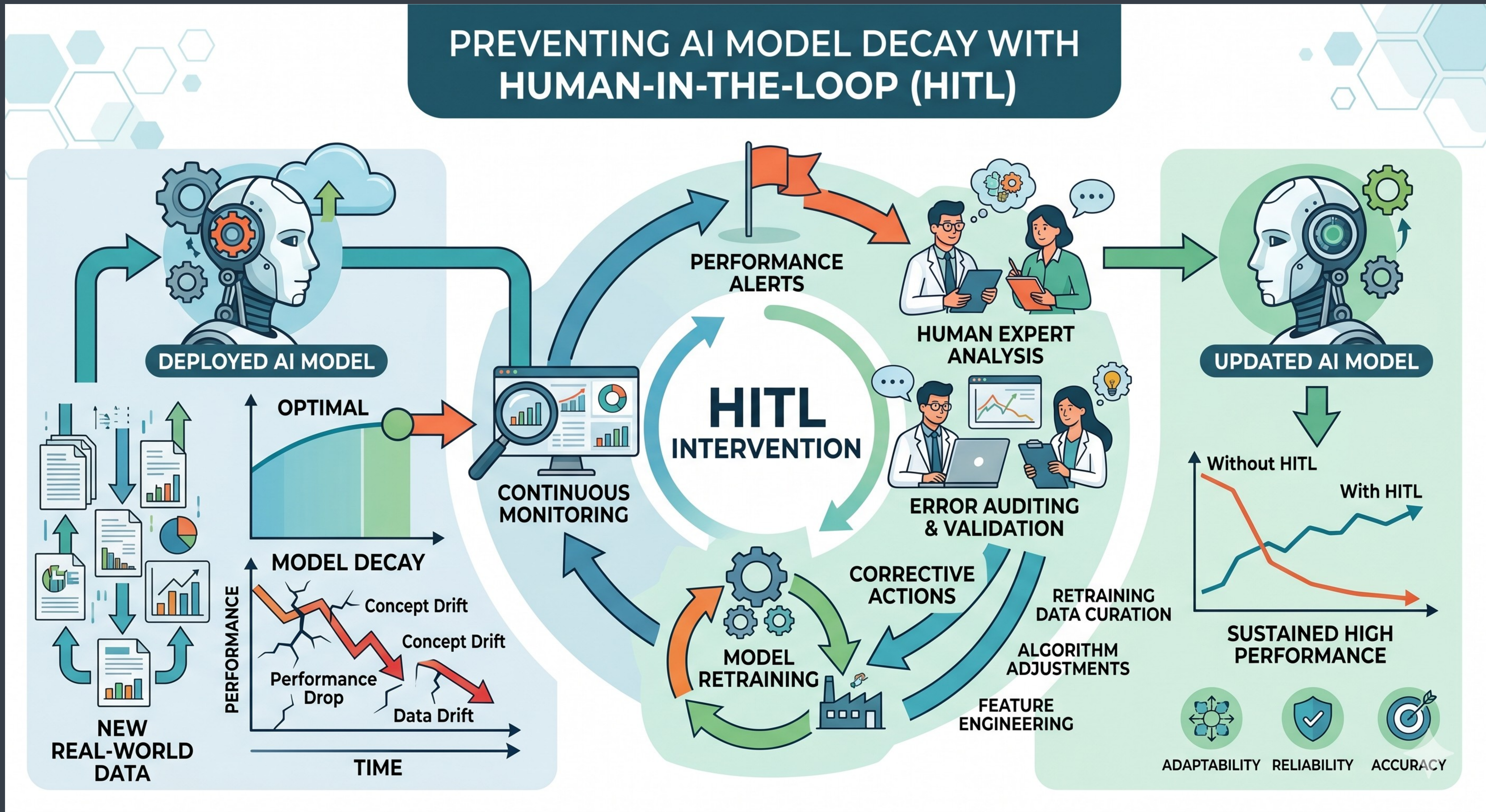
Retraining Pipelines: Transitioning from manual, ad-hoc retraining to automated, trigger-based updates.

Ensemble Weighting: Using multiple models and giving more "weight" to the one trained on the most recent data.



The Entropy of Intelligence (Model Decay)

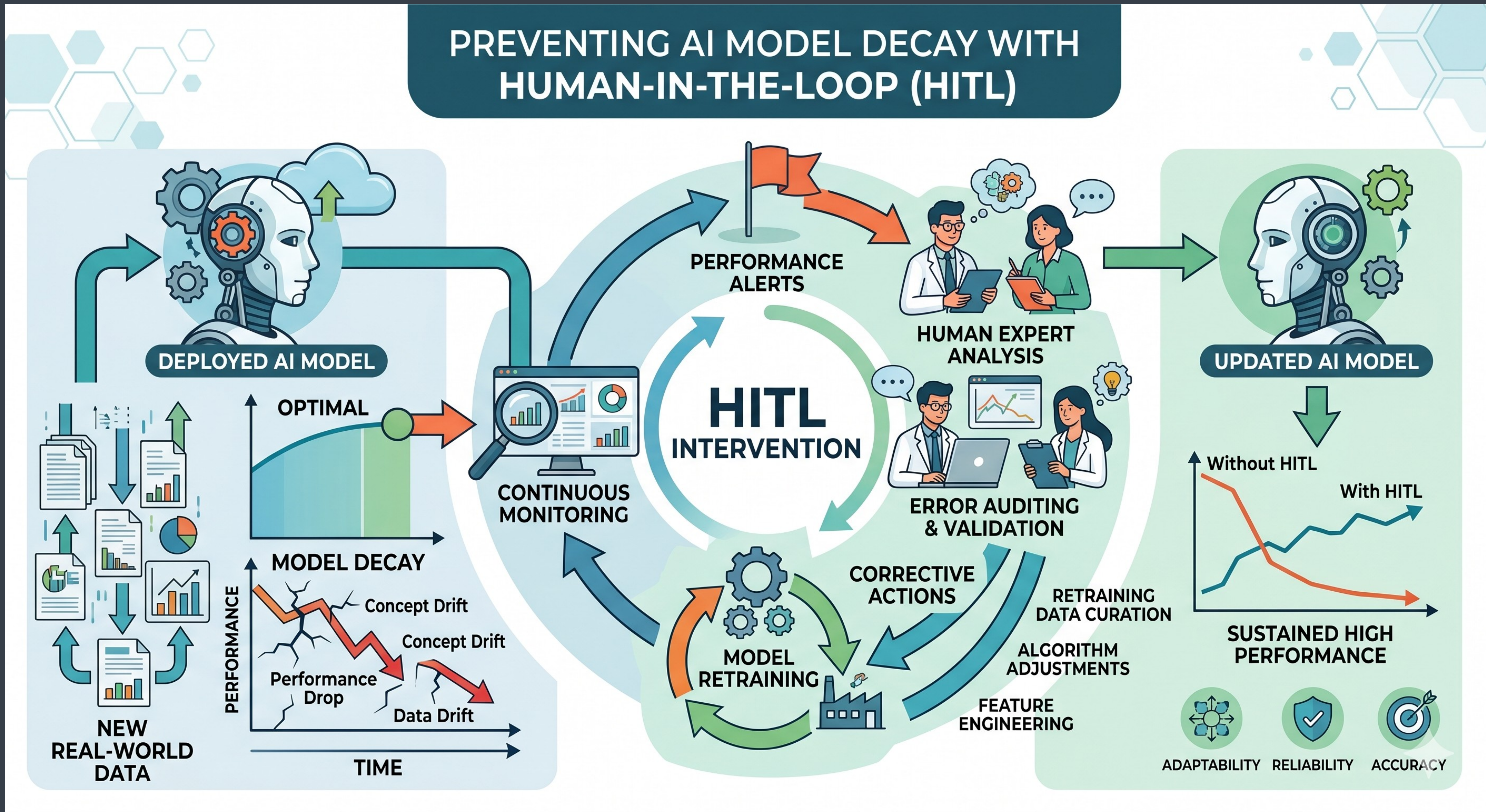
Human-in-the-loop (HITL): Using active learning to have humans label the most "uncertain" or "drifted" cases to provide the model with fresh ground truth.



The Entropy of Intelligence (Model Decay)

Healthcare Example: HITL in Medical Imaging (Radiology)

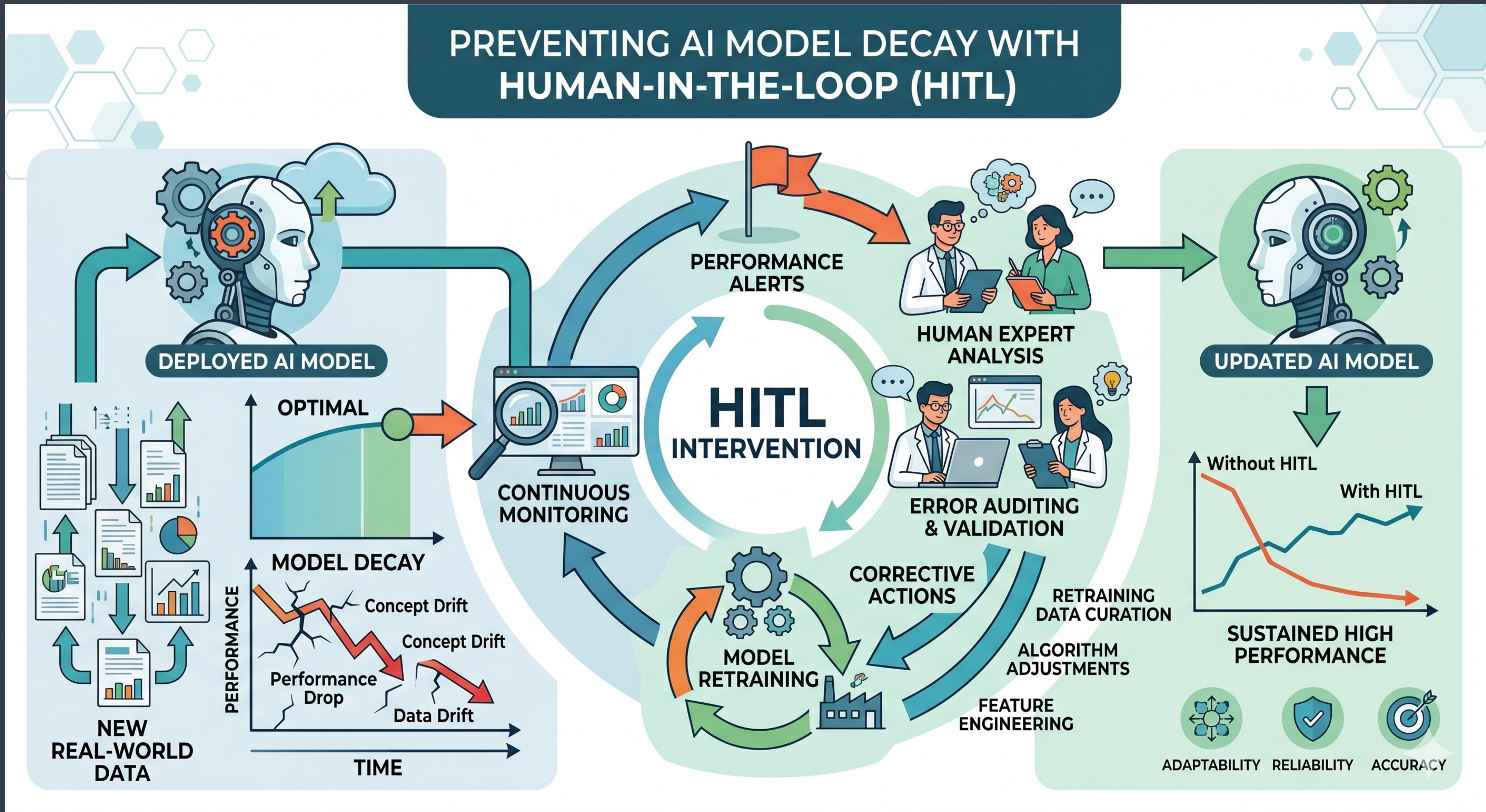
Imagine a hospital uses an AI model to analyze chest X-rays to detect potential signs of pneumonia. When first trained and deployed, the model achieves high accuracy (the "Optimal" phase in the image's top-left graph).



The Entropy of Intelligence (Model Decay)

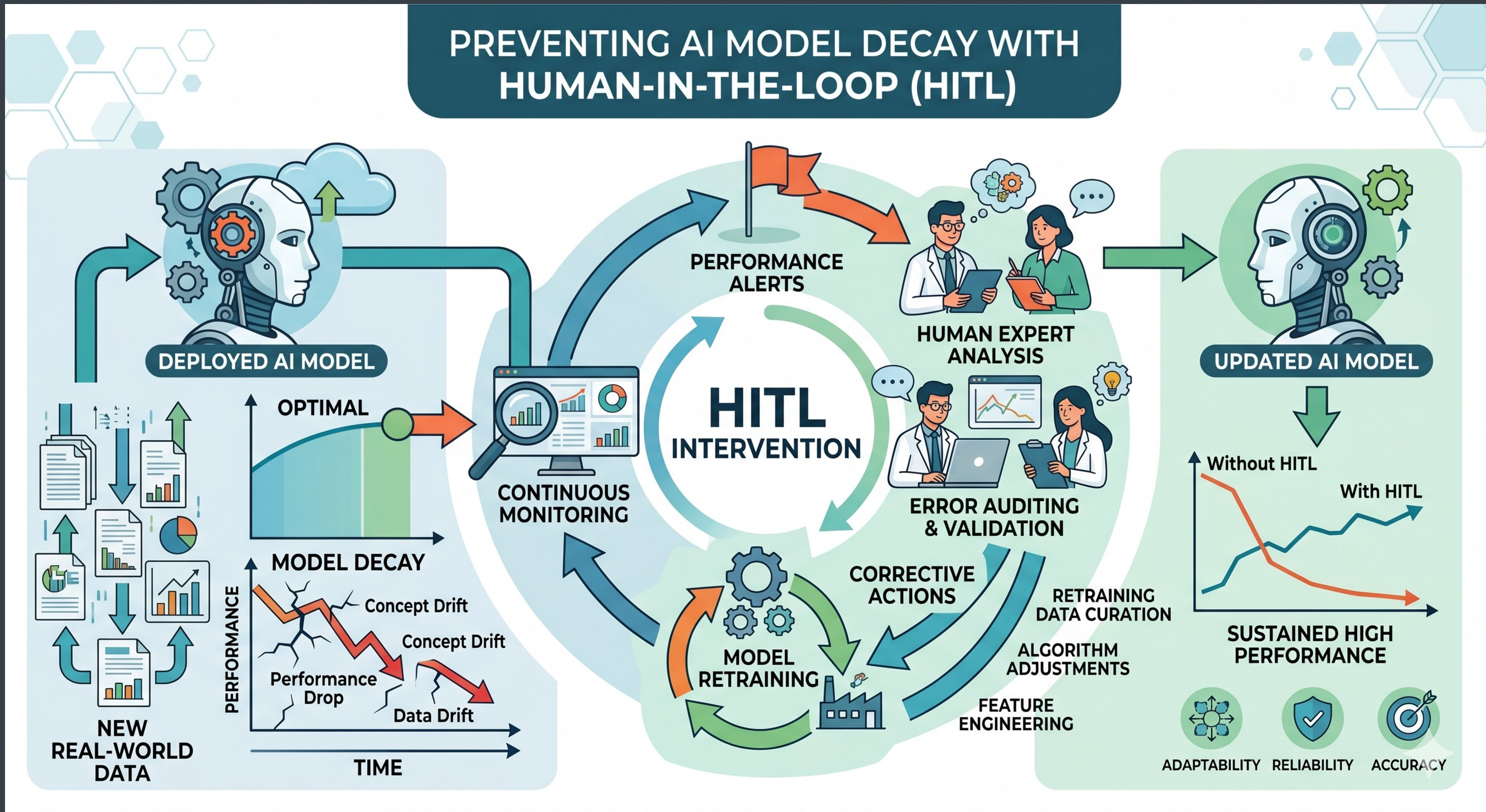
Over time, however, the environment changes, leading to Model Decay.

Data Drift: The hospital updates its X-ray machinery. The new machines use a slightly different resolution or exposure setting than the machines used to generate the original training data. The model, optimized for the older images, struggles to interpret the new ones correctly.



The Entropy of Intelligence (Model Decay)

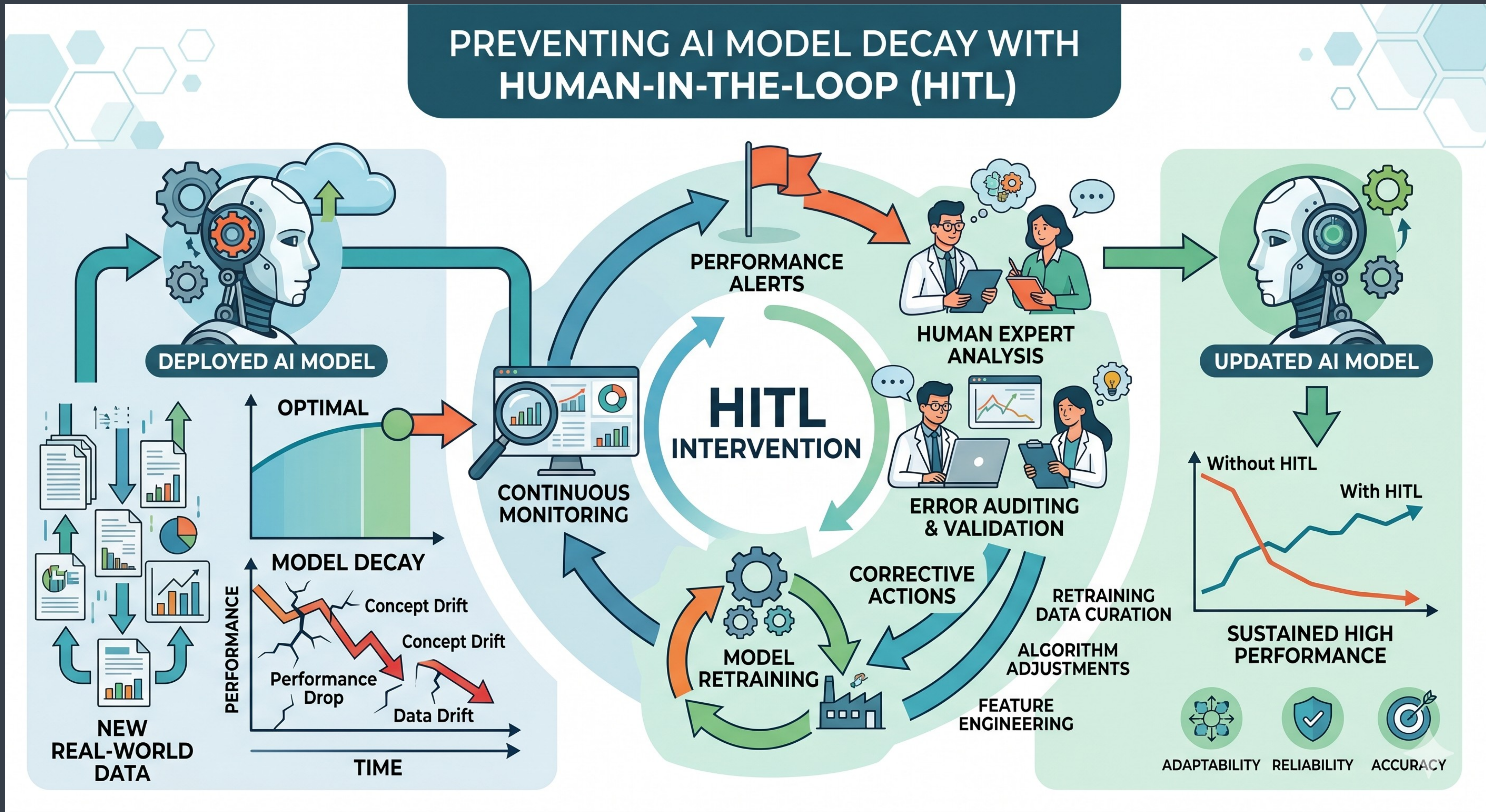
Concept Drift: New clinical guidelines emerge redefining how very early-stage pneumonia appears on scans, or perhaps a new, different type of viral lung infection appears that presents similarly but requires different treatment. The original "concept" of pneumonia the model learned is now outdated.



The Entropy of Intelligence (Model Decay)

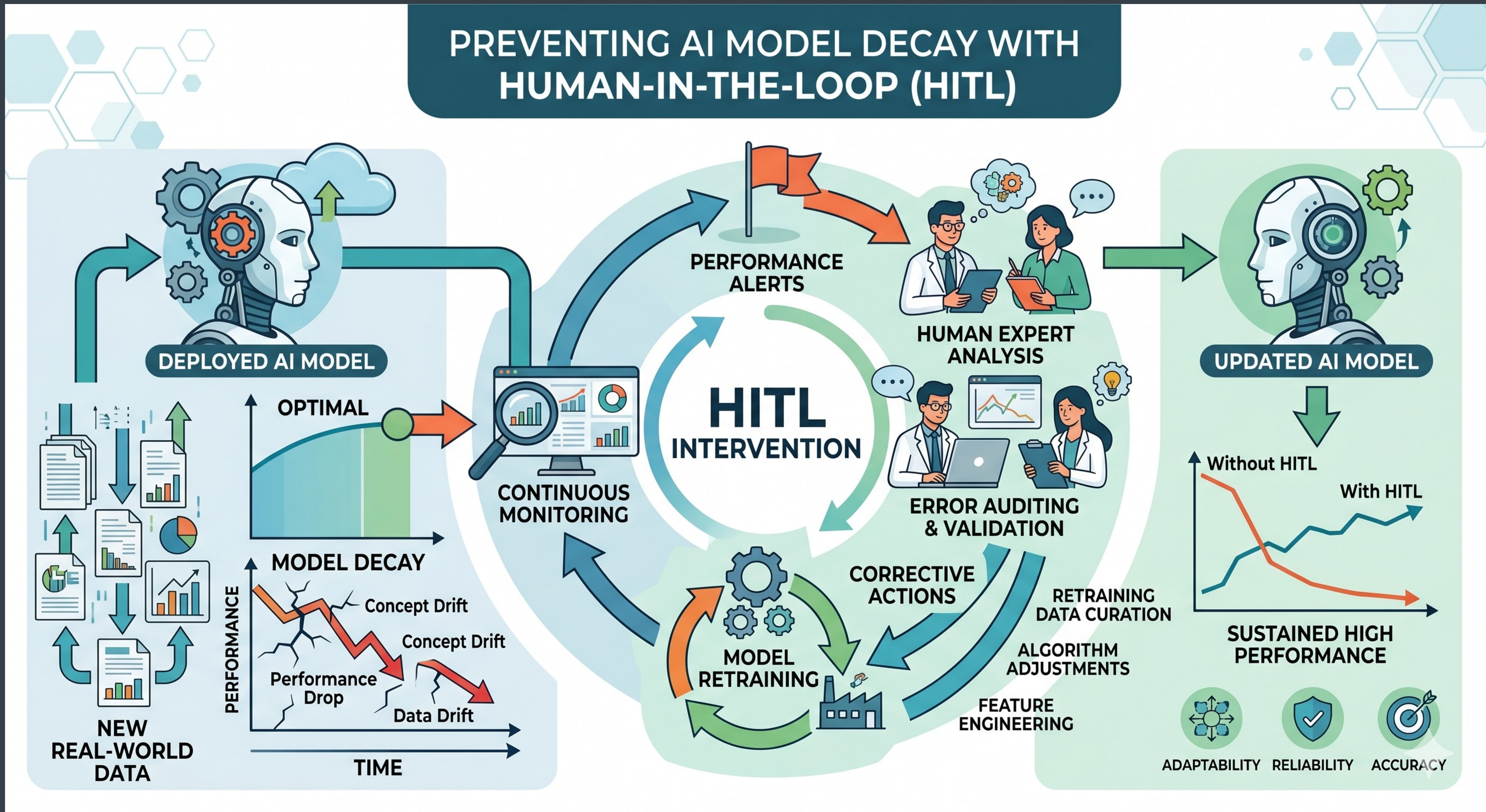
Without intervention, the AI’s performance will drop (the "Without HITL" red line in the final graph).

Continuous Monitoring: A "performance flag" is raised when the system detects that the distribution of new X-ray images (data drift) differs significantly from the training data, or if the model's confidence scores in its predictions have overall decreased.



The Entropy of Intelligence (Model Decay)

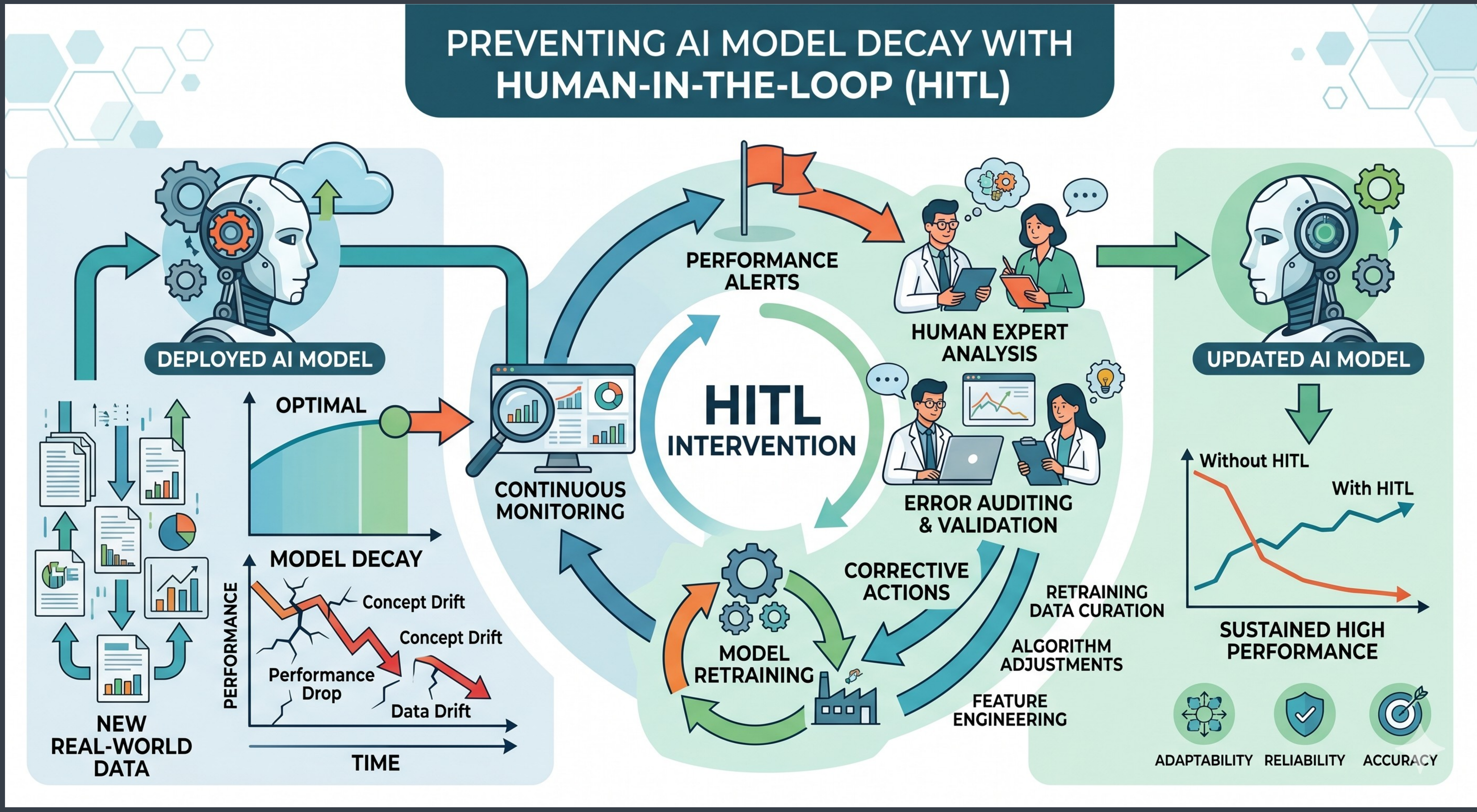
Human Expert Analysis & Error Auditing: A Senior Radiologist (the Human Expert) is brought in-the-loop. The system presents the Radiologist with complex or low-confidence cases. The Radiologist "audits" the AI’s incorrect analyses, marking which ones were misses and diagnosing why—for example, noting that the AI is confused by artifacts from the new scanner.



The Entropy of Intelligence (Model Decay)

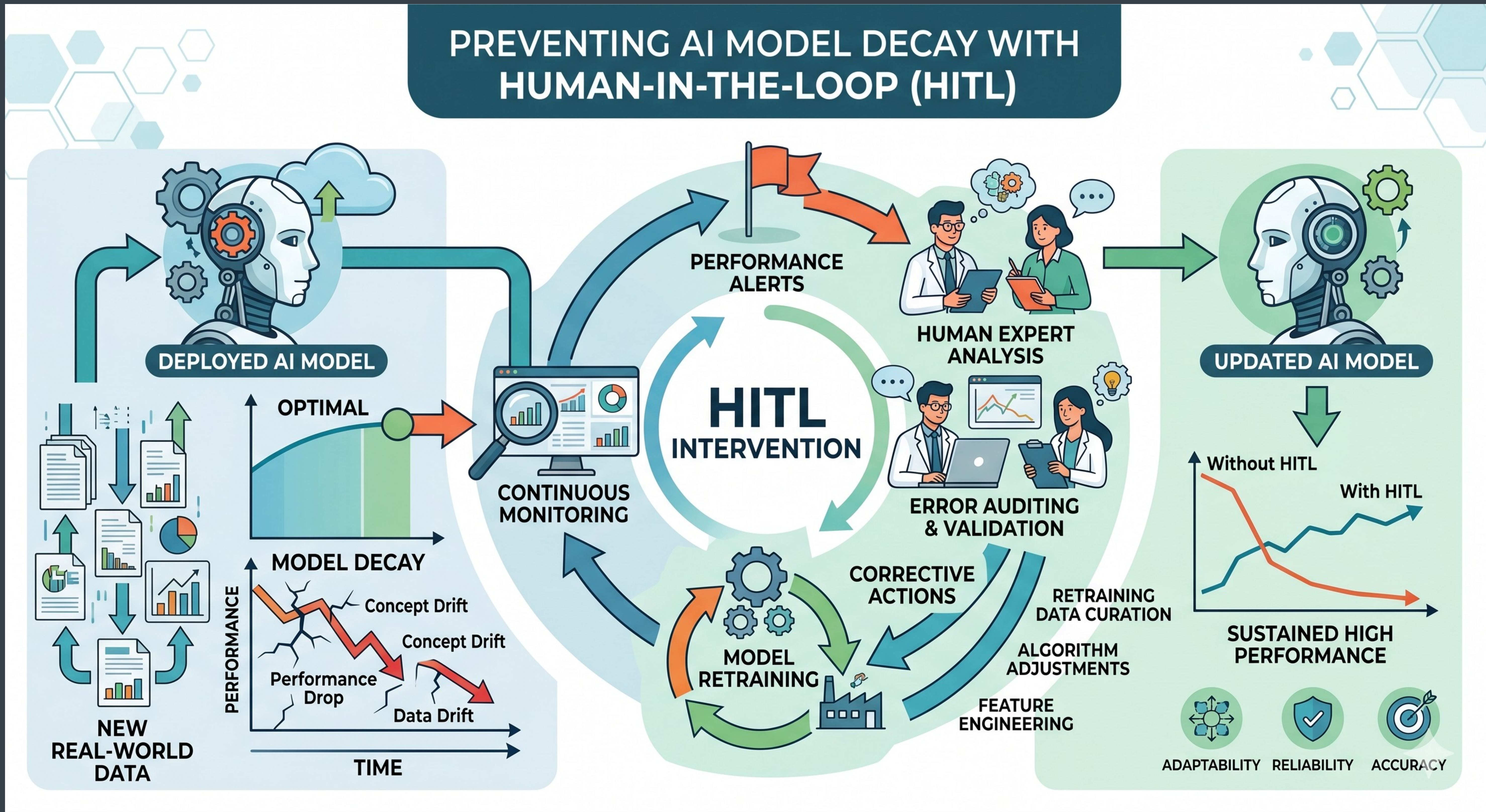
Corrective Actions: These audits provide the insights for corrective actions.

Model Retraining: This newly labeled "corrective" data (images with the updated machine artifacts, re-labeled by the expert) is fed back into the training loop.



The Entropy of Intelligence (Model Decay)

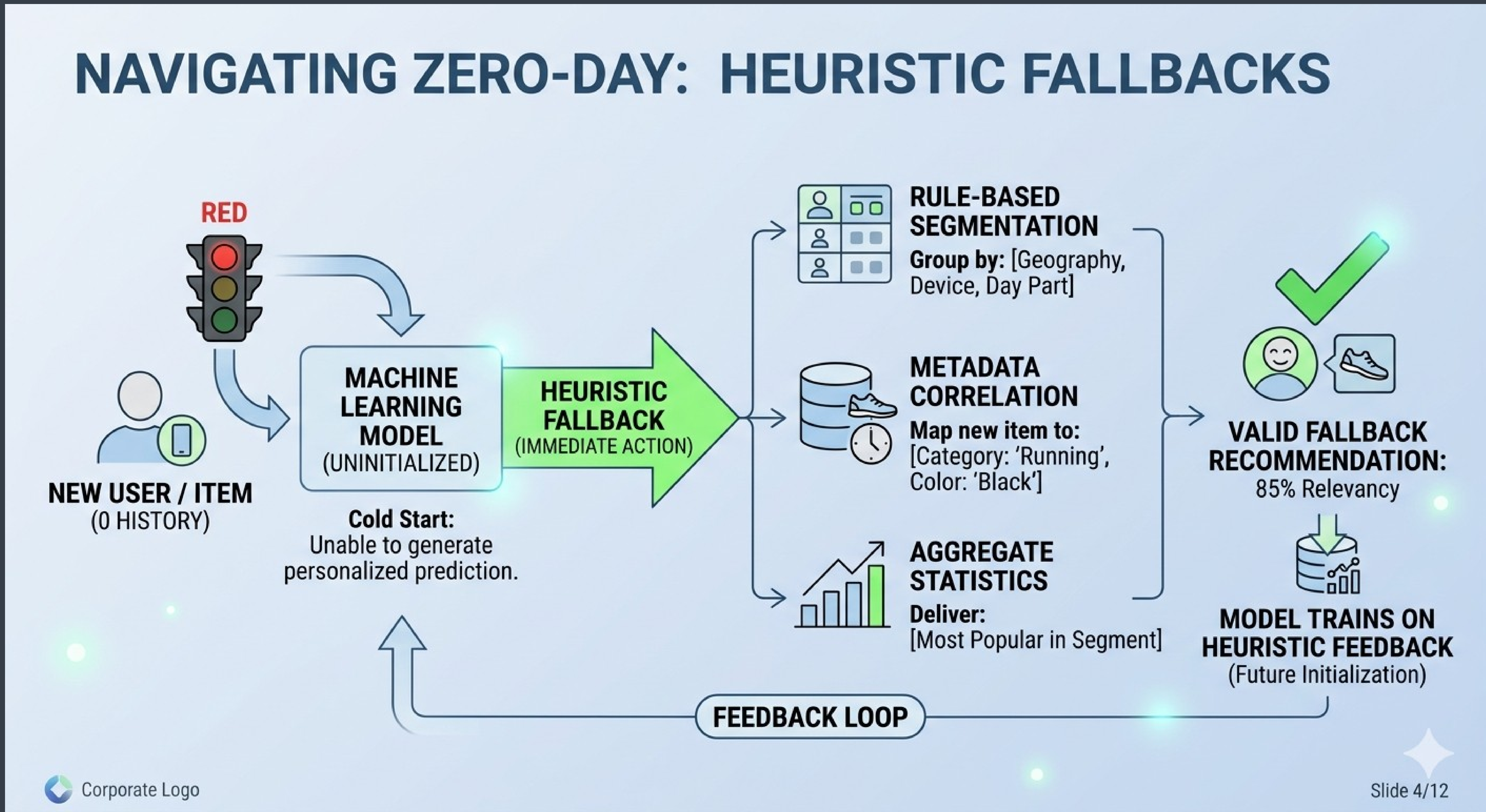
Updated AI Model: The model is retrained and redeployed. It has now learned to understand the new image quality and the refined medical concepts, ensuring Sustained High Performance (the "With HITL" blue line) in a critical healthcare settin



The Zero-Day Crisis (The Cold Start Problem)

Zero-Day State (ie. Cold-Start): A state where a machine learning model is unable to generate a (personalized) prediction because it has zero history for a specific user or item.

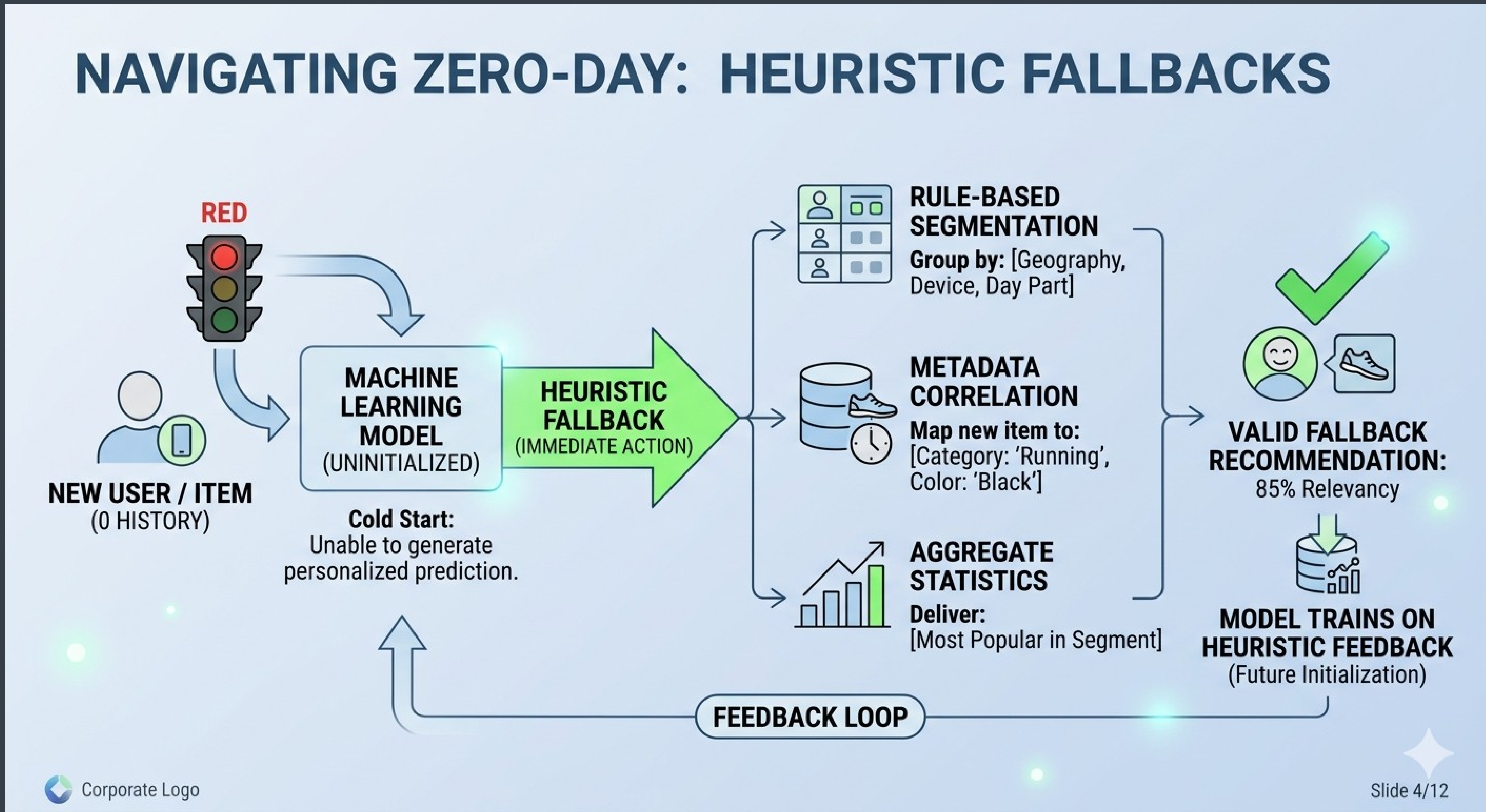
High-performance models require data to function, but new entities enter the system with no data, creating a performance "black hole" on day zero



The Zero-Day Crisis (The Cold Start Problem)

This risks an immediate user churn or system failure if the model simply returns "null" or random results during the first interaction.

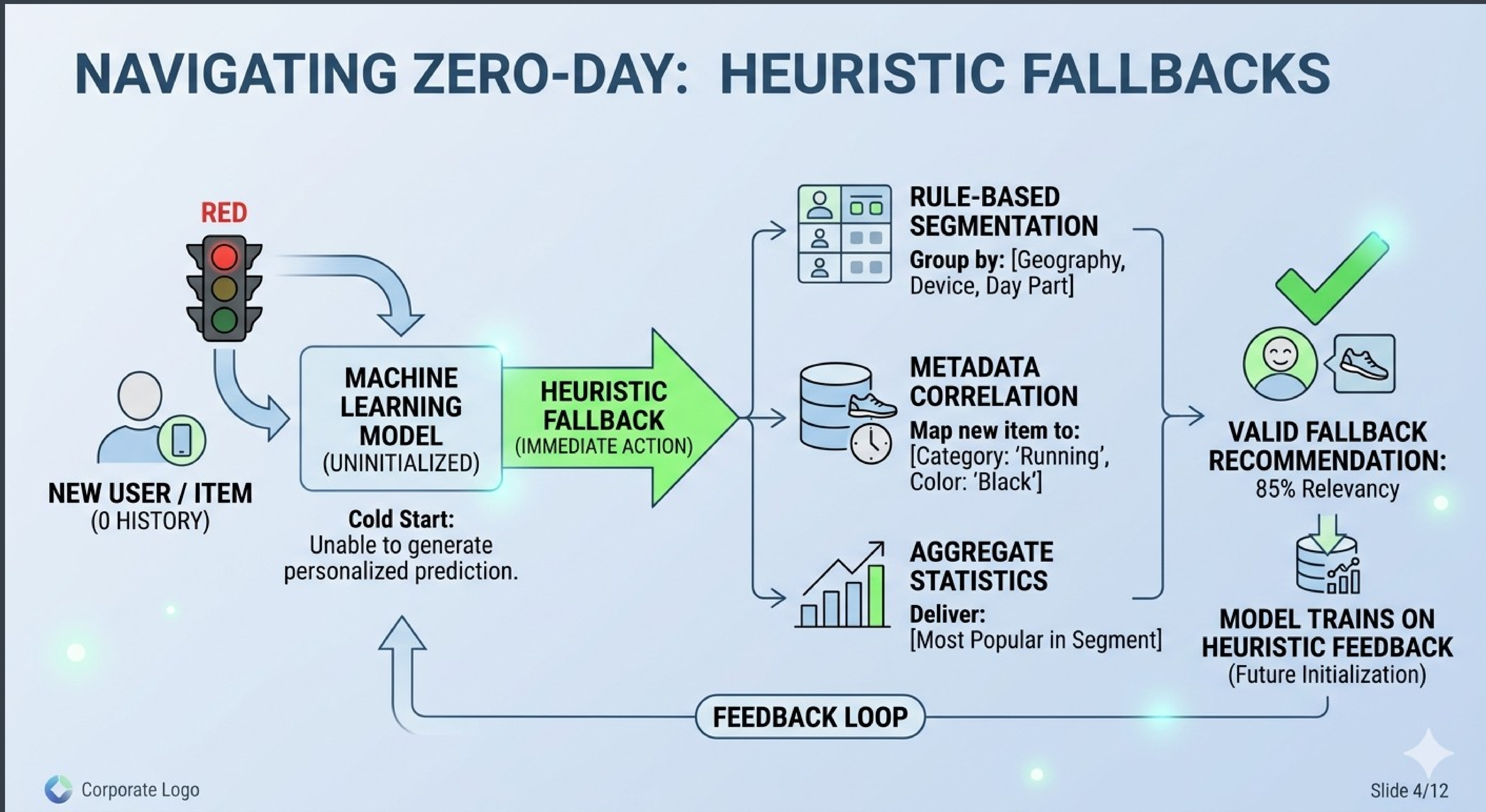
Heuristic Fallback: When the ML model is uninitialized, the system must trigger an immediate heuristic action to maintain service continuity.



The Zero-Day Crisis (The Cold Start Problem)

Instead of predicting individual behavior, the system pivots to broader, data-supported "safety nets" to ensure a valid initial experience.

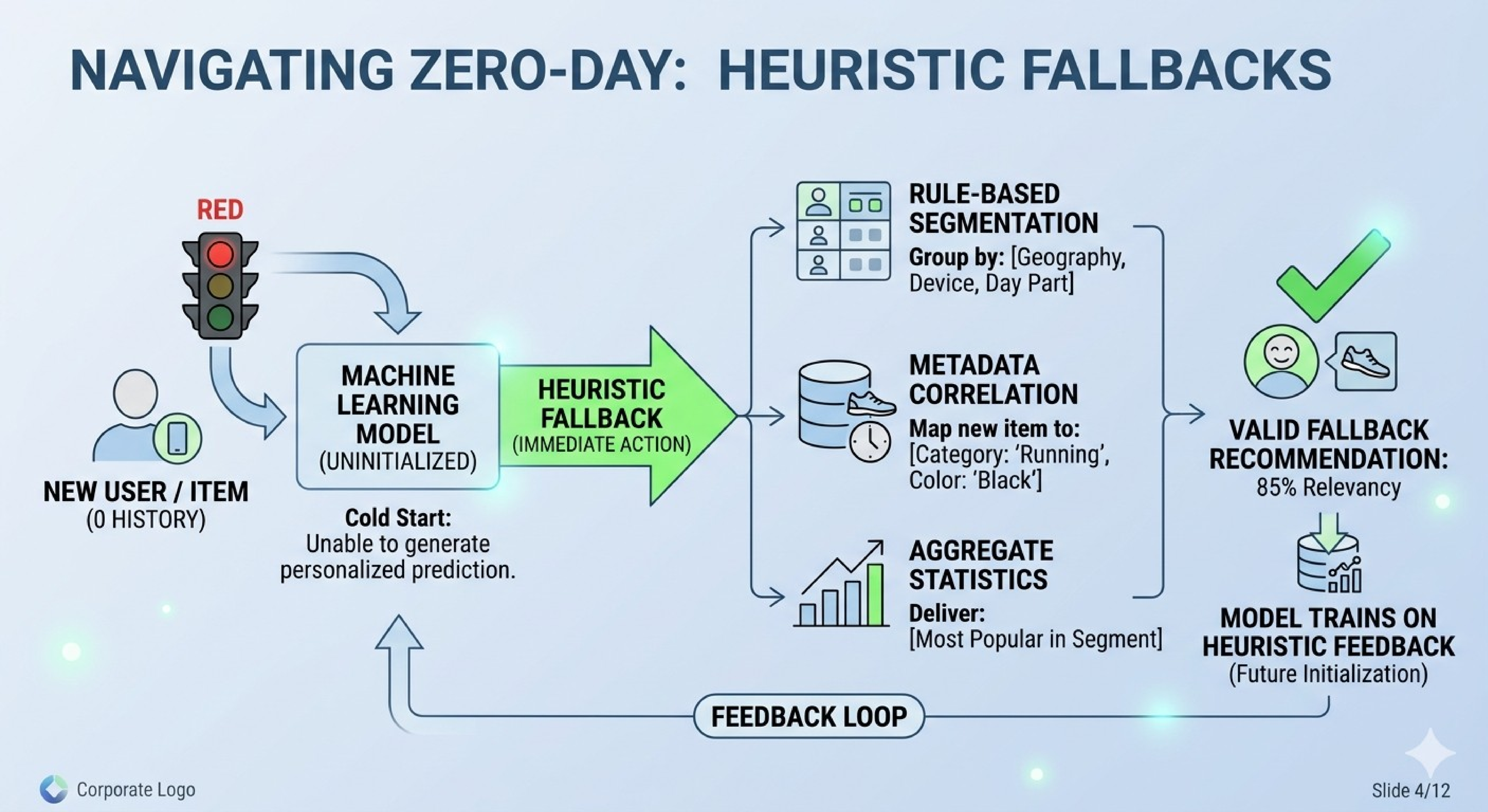
The goal is to deliver a "Valid Fallback Recommendation" with high initial relevancy (e.g., 85%) to bridge the gap until personalization is possible.



The Zero-Day Crisis (The Cold Start Problem)

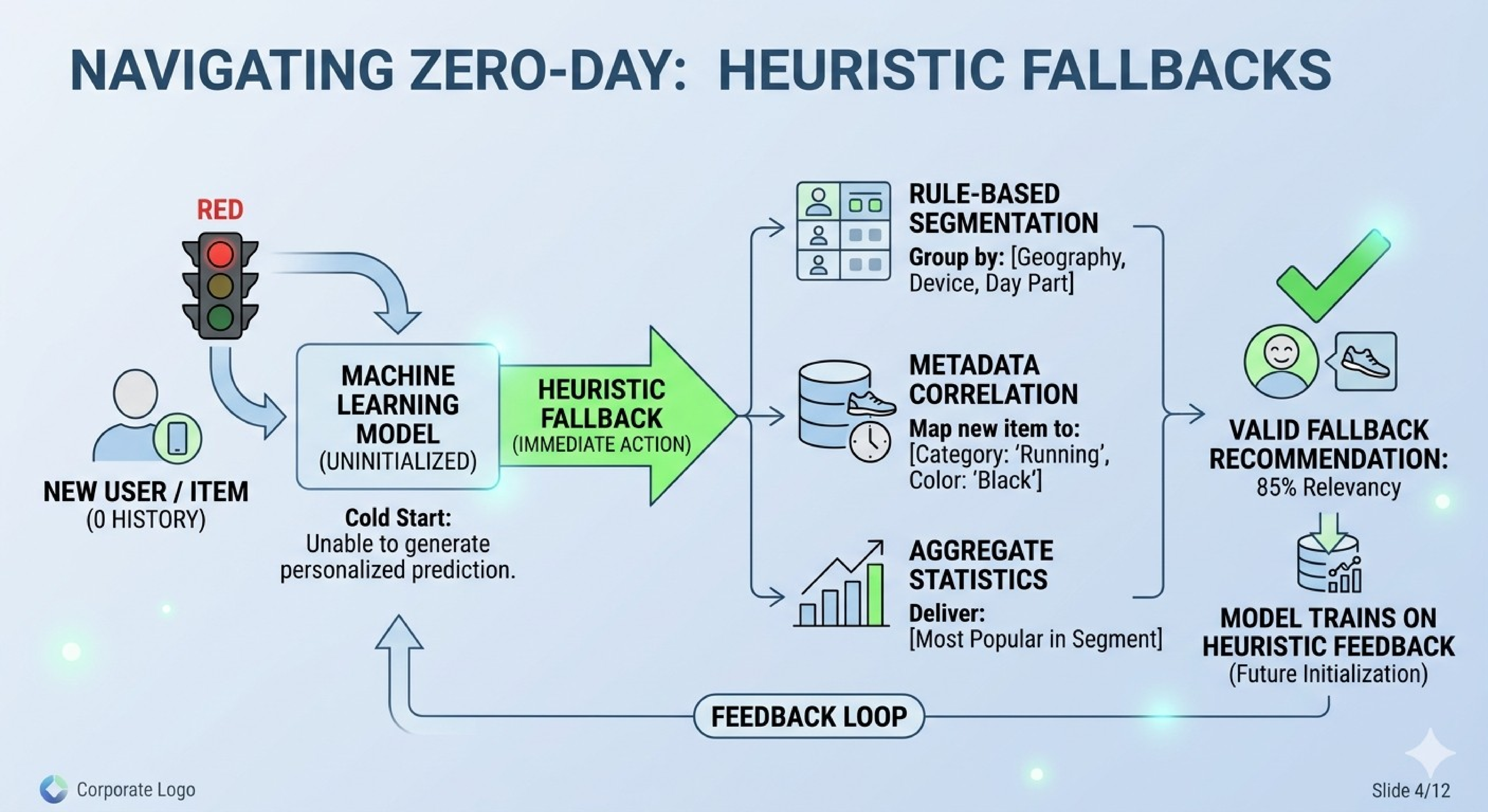
Rule-Based Segmentation: Grouping new users by known environmental metadata such as Geography, Device type, or Day Part.

Metadata Correlation: Mapping a new, unknown item to existing known categories (e.g., categorizing a new shoe as 'Running' or 'Black') to inherit the properties of similar items



The Zero-Day Crisis (The Cold Start Problem)

Aggregate Statistics: Delivering the "Most Popular" items within a specific segment to maximize the probability of relevance before user-specific data arrives

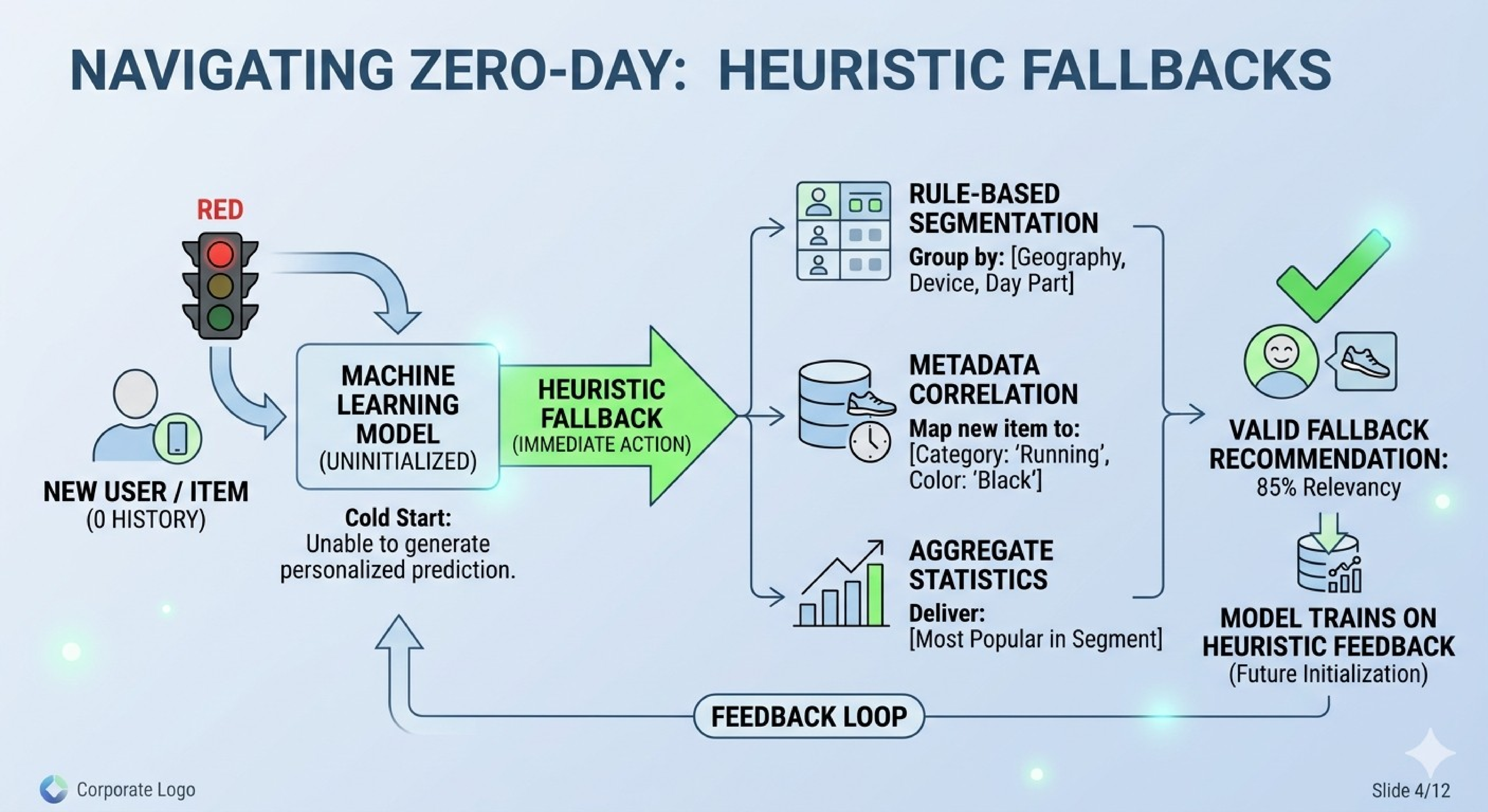


The Zero-Day Crisis (The Cold Start Problem)

The system does not stay in "fallback mode" forever; it uses the results of the heuristic actions as the first training samples.

Fallback models actively train on "Heuristic Feedback"—learning which fallbacks worked best for which segments.

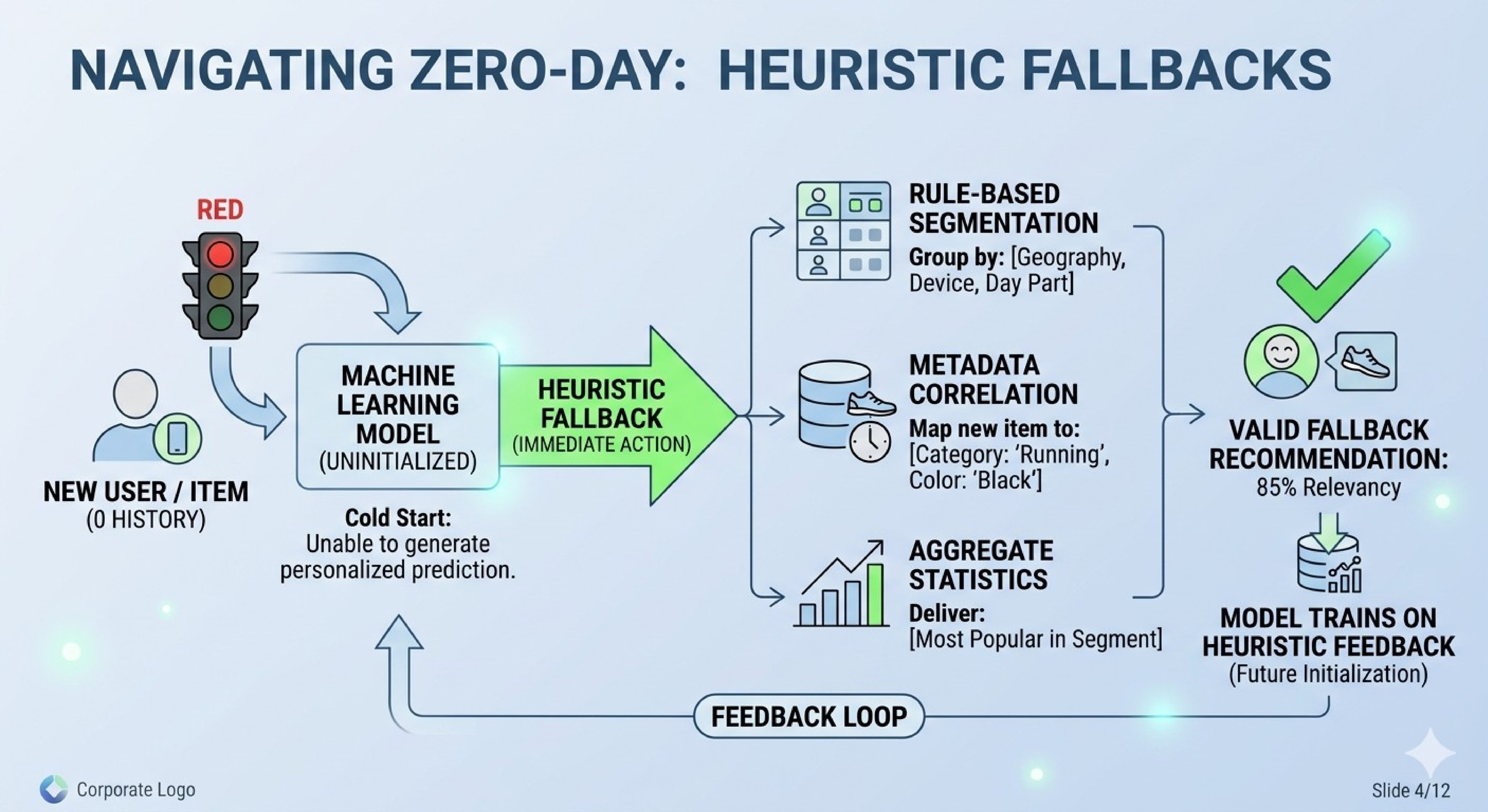
This feedback loop ensures that the "Zero-Day" experience for the next new user is more informed than the last



The Zero-Day Crisis (The Cold Start Problem)

This approach tries to move from random to scientific. We treat the ML Lifecycle as a scientific process where every failure is recorded and every hypothesis is tested.

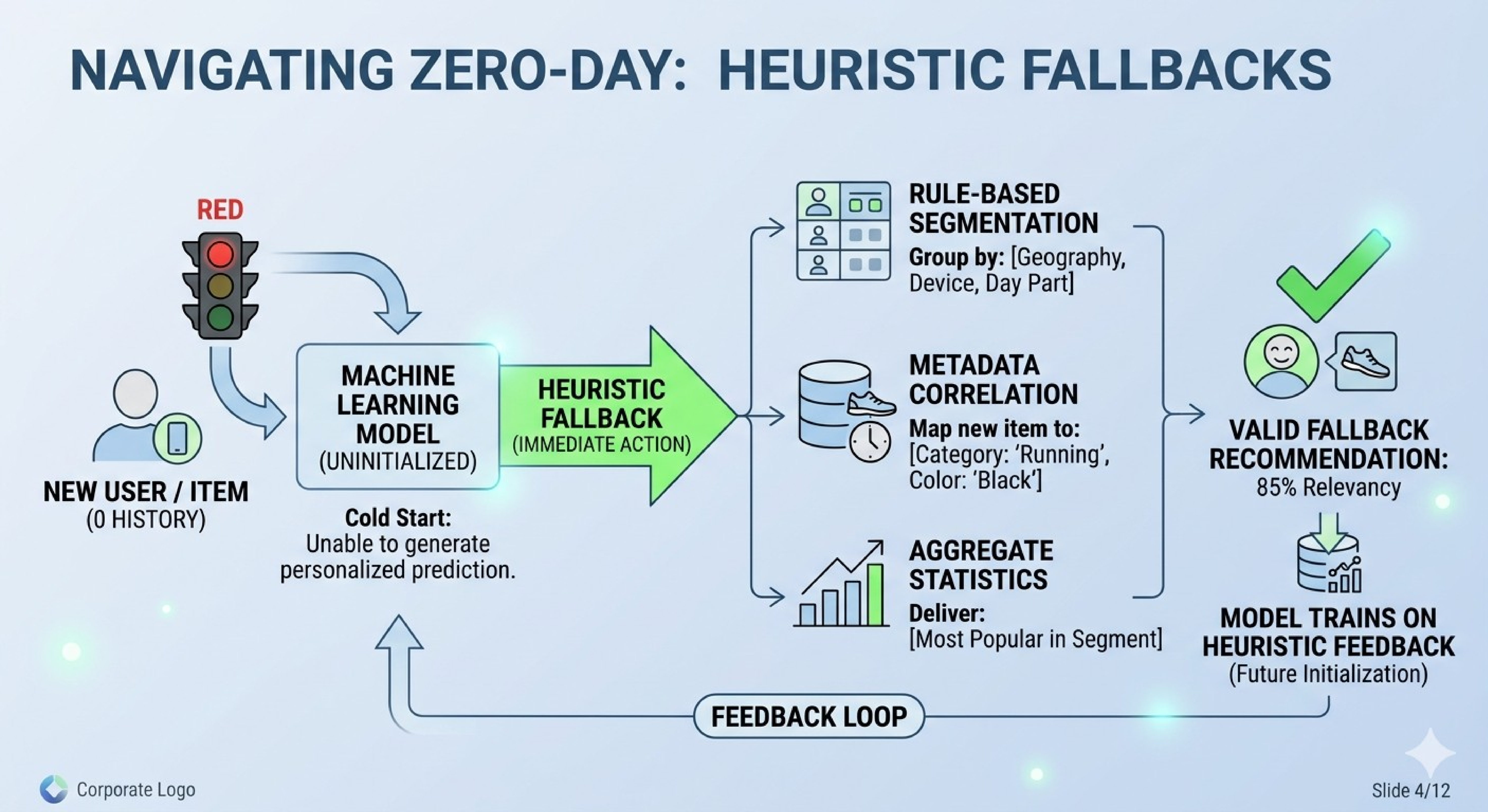
Our “Ledger” approach (last session) is a simplified version of this.



The Zero-Day Crisis (The Cold Start Problem)

Use Case: Personalized "New Player" Storefront

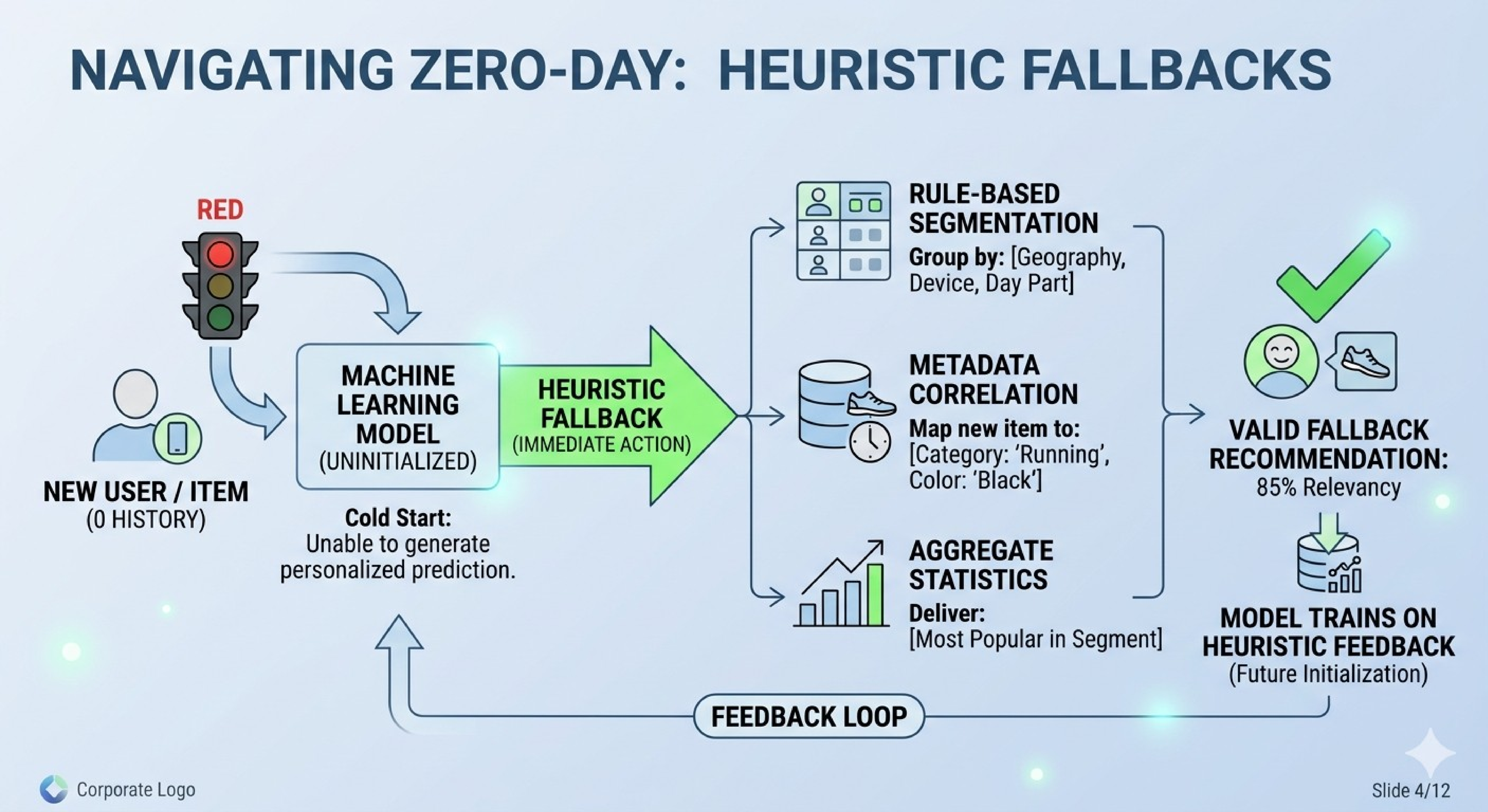
When a player launches a new game for the first time, the machine learning model is uninitialized and unable to generate personalized item predictions based on past behavior. Instead of showing a random or empty storefront, the system uses metadata correlation to bridge the gap.



The Zero-Day Crisis (The Cold Start Problem)

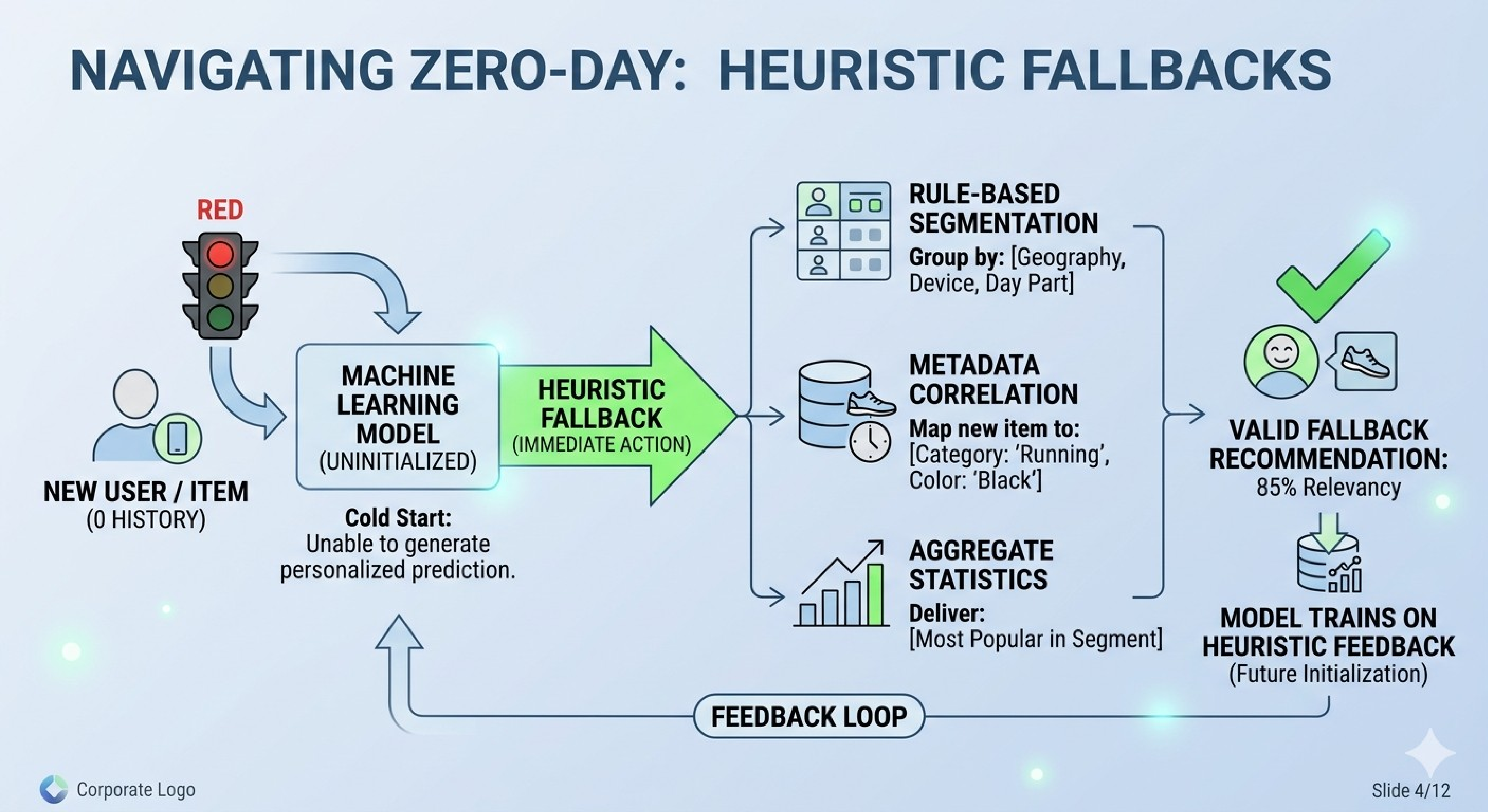
Since there is 0 history for the new user, the system triggers an immediate Heuristic Fallback.

Metadata Mapping: The system maps the new player's account to existing categories based on their platform (e.g., PC vs. Mobile) or the specific marketing campaign that brought them to the game.



The Zero-Day Crisis (The Cold Start Problem)

Segment Correlation: If the player is on a high-end gaming PC, the metadata correlates them with a segment that typically prefers "Core Gamer" cosmetics or high-fidelity textures.

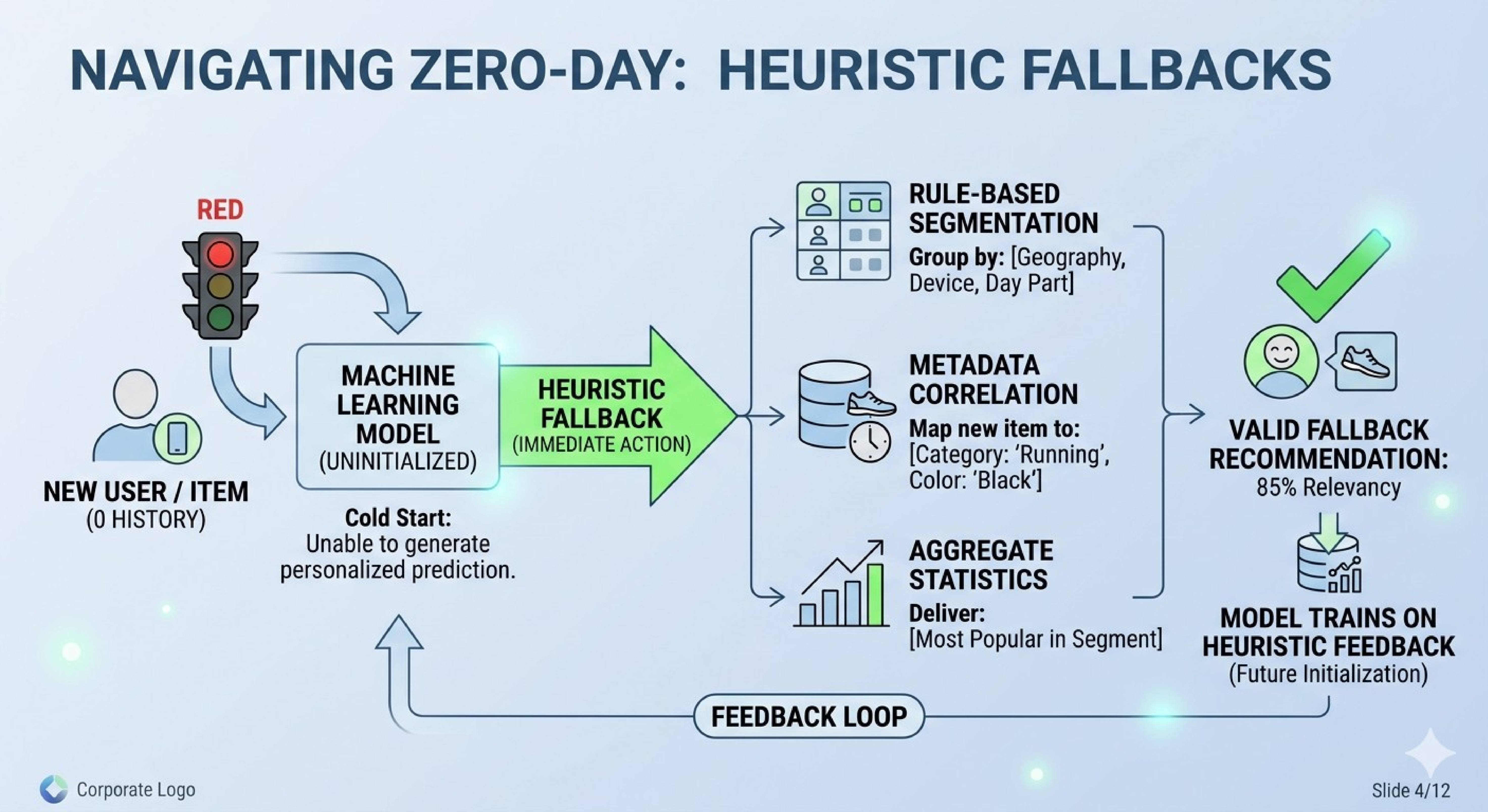


The Zero-Day Crisis (The Cold Start Problem)

The system delivers a Valid Fallback Recommendation that has a high likelihood of relevancy.

Attribute Inheritence: A new "Epic Skin" for a character is mapped to metadata like "Type: Warrior" and "Style: Cyberpunk"

Aggregate Statistics: The storefront displays the "Most Popular" items within the player's specific geographic or device segment.

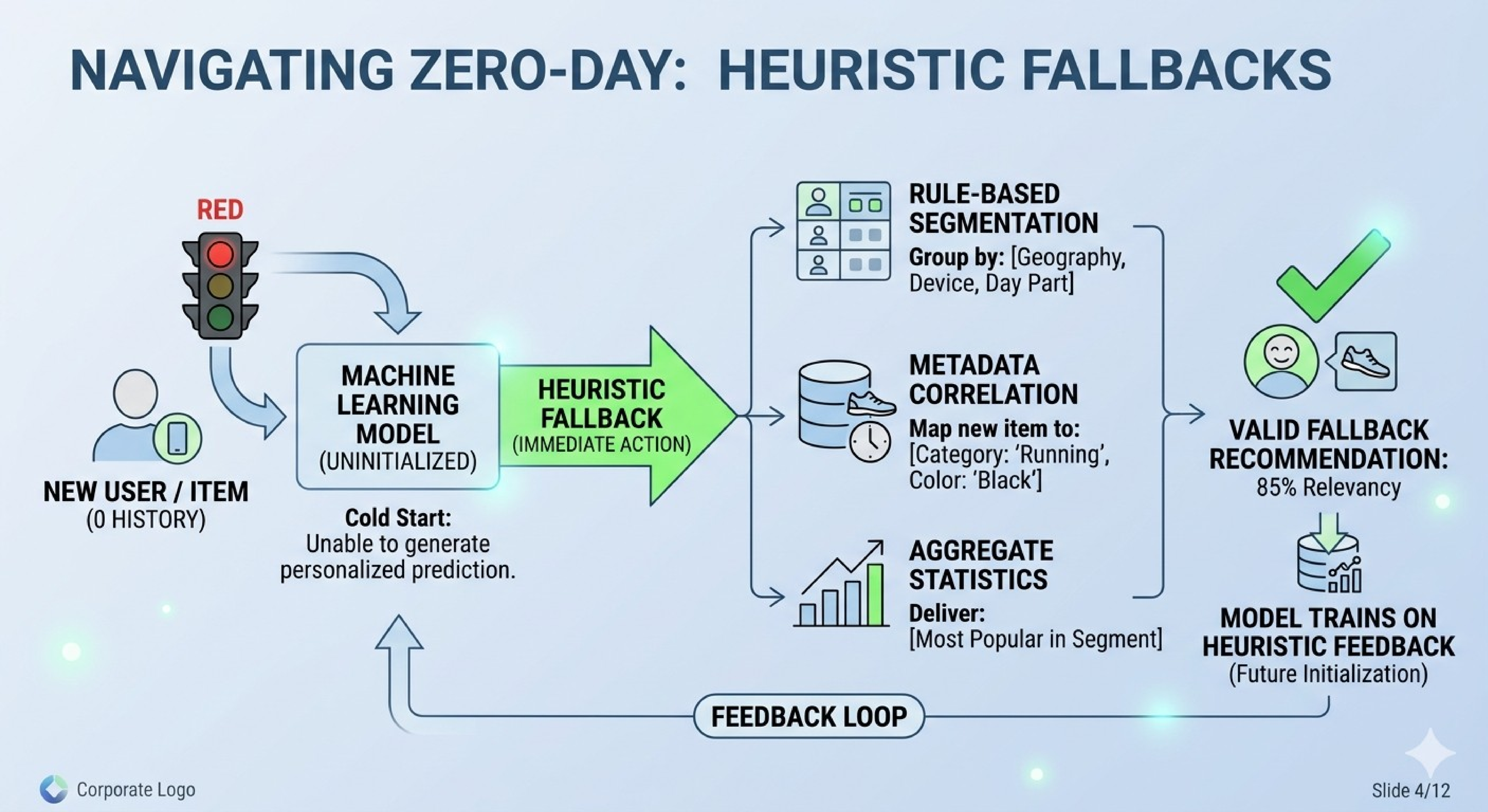


The Zero-Day Crisis (The Cold Start Problem)

The process ensures the model doesn't remain "cold" for long.

The Feedback Loop: As the player interacts with these fallback items, the system logs the results as the first data points in the Feedback Loop.

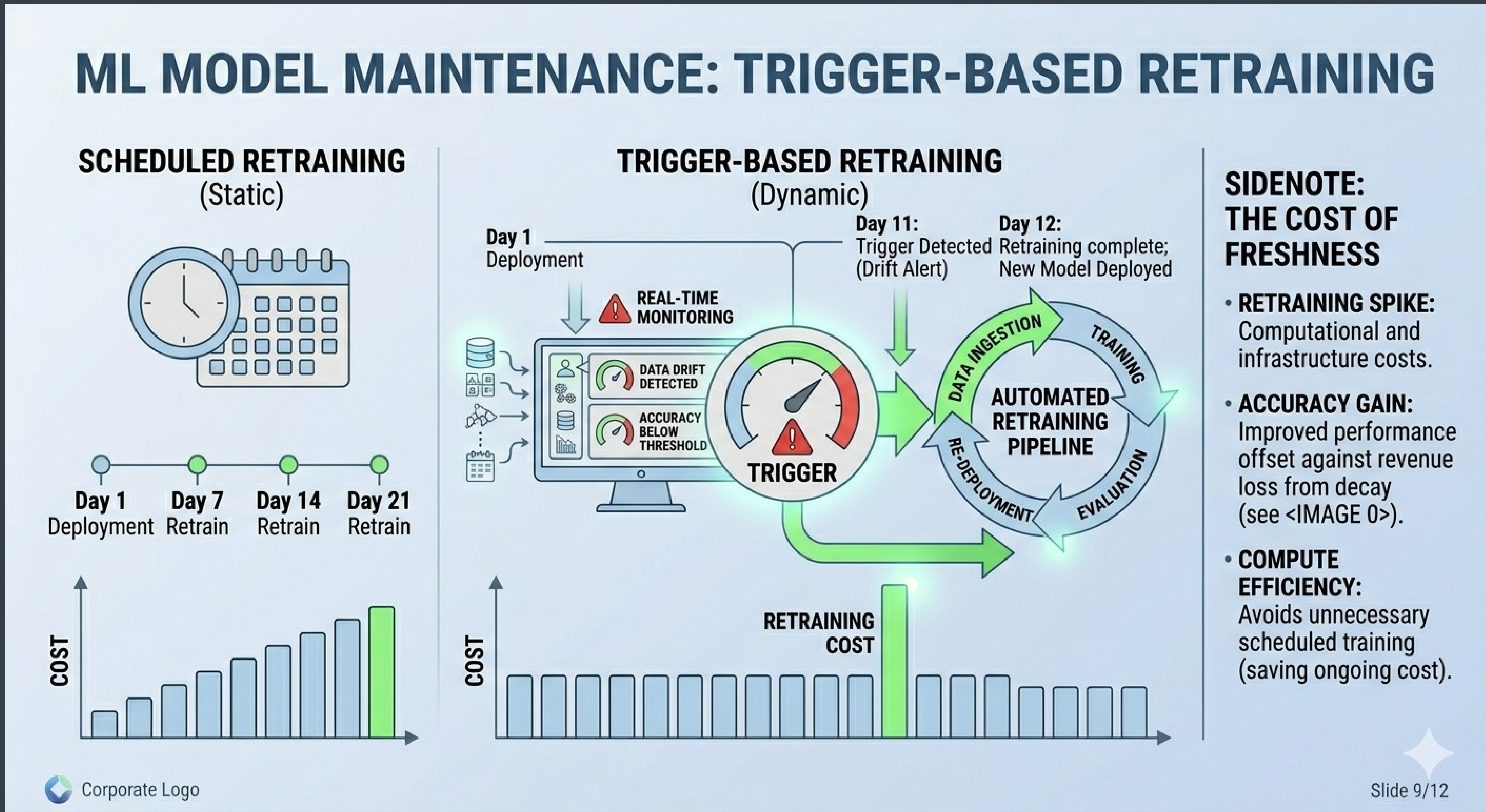
Future Initialization: The model then trains on this heuristic feedback, allowing it to move from general segment-based guesses to personalized predictions for that specific player in the next session.



Maintenance Strategies

Maintenance is an operational reality.

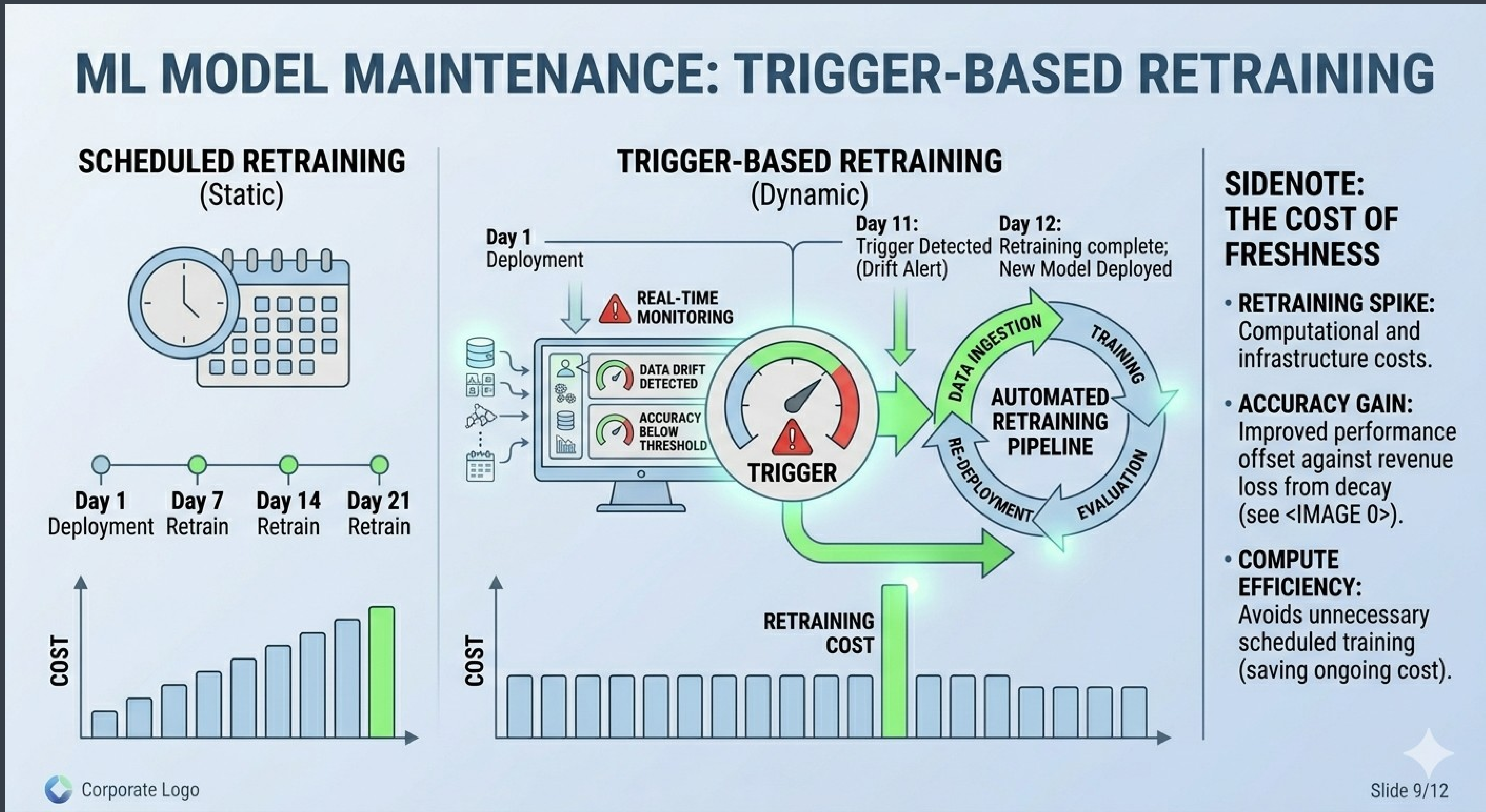
Proactive maintenance is required to protect the business value and ethical integrity of the system.



Maintenance Strategies

Scheduled Retraining (Static): Updates occur at fixed intervals (e.g., every 7 days) regardless of model performance.

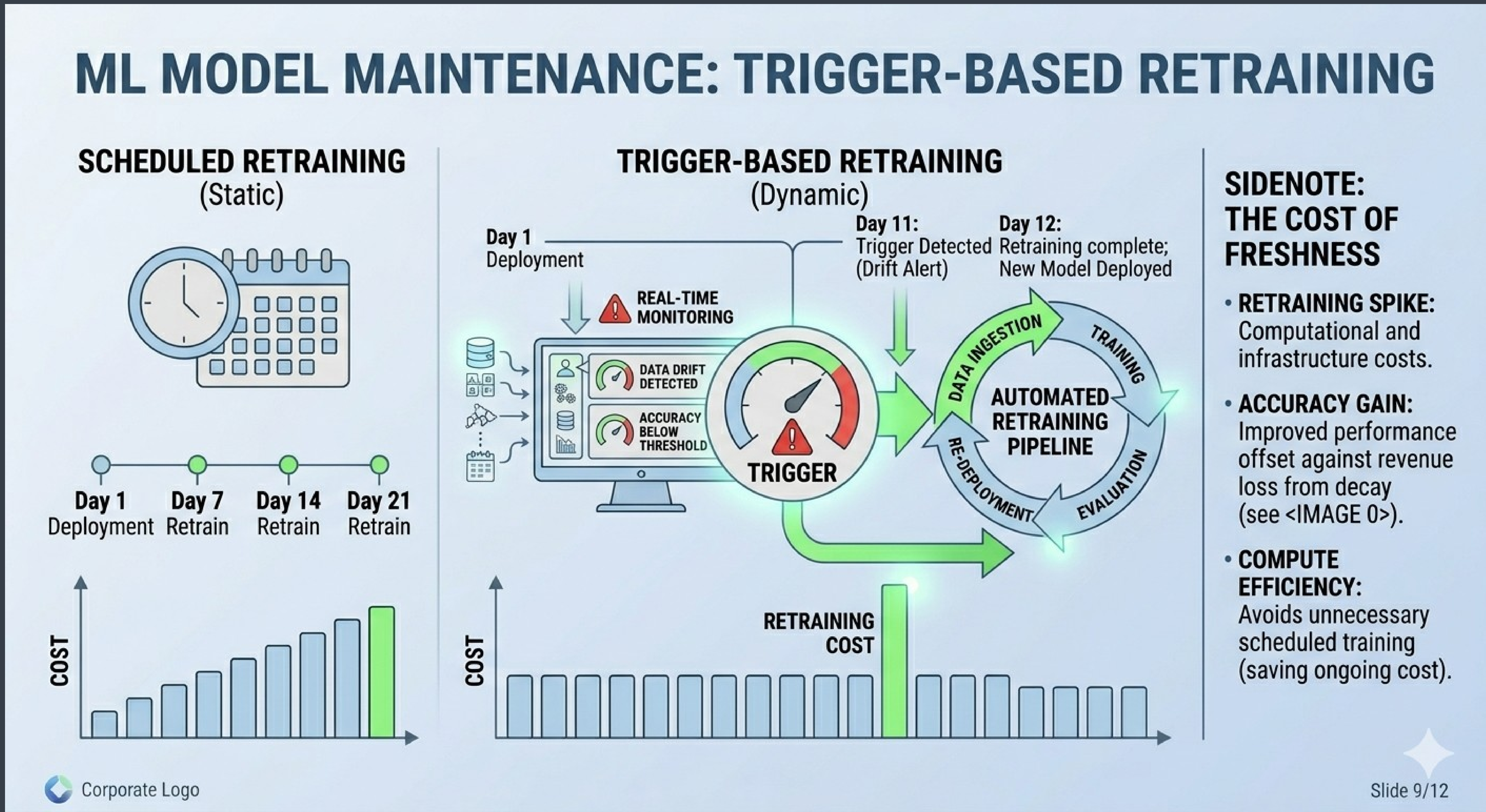
Trigger-Based Retraining (Dynamic): Updates are only initiated when real-time monitoring detects a specific "Drift Alert" or performance drop.



Maintenance Strategies

The Retraining Spike: Dynamic retraining avoids unnecessary scheduled runs, saving ongoing compute costs while providing targeted performance gains.

Accuracy Trade-offs: Improved performance from fresh data is weighed against the infrastructure costs of the retraining pipeline.

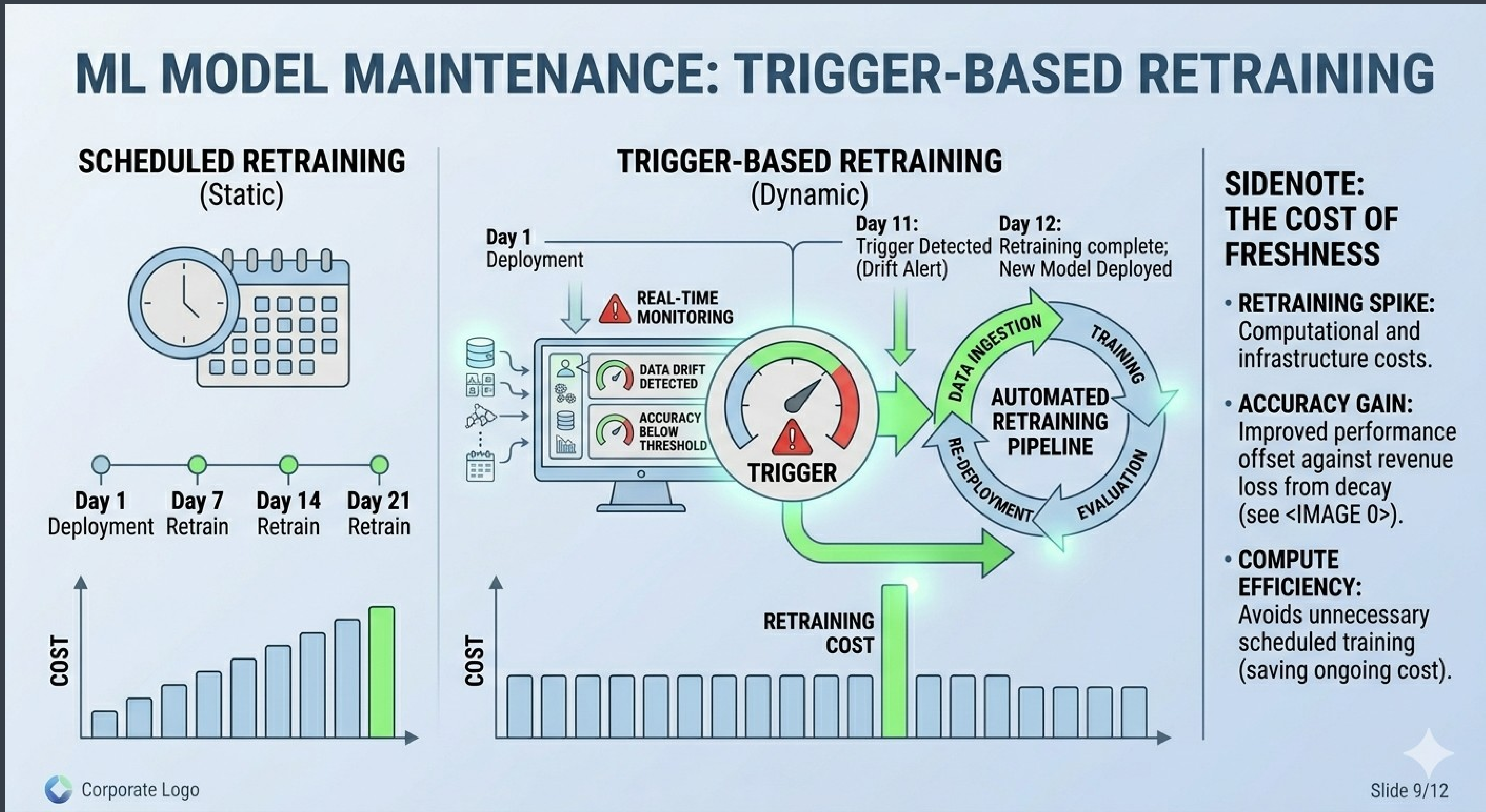


Maintenance Strategies

For high value or high risk models, many deployments focus on ensemble weighting strategies.

Weighted Aggregation: Maintenance involves using multiple "learners" rather than relying on a single model.

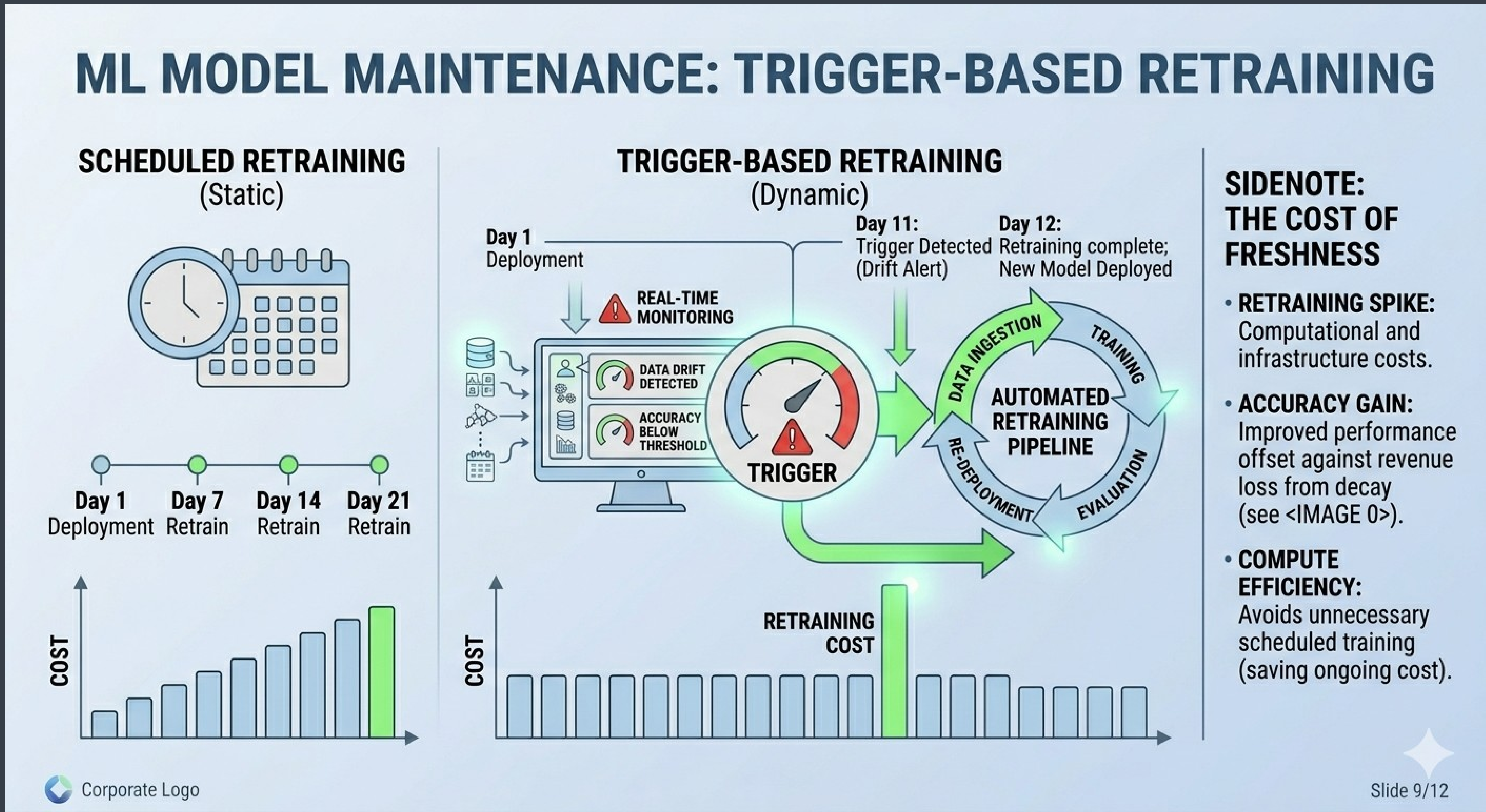
Recency Bias: Greater influence (weight) is assigned to individual models that perform better on the most recent validation data.



Maintenance Strategies

Example Distribution: A primary model might have 60% weight, while secondary models trained on different segments or timeframes provide the remaining 40%.

Optimized Results: This strategy leads to a more accurate and robust overall prediction than any single model could provide alone.

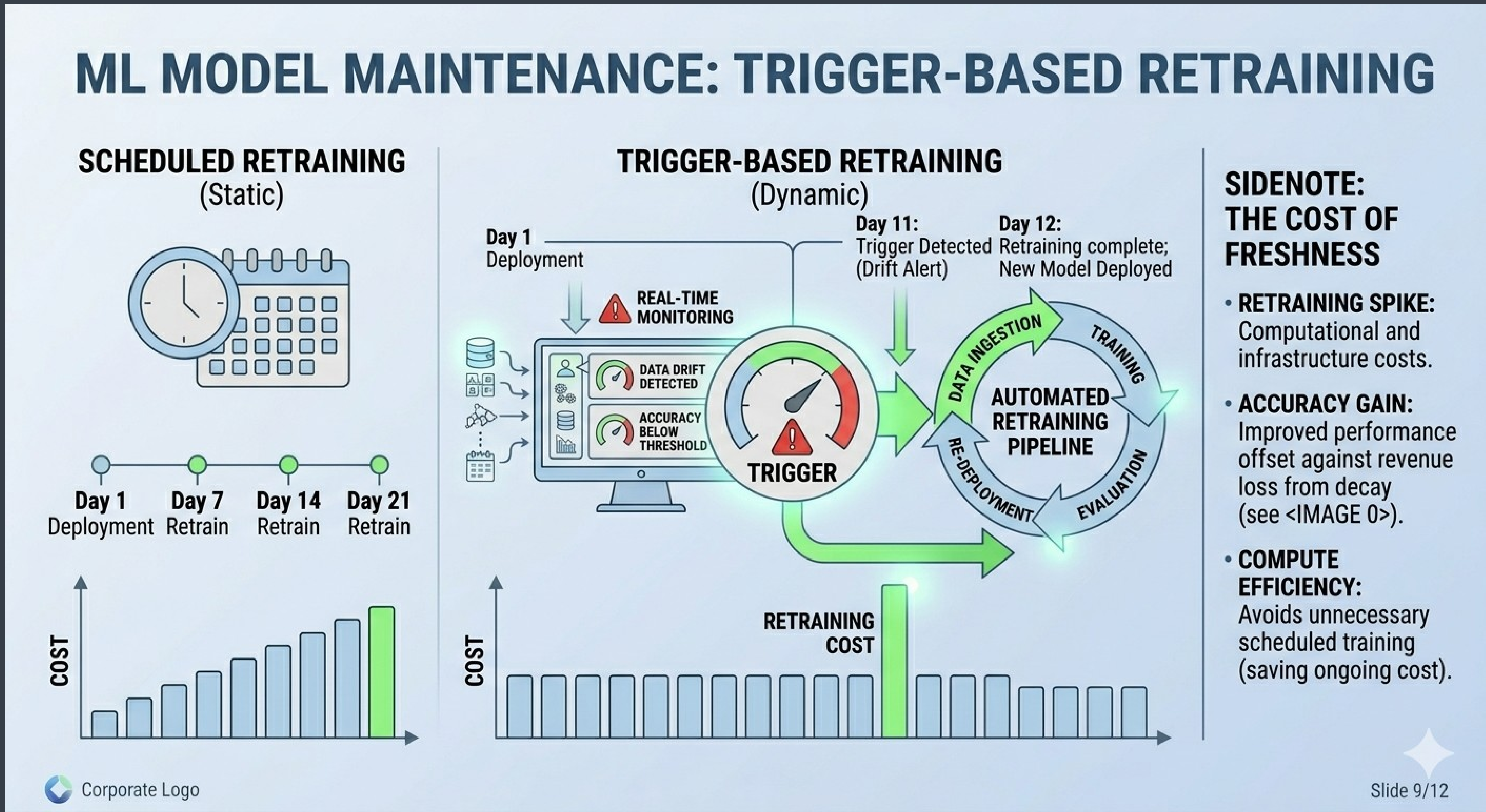


Maintenance Strategies

Automated retraining requires a significant architectural change.

Data Ingestion & Training: Automated pipelines handle the intake of new real-world data and execute the training process without manual intervention.

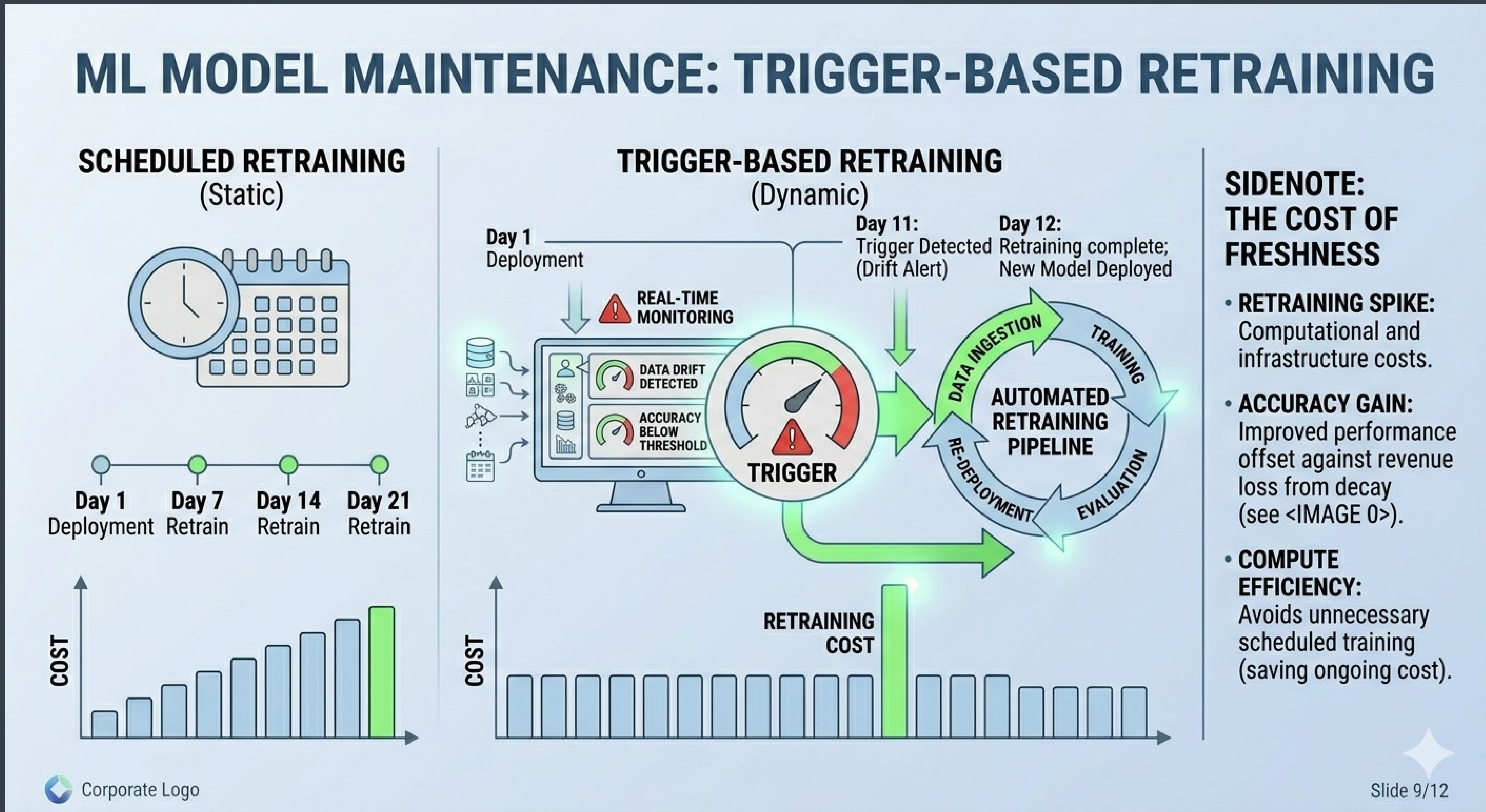
Evaluation & Re-deployment: Models must pass an automated evaluation phase before they are pushed back into the live environment.



Maintenance Strategies

Targeted Retraining: Maintenance can be applied to specific failing business segments (e.g., "Low Income" or "Niche SKUs") rather than the entire population.

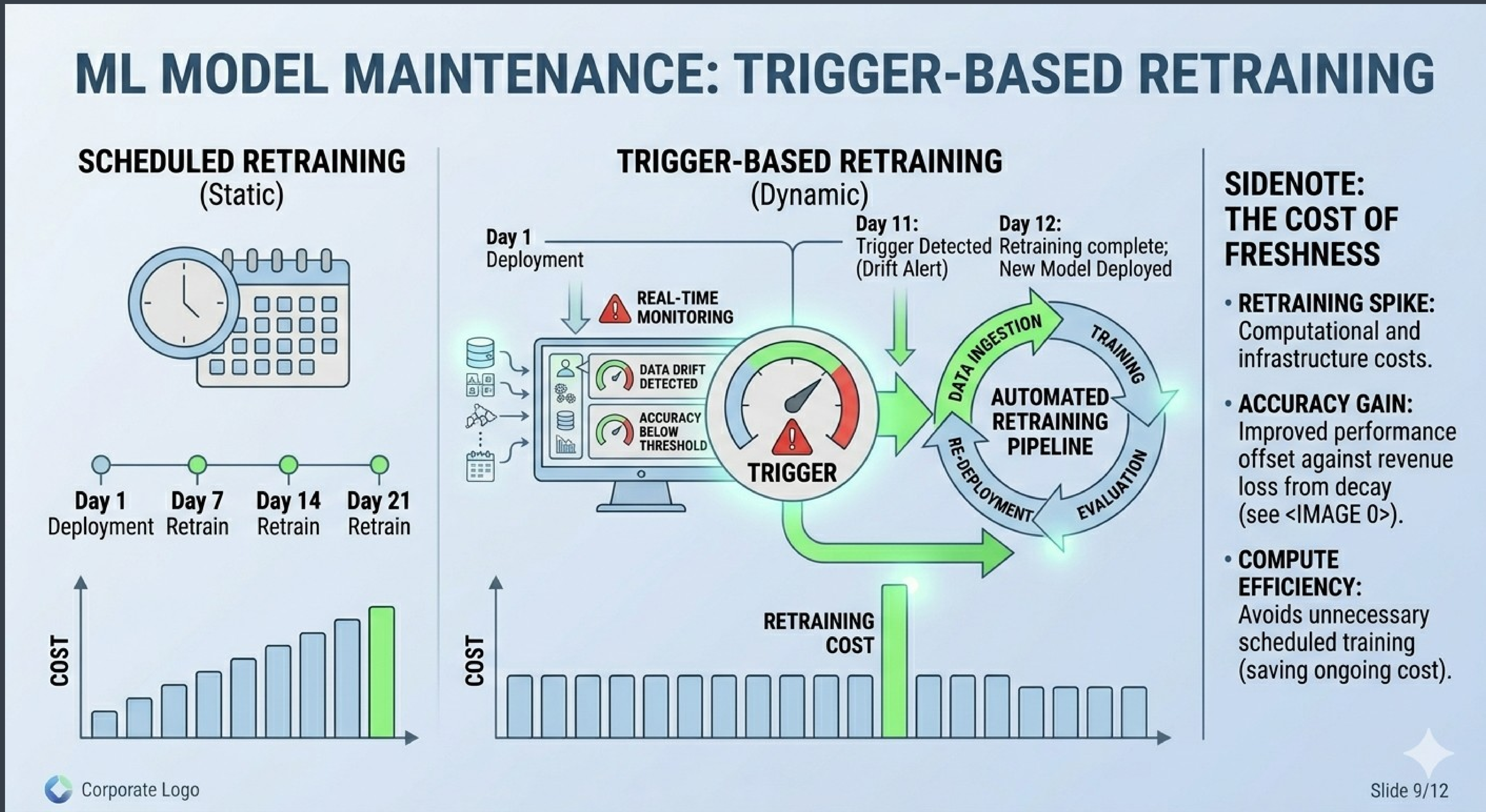
We need tools that can be automated as well as data resources that can feed these tools.



Maintenance Strategies

As of today (2026), the toolset landscape for Automated Retraining Architectures has consolidated into three primary categories: Cloud-Native Ecosystems, Open-Source Stacks, and Specialized Best-of-Breed Platforms.

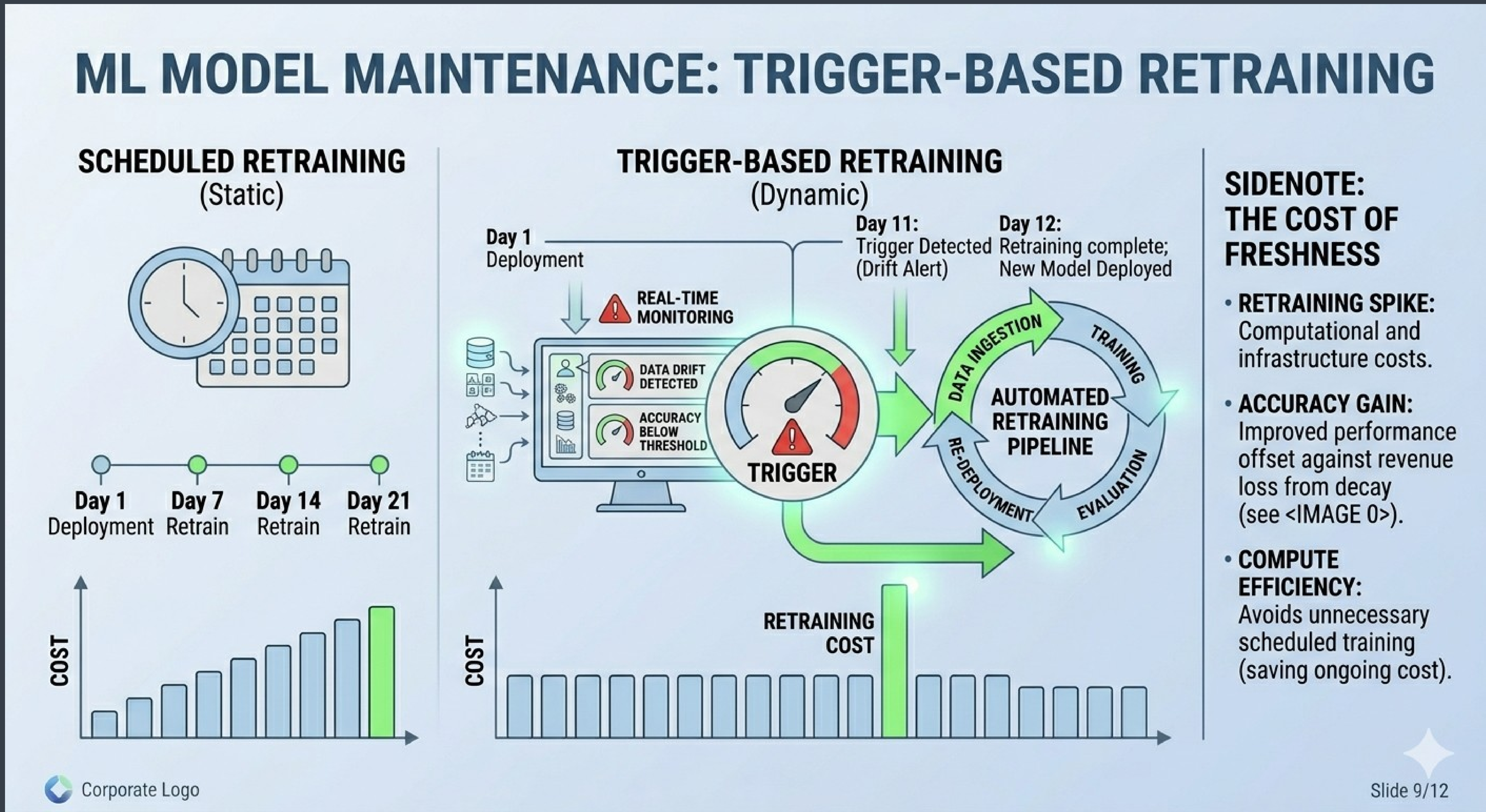
Choosing an alternative depends on your organization's "data gravity"—the principle that compute should be located where your data already lives.



Maintenance Strategies

Cloud-Native Integrated Ecosystems: These platforms offer the most seamless experience for automated retraining because they integrate monitoring, data ingestion, and compute within a single environment.

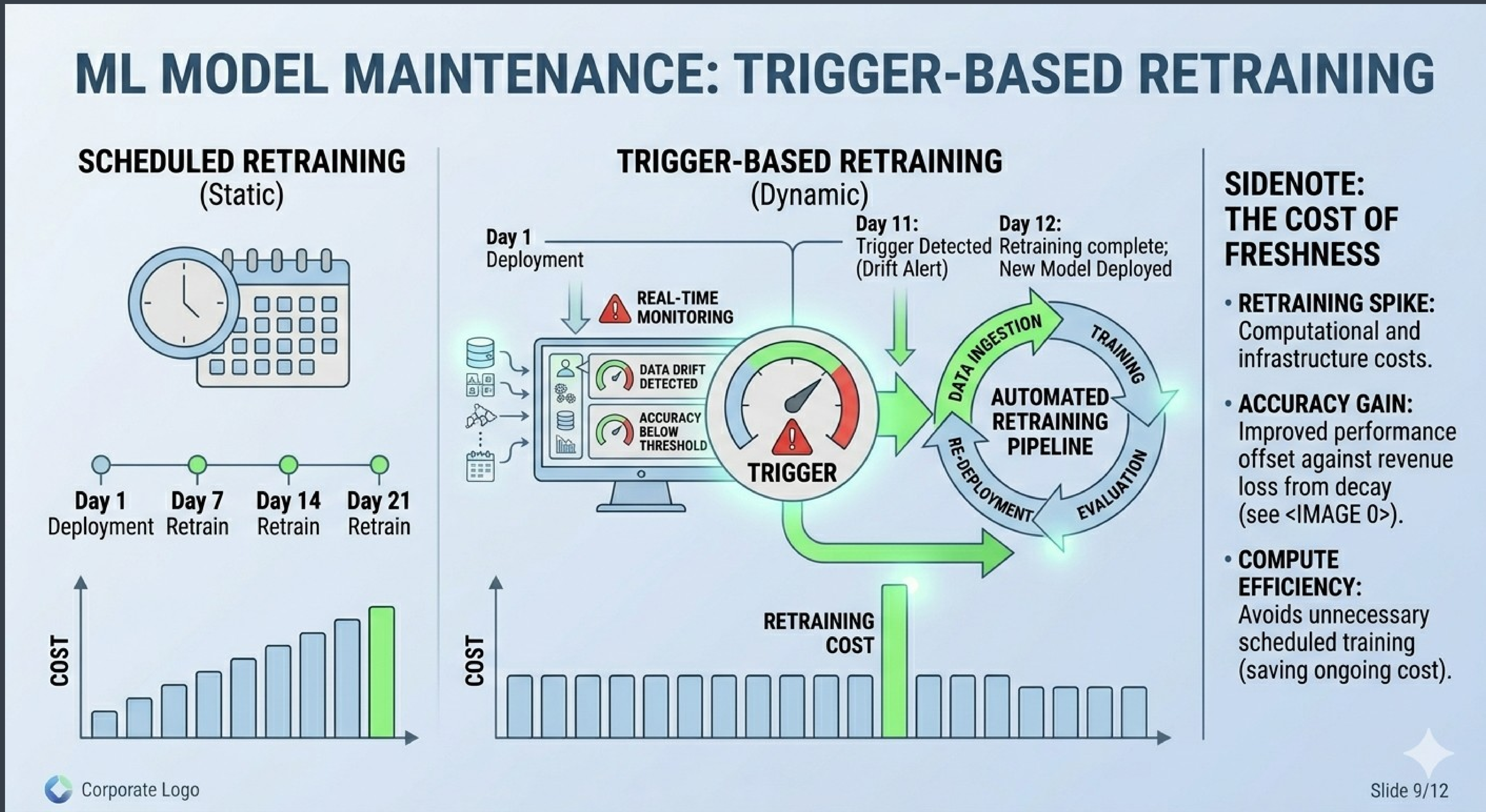
Amazon SageMaker, Google Vertex AI, Azure Machine Learning.



Maintenance Strategies

Open-Source Stacks (High Flexibility): Open-source alternatives provide maximum control and avoid vendor lock-in, but often require higher engineering overhead for maintenance.

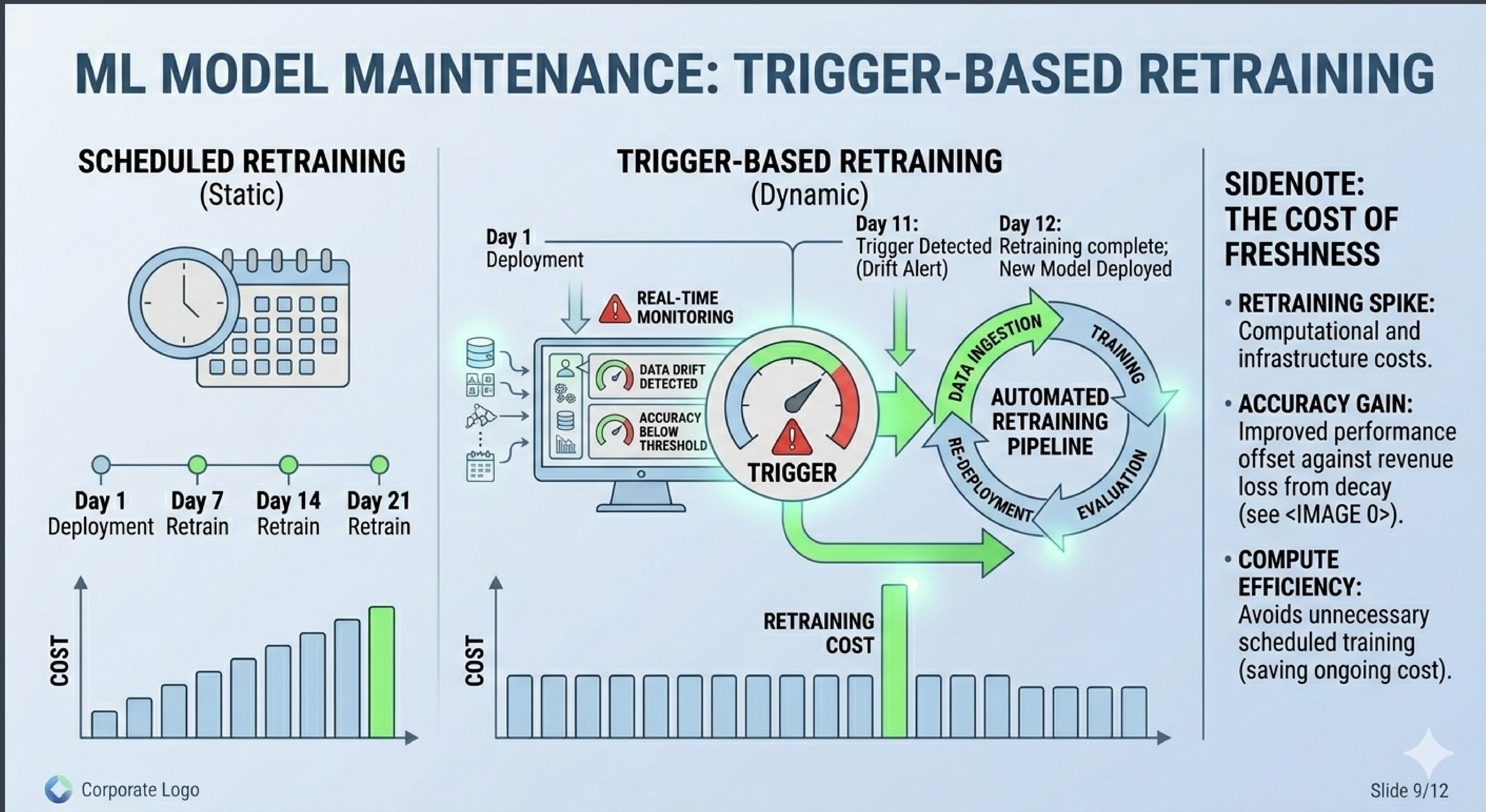
Kubeflow, ML flow, Apache Airflow, ZenML, Apache Spark.



Maintenance Strategies

Specialized Alternatives: These platforms focus on specific components of the retraining loop, such as monitoring or experiment management, and can be plugged into existing architectures.

SAS Viya, IBM Watsonx, Databricks Mosaic AI, Arize AI.



Thank You

Please follow up on Github and your emails (school emails).

Bora GÜNGÖREN